



POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Entanglement-Based Quantum Key Distribution Systems

TESI DI LAUREA MAGISTRALE IN
TELECOMMUNICATION ENGINEERING - INGEGNERIA DELLE
TELECOMUNICAZIONI

Author: **Chiara Di Nardo**

Student ID: 260387

Advisor: Prof. Alberto Gatto

Co-advisors: Dr. Alessandro Gagliano

Academic Year: 2025-2026

Abstract

The advent of quantum computing poses a severe threat to modern cryptographic infrastructure, since algorithms such as Shor’s algorithm are capable of breaking public-key cryptosystems in polynomial time. To overcome this vulnerability, Quantum Key Distribution (QKD) offers a radical paradigm shift, providing information-theoretic security guaranteed by the laws of quantum physics rather than computational complexity. This thesis presents the experimental realization, characterization, and numerical simulation of an entanglement-based QKD system implementing the BBM92 protocol. The architecture uses a source of polarization-entangled photon pairs generated via Spontaneous Parametric Down-Conversion (SPDC) within the C-band. To optimize alignment and manage data acquisition, a dedicated software with a graphical user interface was also developed. The experimental laboratory implementation, carried out using Single-Photon Avalanche Diodes (SPADs), yielded a Secure Key Rate (SKR) of 3.2 bits/s under an optimized coincidence time window of $t_{cc} = 0.68$ ns. Under these conditions, the system recorded a Quantum Bit Error Rate (QBER) of 1.6% in the rectilinear (Z) basis and 10.9% in the diagonal (X) basis; this asymmetry highlights the channel’s susceptibility to phase fluctuations along the fiber link. To evaluate the performance of the infrastructure beyond the physical constraints of the laboratory, a numerical simulation tool was developed and validated. The simulations indicate that upgrading the setup with Superconducting Nanowire Single-Photon Detectors (SNSPDs) would allow the optimal time window to be narrowed to just 25.5 ps, drastically suppressing accidental coincidences. This technological upgrade would achieve a SKR of 602.7 bits/s and allow to extend the potential range of secure quantum communication up to 630 km.

Keywords: Quantum Key Distribution, Entanglement, Quantum communication, Cryptography.

Abstract in lingua italiana

L'avvento del calcolo quantistico rappresenta una grave minaccia per le moderne infrastrutture crittografiche, poiché algoritmi come quello di Shor sono in grado di violare i sistemi a chiave pubblica in tempo polinomiale. Per superare questa vulnerabilità, la Quantum Key Distribution (QKD) offre un radicale cambio di paradigma radicale, garantendo una sicurezza teorico-informativa basata sulle leggi della fisica quantistica anziché sulla complessità computazionale. Questa tesi presenta la realizzazione sperimentale, la caratterizzazione e la simulazione numerica di un sistema QKD basato sull'entanglement che implementa il protocollo BBM92. L'architettura utilizza una sorgente di coppie di fotoni entangled in polarizzazione, generati tramite Spontaneous Parametric Down-Conversion (SPDC) nella banda C. Per ottimizzare l'allineamento e gestire l'acquisizione dei dati, è stato inoltre sviluppata un software dedicato dotato di un'interfaccia grafica (GUI). L'implementazione sperimentale in laboratorio, effettuata utilizzando diodi a valanga a singolo fotone (Single-Photon Avalanche Diodes, SPAD), ha permesso di raggiungere un Secure Key Rate (SKR) di 3.2 bits/s, ottimizzando la finestra di coincidenza temporale a $t_{cc} = 0.68$ ns. In queste condizioni, il sistema ha registrato un Quantum Bit Error Rate (QBER) dell'1.58% nella base rettilinea (Z) e del 10.90% nella base diagonale (X); questa asimmetria evidenzia la suscettibilità del canale alle fluttuazioni di fase lungo il collegamento in fibra ottica. Per valutare le prestazioni dell'infrastruttura oltre i limiti fisici del laboratorio, è stato sviluppato e validato uno strumento di simulazione numerica. Le simulazioni indicano che il passaggio a rivelatori a nanofili superconduttori (SNSPD) consentirebbe di restringere la finestra temporale ottimale a soli 25.47 ps, eliminando drasticamente le coincidenze accidentali. Questo specifico upgrade tecnologico permetterebbe di raggiungere un SKR di 602.66 bits/s e aumenterebbe la perdita di canale massima tollerabile da 5 dB a ben 63 dB, estendendo la portata potenziale della comunicazione quantistica sicura fino a 630 km.

Parole chiave: Distribuzione Quantistica di Chiavi, Entanglement, Comunicazione Quantistica, Crittografia.

Contents

Abstract	i
Abstract in lingua italiana	iii
Contents	v
Introduction	1
1 Quantum information	5
1.1 Principles of Quantum Mechanics	5
1.2 The Quantum Bit	7
1.3 Entanglement in Two-Level Systems	10
1.3.1 From Spin to Qubits	10
1.3.2 Bell's Theorem and CHSH Inequality	11
1.3.3 Bell States and Fidelity	16
1.4 Entanglement distribution	21
2 Quantum Cryptography	23
2.1 Classical Cryptographic Systems	23
2.2 Quantum Key Distribution	25
2.3 BBM92 Protocol	29
2.4 Physical Implementation	31
2.4.1 SPDC Source	32
2.4.2 Single Photon Detectors	33
2.4.3 Time Tagger	35
3 System Parametrization and BBM92 Numerical Modeling	37
3.1 Source Characterization	37
3.1.1 The Sagnac Interferometer Source	37
3.1.2 Optical Power Characterization	38

3.1.3	Spectral Analysis	42
3.1.4	Coincidence Peak Identification	46
3.2	Optimization Model of BBM92	51
4	Interface Tool for Experimental Setup Management	59
4.1	Software Architecture	60
4.2	Graphical User Interface (GUI)	62
4.2.1	Tab 1: SPAD Parameters Configuration	62
4.2.2	Tab 2: Dark Counts and Background Calibration	63
4.2.3	Tab 3: Live Counts, Coincidences, and Data Export	64
5	BBM92 Implementation and Results	69
5.1	Experimental Setup	69
5.2	Experimental Procedure and Measurement Phases	71
5.2.1	Initial Hardware Configuration	71
5.2.2	Noise Calibration and Polarization Alignment	72
5.2.3	Data Analysis and Key Rate Evaluation	77
5.3	Simulation Framework and Comparative Analysis	85
5.3.1	Parameter Estimation and Model Validation	85
5.3.2	Performance Evaluation of a BBM92 System	89
6	Conclusions and Future Developments	97
	Bibliography	101
	List of Figures	105
	List of Tables	109
	Acknowledgements	111

Introduction

¹ The statement “God does not play dice,” formulated by Albert Einstein in a 1926 letter to Max Born [1], reflects his skepticism toward the probabilistic interpretation of quantum mechanics. This position highlights a broader conceptual tension with a theory that, despite its remarkable predictive success, appeared counterintuitive and, in some respects, incomplete in its description of physical reality.

This critical perspective was systematically formulated in the Einstein–Podolsky–Rosen paradox (1935)[2], born to prove that quantum mechanics was an incomplete theory. Nevertheless, these considerations contributed opened the doors to understanding the most extraordinary and revolutionary phenomenon in quantum physics: **entanglement**.

In modern quantum information science, these principles are used as fundamental resources for secure communication. By using unbreakable laws of physics (such as the no-cloning theorem and measurement-induced disturbance) Quantum Key Distribution (QKD) allows the sharing of secure cryptography keys without relying on mathematical complexity. This aspect has become particularly relevant with the development of quantum computing. Quantum algorithms, such as the one proposed by Peter Shor, can solve these problems in polynomial time, thereby compromising the security of many widely used cryptographic schemes. On the other hand, QKD solves this vulnerability by offering information-theoretic security guaranteed by the laws of nature itself.

The present thesis is situated within this framework, focusing on the experimental realization and characterization of an entanglement-based quantum communication system and on the evaluation of its performance by suitable numerical simulations. In particular, an experimental setup implementing the BBM92 protocol has been developed, based on the use of polarization-entangled photon pairs to produce cryptographic keys. The system has been designed, implemented, and experimentally characterized in order to verify its correct operation and to evaluate its performance under laboratory conditions.

¹Editorial Note: To refine the clarity and linguistic consistency of this manuscript, the author used Google Gemini language model as a writing assistant. The tool was employed exclusively for grammar correction, style refinement and improving textual fluency. All scientific concepts, theoretical models, analyses, results and discussions remain entirely the original work of the author.

Carrying these theoretical protocols into a concrete experimental framework required a substantial effort in setting up the apparatus and calibrating the laboratory instrumentation. A central challenge of this implementation involved the developing of precise methodologies to stabilize and control the polarization of light along the optical path, since polarization fluctuations can drastically degrade both the quality of the quantum state and the resulting key rate. To ensure efficient control over the setup, a custom software architecture and Graphical User Interface (GUI) were developed enabling automated data acquisition, real-time monitoring of photon counts, and live coincidence processing. Finally, the experimental framework is complemented by a numerical tool, developed to perform a sensitivity analysis of the system under optimal conditions. This allows for the evaluation of a potential technological upgrade of the detection setup, demonstrating in particular how the upgrade can maximize both the secure key rate and the overall robustness of the system.

The thesis is organized as follows:

- **Chapter 1:** an overview of quantum mechanics up to the introduction of the concept of quantum bit. Following this, quantum entanglement is explored alongside the key theorems that characterize it, with a specific focus on Bell's theorem and the CHSH inequality.
- **Chapter 2:** it presents the state of the art of Quantum Key Distribution focusing on entanglement-based protocols. In particular, it provides a detailed analysis of the BBM92 protocol, introducing the physical hardware required for its implementation: the SPDC source, Single Photon Detectors, and the Time Tagger.
- **Chapter 3:** it provides a characterization of the source used to implement the protocol, analyzing in detail how entangled photons are generated through a Sagnac interferometer. Furthermore, it covers the evaluation of the emitted source power, as well as the methods used for coincidence peak identification and emission spectrum analysis.
- **Chapter 4:** it presents the developed Python software and its Graphical User Interface (GUI).
- **Chapter 5:** it is dedicated to the final experimental outcomes. It outlines the configuration of the setup, illustrates the measurement phases, and provides a detailed evaluation of the Secure Key Rate (SKR). Furthermore, these experimental results are utilized to validate a numerical simulation tool, which evaluates the system's behavior under favorable conditions and offers a prospective outlook on fu-

ture hardware upgrades, such as Superconducting Nanowire Single-Photon Detector (SNSPD).

- **Chapter 6:** it outlines potential future developments and research directions in the field of entanglement-based QKD and entanglement distribution.

1 | Quantum information

In quantum mechanics, the physical state of a system is described by a well-defined mathematical formalism, proving a rigorous framework for the description of microscopic physical phenomena. This chapter introduces the fundamental concepts required to describe quantum states and their properties, which will be used in the discussion of quantum key distribution (QKD) protocols. In particular, the phenomenon of quantum entanglement is examined with its properties and foundational theorems that govern it.

1.1. Principles of Quantum Mechanics

In contrast to classical physics, where the state of a physical system is described by quantities with precise and simultaneously measurable values, quantum mechanics is based on a probabilistic description of physical phenomena.

The mathematical representation of quantum states is formalized through the bra-ket notation introduced in 1939 by P. A. M. Dirac [3]. Within this formalism, a quantum state is represented by a unit vector within a complex vector space equipped with a scalar product, known as Hilbert space. In particular, a state is denoted by a *ket* $|\psi\rangle$, while the associated dual vector $\langle\psi|$ is called *bra*. This vector contains all the information about the physical system which remains indivisible until the moment of measurement.

In quantum mechanics, measurement corresponds to choosing a specific property of the physical system, known as an observable [4]. Mathematically, each observable is associated with a Hermitian operator $\hat{A}$, whose properties ensure physical coherence:

- its eigenvalues $\{a_k\}$ are real numbers, corresponding to the possible physical outcomes of the measurement;
- its eigenvectors $\{|\psi_k\rangle\}$ are mutually orthogonal and form an orthonormal basis for the Hilbert space.

Due to the completeness of the data, every vector $|\psi\rangle$ can be expanded as:

$$|\psi\rangle = \sum_k c_k |\psi_k\rangle \quad (1.1)$$

where $c_k = \langle \psi_k | \psi \rangle$ is a complex coefficient that quantifies the contribution of each eigenvector to the global state, and $\{|\psi_k\rangle\}$ is the eigenvectors that denotes the set of all possible directions onto which the system can project itself at the moment of measurement. This linear combination is the formal expression of the principle of superposition: the system is not confined to a single eigenstate; instead, it is distributed across all configurations permitted by the basis $\{|\psi_n\rangle\}$ according to the weights c_k . At the moment of the experimental measurement, the state vector instantaneously collapses onto one of the eigenvectors of the chosen operator $\hat{A}$

$$\hat{A} |\psi_k\rangle = a_k |\psi_k\rangle. \quad (1.2)$$

The value obtained is the eigenvalue a_k corresponding to that eigenvector and represents the result of the measurement operation. The probability of this collapse is governed by Born's Law, introduced by Max Born in 1926, according to which:

$$P(a_k) = |c_k|^2 = \|\langle \psi_k | \psi \rangle\|^2 \quad (1.3)$$

In conclusion, the measurement process can be regarded as a black box (Fig. 1.1): given an input quantum state $|\psi\rangle$, it returns an eigenvalue of the observable being measured with probability $P(a_k)$.

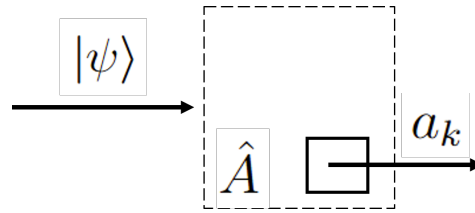


Figure 1.1: Representation of measure

In physical terms, measurement can be interpreted as an interaction between the quantum state and a macroscopic apparatus. This interaction induces an unavoidable perturbation of the quantum state, leading to the selection of a particular observable ($\hat{A}$). Consequently, a state initially described as a superposition of multiple configurations evolves, upon measurement, into the collapse of a single outcome: $|\psi\rangle \rightarrow |\psi_k\rangle \rightarrow a_k$.

In addition, every measurement process alters the system, making it impossible to si-

multaneously determine all physical quantities with arbitrary precision. This limitation is formalized by the Heisenberg uncertainty principle, according to which pairs of conjugate observables, such as position and momentum, cannot be simultaneously determined with arbitrary precision. The interaction required to measure one quantity inevitably introduces an uncertainty in the conjugate variable.

This framework differs fundamentally from classical physics, where measurement is generally regarded as a passive process that does not affect the state of the observed system.

1.2. The Quantum Bit

In classical communication, the fundamental unit of information is the bit, which can take only the logical values 0 and 1. In the quantum setting, however, information is encoded in quantum bits, or Qubits, which represent the quantum generalization of the classical bit [5]. Unlike the classical case, a Qubit is not restricted to a definite state, but can exist in a coherent superposition of its basis states. In particular, a general Qubit state can be written as a linear superposition of two states forming an orthonormal basis on Hilbert space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle. \quad (1.4)$$

The complex coefficients α and β satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$, while $|0\rangle$ and $|1\rangle$ form the computational basis of the system. These states are chosen to be orthonormal, i.e. $\langle 0|1\rangle = 0$ and $\langle 0|0\rangle = \langle 1|1\rangle = 1$, ensuring a consistent probabilistic description of measurement outcomes. This structure highlights that the Qubit represents the simplest quantum mechanical system, described by a two-dimensional state space spanned by the orthonormal basis $|0\rangle, |1\rangle$. In this context, quantum information is not associated with a deterministic value, but rather with a coherent superposition of basis states, from which probabilistic outcomes emerge until measurement.

A particularly useful representation of a Qubit state is the Bloch sphere, which provides a geometric interpretation of the state space of a two-level quantum system. Any pure Qubit state can be mapped onto a point on the surface of a unit sphere in three-dimensional space. This representation exploits the fact that a general Qubit state is fully determined, up to a global phase, by two real parameters. Specifically, the state, by using the polar coordinates, can be rewritten in the form:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle, \quad (1.5)$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ are spherical angular parameters that uniquely define a point on the sphere. In this representation, the basis states $|0\rangle$ and $|1\rangle$ correspond to the north and south poles of the Bloch sphere, respectively, while superposition states lie on intermediate points of the surface.

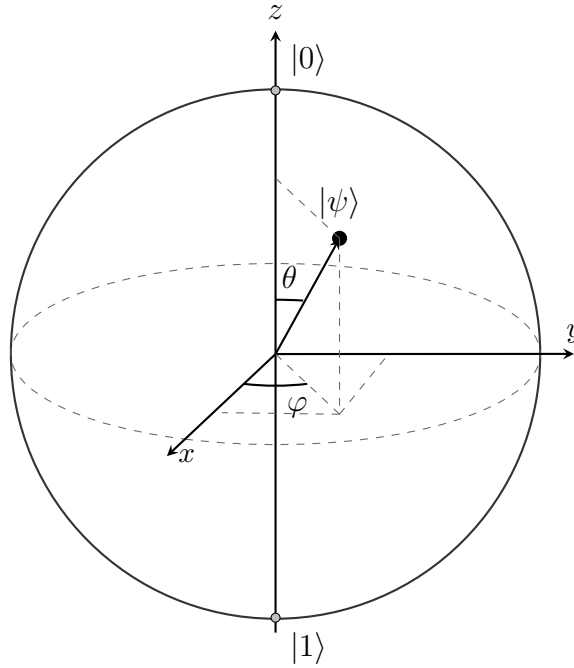


Figure 1.2: Bloch sphere representation of a Qubit adapted from [6].

The Bloch sphere provides not only an intuitive visualization of Qubit states, but also a powerful geometric interpretation of quantum operations. In fact, unitary transformations acting on a Qubit correspond to rotations of the state vector on the surface of the sphere, which is particularly useful for understanding quantum gates and state evolution in quantum information processing.

The description of a Qubit through the state vector $|\psi\rangle$ and its mapping on the surface of the Bloch sphere refers to what is known as a **pure state**. A pure state represents the most complete possible description of a quantum system. The system is in a superposition of fundamental states between which there is a well-defined phase relationship. This relationship, known as quantum coherence, is responsible for the interference phenomena. The uncertainty of the measurement outcome does not derive from a lack of information, but is intrinsic to the nature of the quantum system.

In contrast, a mixed state describes a situation in which the information about the system is incomplete or degraded, for example, under a realistic experimental condition where the system is rarely perfectly isolated. In this case, the state is described as a statistical combination of pure states. The uncertainty associated with a mixed state is of a classical nature: the system is in a precise state (not a superposition), but the observer does not know which one. There are no phase relationships between the components.

To describe both pure and mixed states, it is useful to introduce a formalization [4] using a density matrix ρ :

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i| \quad (1.6)$$

where $|\psi_i\rangle$ denotes the set of possible pure states in the ensemble, and p_i represents the classical probability associated with each state $|\psi_i\rangle$.

In quantum communication, the **polarization of light** is widely used as a physical degree of freedom for encoding quantum information. In particular, a Qubit can be physically implemented by the polarization state of a single photon. In classical optics, light is described as an electromagnetic wave, and its polarization is a property of the wave that characterizes the temporal evolution of the electric field in the plane orthogonal to the direction of propagation. The components of the electric field can be conveniently represented using the Jones vector formalism, which provides a compact description of the polarization state in terms of a two-dimensional complex vector. This formalism is used also in the quantum case. So, there is a correspondence between the state of polarization and a computational basis vector:

$$|H\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1.7)$$

$$|V\rangle = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.8)$$

For polarization encoded Qubits there are additional states to explore: diagonal, anti-diagonal, right-circular, and left-circular.

$$|D\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.9)$$

$$|A\rangle = \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.10)$$

$$|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (1.11)$$

$$|L\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (1.12)$$

1.3. Entanglement in Two-Level Systems

Consider a system composed of two quantum states. The overall system is described by introducing a new vector space, defined as the tensor product of the respective Hilbert spaces:

$$\mathcal{H}_{tot} = \mathcal{H}_A \otimes \mathcal{H}_B$$

Within this space, it is possible to describe all potential configurations of the system. This includes not only the combinations of the basis states of the individual systems, but also every possible superposition between them.

In this vector space, we can distinguish between two types of state: separable states and entangled states. A state is said to be **separable** when it can be expressed as the tensor product of the two individual states. In this scenario, the two subsystems are independent, and their properties can be described individually. On the other hand, we speak of **entangled** states when this factorization is not possible. In this case, there are quantum correlations between the states that cannot be attributed to simple classical correlations. When a system is entangled, it no longer makes sense to describe the individual states separately. Instead, the system must be considered as a single entity since the properties of the individual states are no longer independent, but intrinsically linked to each other.

1.3.1. From Spin to Qubits

A fundamental physical example of such a composite quantum system is the spin of two elementary particles. In this case, the tensor product structure describes the combined spin state of the system and naturally provides a situation where entanglement can occur. In the case of the electron, a measurement of the spin component along a given direction always yields only two possible discrete values: spin-up and spin-down, corresponding respectively to $+\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$. This binary nature shows that, in quantum mechanics, spin is quantized.

This quantized nature is formally described by introducing the spin observable operator $\hat{S}_z$, whose eigenvalues are $+\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$, associated respectively with the eigenstates $|\uparrow\rangle$ and $|\downarrow\rangle$:

$$\hat{S}_z|\uparrow\rangle = +\frac{\hbar}{2}|\uparrow\rangle, \quad \hat{S}_z|\downarrow\rangle = -\frac{\hbar}{2}|\downarrow\rangle.$$

These eigenstates form a complete orthonormal basis of a two-dimensional Hilbert space describing the spin degree of freedom of the electron. This implies that any generic spin state can be expressed as a linear combination of these basis states:

$$|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle, \quad \alpha, \beta \in \mathbb{C}.$$

The coefficients α and β determine the probabilities of obtaining the corresponding measurement outcomes along the chosen direction.

We now consider the case in which a particle with zero total spin decays into two electrons. Due to the conservation of angular momentum, the sum of the spins of the two electrons must remain zero. It follows that, if the spin of one of the particles is measured and a given value is obtained, the spin of the other must necessarily be opposite. However, prior to measurement, the system cannot be described by assigning definite states to the individual particles. Instead, it is represented by a single global quantum state. In this situation, the electrons do not possess well-defined individual properties, but are described by a coherent superposition of joint states. Only at the moment of measurement does the system reduce to one of the possible outcomes, and a measurement performed on one particle immediately determines the result of the other. This occurs independently of the distance separating them and in the absence of any subsequent interaction. This statistical dependence between measurement outcomes is the manifestation of quantum correlations.

The situation described above is an example of **quantum entanglement**; a system consisting of two spin-entangled electrons represents the simplest example of a two-level quantum system. It provides the fundamental model used to describe Qubits in quantum information theory.

1.3.2. Bell's Theorem and CHSH Inequality

In the famous EPR paradox (Einstein, Podolsky, Rosen) [2] of 1935, the authors questioned the completeness of quantum mechanics. This doubt originated from the analysis of a phenomenon that was later called Entanglement by Erwin Schrödinger in the same

year, in response to the EPR paradox, and which Einstein described as ‘spooky action at a distance’ [1]. This phenomenon allows two particles to remain correlated even when separated by arbitrarily large distances. In fact, a measurement performed on one of the two particles instantly determines the outcome of the measurement on the other. According to Einstein, this behavior could be explained in only two ways:

1. **The violation of Relativity:** admitting that information is transmitted between particles at a speed greater than that of light in a vacuum ($v > c$). This would contradict the theory of Special Relativity, which imposes c as an insurmountable speed limit.
2. **The existence of hidden variables:** to respect the principle of locality (the idea that an object can only be influenced by its immediate surroundings, and that influencing something far away requires a signal traveling no faster than the speed of light), Einstein hypothesized that particles already possessed well-defined properties (**hidden variables**) before being measured; the measurement, therefore, only reveals these pre-existing properties (*Realism*).

The EPR argument therefore suggests that quantum mechanics is incomplete: if locality (and thus Relativity) is assumed to be valid, the results of measurements must be determined by pre-existing hidden variables. The absence of these variables within quantum theory implies that it does not provide a complete description of the system.

Later, Bell responded to this paradox by proving in his paper [7] that quantum mechanics violates the principle of local realism. He demonstrated that if local hidden variables existed, the results of physical experiments would have to satisfy strict numerical constraints, known as Bell’s inequalities. However, quantum mechanics predicts correlations that violate these limits. Consequently, the argument that quantum mechanics is an incomplete theory (as suggested by EPR) is no longer tenable. Bell’s theorem proves that no theory based on local hidden variables can reproduce quantum mechanics, establishing it instead as a fundamentally non-local theory.

To experimentally verify the validity of the principle of locality and test for the existence of hidden variables, various experiments have been proposed [8, 9]. These include tests based on the CHSH inequality, named after Clauser, Horne, Shimony and Holt which provides a practical framework for testing local realism through polarization measurements on pairs of entangled photons [9]. In this experimental setup, two distant observers, Alice and Bob, perform measurements on the polarization of incoming entangled photon pairs. Each observer independently chooses between two possible measurement directions, representing the geometric orientations of their respective polarizers:

- Alice chooses between directions $\hat{n}_A$ (associated with an angle α) or $\hat{n}'_A$ (associated with an angle α').
- Bob chooses between directions $\hat{n}_B$ (associated with an angle β) or $\hat{n}'_B$ (associated with an angle β').

Each measurement directly projects the photon's state onto the selected axis. The result is recorded as a Vertical outcome (V) if the photon's polarization is parallel to the polarizer's direction, or as a Horizontal outcome (H) if it is perpendicular. Assuming the source generates photon pairs in a maximally entangled singlet state, quantum mechanics predicts that the detection probabilities are governed by Born's law. Specifically, the probability of recording a Vertical state is proportional to $\cos^2(\theta)$, while a Horizontal state is proportional to $\sin^2(\theta)$. Here, the relative angle θ represents the geometric difference between the orientation chosen by Bob (β) and the one chosen by Alice (α):

$$\theta = \beta - \alpha \quad (1.13)$$

To generalize this approach, the joint probability P_{ij} of Alice detecting outcome i and Bob detecting outcome j (where $i, j \in \{V, H\}$) can be expressed through conditional probability as:

$$P_{ij} = P(i_A) \cdot P(j_B | i_A) \quad (1.14)$$

Here, $P(i_A)$ represents the probability of Alice's photon collapsing into state i , while $P(j_B | i_A)$ is the conditional probability that Bob's photon collapses into state j given Alice's result. Applying this general relation to the singlet state yields the specific joint probabilities:

- **Coincident outcome** (P_{VV} and P_{HH}): the probabilities that both photons are detected with the same polarization relative to the chosen axes.

$$P_{VV}(\hat{n}_A, \hat{n}_B) = \frac{1}{2} \cos^2(\theta) \quad (1.15)$$

$$P_{HH}(\hat{n}_A, \hat{n}_B) = \frac{1}{2} \cos^2(\theta) \quad (1.16)$$

- **Anti-coincidence outcomes** (P_{VH} and P_{HV}): the probabilities that the photons exhibit opposite polarizations.

$$P_{VH}(\hat{n}_A, \hat{n}_B) = \frac{1}{2} \sin^2(\theta) \quad (1.17)$$

$$P_{HV}(\hat{n}_A, \hat{n}_B) = \frac{1}{2} \sin^2(\theta) \quad (1.18)$$

These values allow us to define the correlation coefficient $E(\hat{n}_A, \hat{n}_B)$ [4], which quantifies the degree of statistical dependence between the results of the two observers:

$$E(\hat{n}_A, \hat{n}_B) = P_{VV}(\hat{n}_A, \hat{n}_B) + P_{HH}(\hat{n}_A, \hat{n}_B) - P_{VH}(\hat{n}_A, \hat{n}_B) - P_{HV}(\hat{n}_A, \hat{n}_B) \quad (1.19)$$

A value of $E = 1$ indicates perfect correlation, -1 indicates perfect anti-correlation, while 0 indicates that the results are entirely independent. Substituting the probabilistic definitions derived from Born's law into equation (1.19) yields:

$$E(\hat{n}_A, \hat{n}_B) = \left[\frac{1}{2} \cos^2(\theta) + \frac{1}{2} \cos^2(\theta) \right] - \left[\frac{1}{2} \sin^2(\theta) + \frac{1}{2} \sin^2(\theta) \right] \quad (1.20)$$

Summing the identical fractions, the expression reduces to:

$$E(\hat{n}_A, \hat{n}_B) = \cos^2(\theta) - \sin^2(\theta) \quad (1.21)$$

Using the cosine double-angle trigonometric identity ($\cos^2 \theta - \sin^2 \theta = \cos(2\theta)$), the quantum correlation between the two angles can be expressed in its well-known compact form.

$$E(\hat{n}_A, \hat{n}_B) = \cos(2\theta) = \cos(2(\beta - \alpha)) \quad (1.22)$$

To verify the violation of local realism, the correlations obtained from the four possible combinations of measurement directions are summed into the parameter S according to the CHSH inequality:

$$S = E(\hat{n}_A, \hat{n}_B) - E(\hat{n}_A, \hat{n}'_B) + E(\hat{n}'_A, \hat{n}_B) + E(\hat{n}'_A, \hat{n}'_B) \quad (1.23)$$

According to theories based on local hidden variables, the value of S must satisfy the strict classical bound:

$$-2 \leq S \leq 2 \quad (1.24)$$

This range represents the insurmountable limit imposed by classical physics and the logic of local realism. However, quantum mechanics allows this threshold to be exceeded. To obtain the maximum theoretical violation possible, four specific angles are chosen, offset at regular intervals of 22.5° :

- Alice: $\alpha = 0^\circ$, $\alpha' = 45^\circ$
- Bob: $\beta = 22.5^\circ$, $\beta' = 67.5^\circ$

The relative angle θ and the respective correlation $E(\theta) = \cos(2\theta)$ (Figure 1.3) are calculated for each of the four combinations in formula (1.23):

1. First pair $(\hat{n}_A, \hat{n}_B)$: Relative angle $\theta_1 = \beta - \alpha = 22.5^\circ - 0^\circ = 22.5^\circ$

$$E(\hat{n}_A, \hat{n}_B) = \cos(2 \cdot 22.5^\circ) = \cos(45^\circ) = \frac{\sqrt{2}}{2} \approx 0.707 \quad (1.25)$$

2. Second pair $(\hat{n}'_A, \hat{n}_B)$: Relative angle $\theta_2 = \beta - \alpha' = 22.5^\circ - 45^\circ = -22.5^\circ$

$$E(\hat{n}'_A, \hat{n}_B) = \cos(2 \cdot (-22.5^\circ)) = \cos(-45^\circ) = \frac{\sqrt{2}}{2} \approx 0.707 \quad (1.26)$$

3. Third pair $(\hat{n}'_A, \hat{n}'_B)$: Relative angle $\theta_3 = \beta' - \alpha' = 67.5^\circ - 45^\circ = 22.5^\circ$

$$E(\hat{n}'_A, \hat{n}'_B) = \cos(2 \cdot 22.5^\circ) = \cos(45^\circ) = \frac{\sqrt{2}}{2} \approx 0.707 \quad (1.27)$$

4. Fourth pair $(\hat{n}_A, \hat{n}'_B)$: Relative angle $\theta_4 = \beta' - \alpha = 67.5^\circ - 0^\circ = 67.5^\circ$

$$E(\hat{n}_A, \hat{n}'_B) = \cos(2 \cdot 67.5^\circ) = \cos(135^\circ) = -\frac{\sqrt{2}}{2} \approx -0.707 \quad (1.28)$$

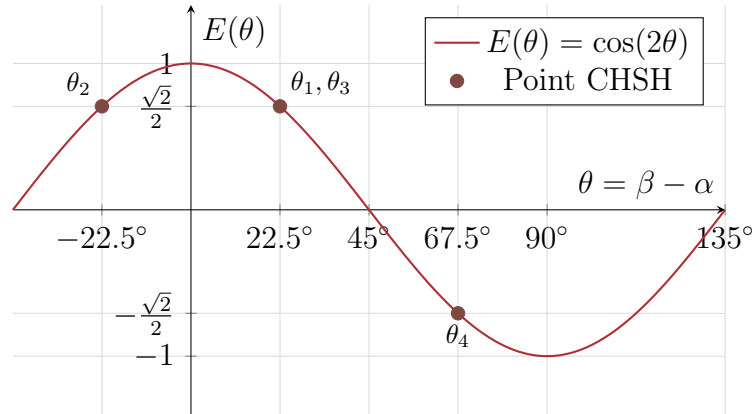


Figure 1.3: Quantum correlation coefficient as a function of the relative angle θ . The dots indicate the four geometric configurations used for the maximum violation of the CHSH inequality.

Substituting these numerical values into the parameter S gives:

$$S = \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right) + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = 4 \cdot \frac{\sqrt{2}}{2} = 2\sqrt{2} \approx 2.82 \quad (1.29)$$

This numerical value establishes the Tsirelson limit [10]. Since $2\sqrt{2} > 2$, the quantum formalism based on the probabilities of Born's law mathematically demonstrates the violation of local realism. The experimental confirmation of a value $S > 2$ demonstrates that particles do not possess definite properties prior to measurement and that quantum correlations exceed the possibilities of any local theory.

To visualize the continuous behavior of the parameter S as a function of a single angular shift θ , the measurement directions of the first observer (Alice) are held fixed at the optimal values $\alpha = 0^\circ$ and $\alpha' = 45^\circ$. In this way, the entire set of four CHSH angular combinations is made to depend exclusively on the continuous rotation of Bob's polarizers, defined as $\beta = \theta$ and $\beta' = \theta + 45^\circ$ (Fig. 1.4). Under this geometric constraint, the parameter reduces to the one-dimensional wave function $S(\theta) = 2 \cos(2\theta) + 2 \sin(2\theta)$.

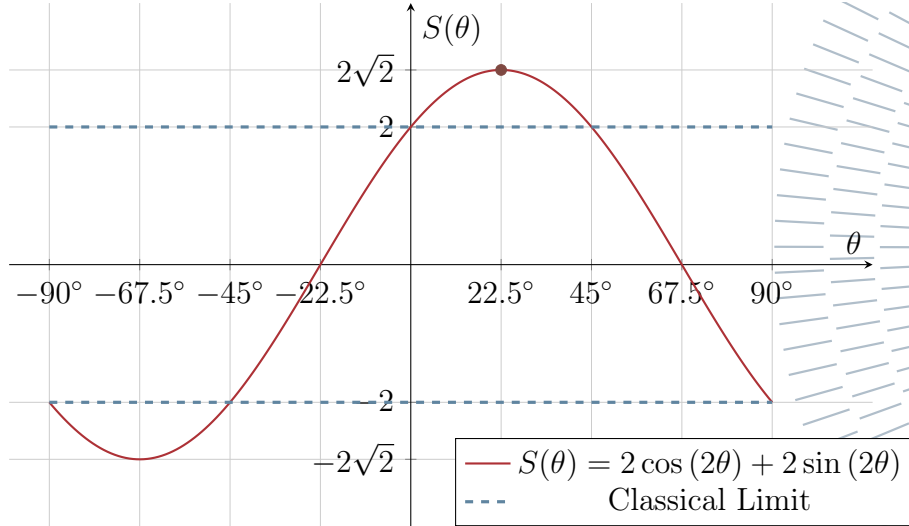


Figure 1.4: CHSH parameter S as a function of Bob's relative angle θ (with fixed $\alpha = 0^\circ, \alpha' = 45^\circ$). The quantum curve violates the classical local-realistic bounds (dashed lines, ± 2), reaching the Tsirelson limit ($2\sqrt{2}$) at $\theta = 22.5^\circ$.

1.3.3. Bell States and Fidelity

The states that allow for this maximum violation are known as Bell states, which represent examples of maximally entangled states. In quantum information, each Qubit is represented by a two-dimensional base $\{|1\rangle, |0\rangle\}$ which corresponds, in terms of polarization encoding, to $\{|V\rangle, |H\rangle\}$. In a two-dimensional space, the tensor product allows us to combine these basis vectors to form a new orthonormal basis $\{|HH\rangle, |VV\rangle, |VH\rangle, |HV\rangle\}$ and each state can be written as:

$$|\psi\rangle = \alpha|HH\rangle + \beta|HV\rangle + \gamma|VH\rangle + \eta|VV\rangle. \quad (1.30)$$

where $\alpha, \beta, \gamma, \eta$ are complex coefficients such that $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\eta|^2 = 1$

In this notation, the bell states are defined as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle) \quad (1.31)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|HH\rangle - |VV\rangle) \quad (1.32)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|HV\rangle + |VH\rangle) \quad (1.33)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|HV\rangle - |VH\rangle) \quad (1.34)$$

These states describe pairs of particles in which the measurement results are strongly correlated. To formalize these correlations, the measurement probabilities in different bases can be computed by applying Born's rule (1.3), which are subsequently combined through the definition of the correlation function (1.19).

Let us consider the entangled Bell state $|\psi^+\rangle$. To explicitly evaluate the joint probabilities, the four-dimensional basis vectors for the rectilinear framework are obtained by applying the tensor product ($\otimes$) to the single-photon states:

$$|HH\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \text{with} \quad \langle HH| = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \quad (1.35)$$

$$|VV\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \text{with} \quad \langle VV| = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} \quad (1.36)$$

$$|HV\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \text{with} \quad \langle HV| = \begin{pmatrix} 0 & 1 & 0 & 0 \end{pmatrix} \quad (1.37)$$

$$|VH\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \text{with} \quad \langle VH| = \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix} \quad (1.38)$$

Accordingly, the state $|\psi^+\rangle$ can be written in its vector form as:

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}. \quad (1.39)$$

By applying Born's rule, the probability $P_{HH} = |\langle HH|\psi^+\rangle|^2$ is computed by multiplying the row vector and the column vector:

$$\langle HH|\psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}(0) = 0 \implies P_{HH} = 0 \quad (1.40)$$

Following the same exact procedure for the parallel state $|VV\rangle$, we obtain:

$$\langle VV|\psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}(0) = 0 \implies P_{VV} = 0 \quad (1.41)$$

Conversely, the cross-polarized outcomes $|HV\rangle$ and $|VH\rangle$ lie precisely within the state's superposition channels. Performing the row-by-column multiplication yields:

$$\langle HV|\psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \implies P_{HV} = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (1.42)$$

$$\langle VH|\psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \implies P_{VH} = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (1.43)$$

Finally, substituting these four joint probabilities into the correlation function E gives:

$$E_Z = P_{VV} + P_{HH} - P_{VH} - P_{HV} = 0 + 0 - \frac{1}{2} - \frac{1}{2} = -1 \quad (1.44)$$

This result demonstrates a perfect anti-correlation in the Z basis, meaning that whenever Alice detects a specific polarization, Bob will observe the orthogonal one.

To evaluate the joint probabilities and the resulting correlation in the diagonal basis X , the tensor product is applied to obtain the four-dimensional basis vectors of the composite system:

$$|DD\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad \text{with} \quad \langle DD| = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} \quad (1.45)$$

$$|AA\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \quad \text{with} \quad \langle AA| = \frac{1}{2} \begin{pmatrix} 1 & -1 & -1 & 1 \end{pmatrix} \quad (1.46)$$

$$|DA\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \quad \text{with} \quad \langle DA| = \frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & -1 \end{pmatrix} \quad (1.47)$$

$$|AD\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \quad \text{with} \quad \langle AD| = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 & -1 \end{pmatrix} \quad (1.48)$$

By applying Born's rule, the probability $P_{DD} = |\langle DD|\Psi^+\rangle|^2$ is computed by multiplying the

row vector $\langle DD|$ and the column vector representing $|\Psi^+\rangle$:

$$\langle DD|\Psi^+\rangle = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \implies P_{DD} = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (1.49)$$

Following the same exact procedure for the parallel diagonal state $|AA\rangle$, the row-by-column multiplication yields:

$$\langle AA|\Psi^+\rangle = \frac{1}{2} \begin{pmatrix} 1 & -1 & -1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = -\frac{2}{2\sqrt{2}} = -\frac{1}{\sqrt{2}} \implies P_{AA} = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (1.50)$$

Conversely, for the cross-polarized diagonal outcomes $|DA\rangle$ and $|AD\rangle$, the projections onto the state $|\Psi^+\rangle$ lead to destructive interference:

$$\langle DA|\Psi^+\rangle = \frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = 0 \implies P_{DA} = 0 \quad (1.51)$$

$$\langle AD|\Psi^+\rangle = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = 0 \implies P_{AD} = 0 \quad (1.52)$$

Finally, substituting these four joint probabilities into the correlation function adapted for the diagonal basis $E = P_{AA} + P_{DD} - P_{AD} - P_{DA}$ gives:

$$E = \frac{1}{2} + \frac{1}{2} - 0 - 0 = +1 \quad (1.53)$$

This result demonstrates that while the Bell state $|\Psi^+\rangle$ exhibits perfect anti-correlation ($E = -1$) in the computational basis, it transitions to a state of perfect correlation ($E = +1$).

By systematically repeating the same matrix calculations for the remaining Bell states, the correlation values across different measurement bases can be derived. The behavior of each state depends strictly on the chosen reference frame, namely the rectilinear basis (Z -basis) and the diagonal basis (X -basis).

The complete classification of these quantum correlations is summarized in Table 1.1, which maps each Bell state to its corresponding correlation or anti-correlation outcome.

Bell State	Basis Z (E_Z)	Behavior in Z	Basis X (E_X)	Behavior in X
$ \phi^+\rangle$	+1	Correlation	+1	Correlation
$ \phi^-\rangle$	+1	Correlation	-1	Anti-correlation
$ \psi^+\rangle$	-1	Anti-correlation	+1	Correlation
$ \psi^-\rangle$	-1	Anti-correlation	-1	Anti-correlation

Table 1.1: Quantum correlation and anti-correlation behavior for the four Bell states in the Z and X bases.

As discussed in section 1.2, pure states may be degraded during experiments or generated imperfectly, transforming into a mixed state. In such scenarios, it is necessary to quantify this degradation using fidelity. Fidelity measures how close two quantum systems are. In the case of two pure states $|\psi\rangle$ and $|\phi\rangle$ [4]:

$$F(|\psi\rangle, |\phi\rangle) = |\langle\psi|\phi\rangle| \quad (1.54)$$

If, on the other hand, we consider a real state described by the density matrix (1.6) and compare it with the ideal pure state $|\psi\rangle$, the fidelity is expressed as:

$$F(|\psi\rangle, \hat{\rho}) = \sqrt{\langle\psi|\hat{\rho}|\psi\rangle} \quad (1.55)$$

The fidelity value lies within the interval $[0, 1]$, where $F = 1$ indicates that the two states are identical, whilst $F = 0$ indicates that the two states are orthogonal to one another.

In the field of quantum communication, fidelity is used to quantify the degree to which an experimentally obtained state differs from a Bell state (a maximally entangled state). In this context, fidelity serves as an indicator of the efficiency of quantum correlations: a high fidelity value relative to a Bell state ensures that the system retains the entanglement properties necessary for protocols such as quantum key distribution.

1.4. Entanglement distribution

Quantum entanglement is the key to the development of quantum communication technologies, particularly for the realization of the future Quantum Internet. This network infrastructure is essential for interconnecting multiple remote quantum computers. Since scaling a single device to a high number of Qubits remains heavily constrained by environmental noise and hardware limitations, a simpler solution is to connect multiple smaller computers, each with fewer Qubits.

By distributing entanglement across these separate units, they can exchange quantum states and operate synergistically as a single, highly powerful distributed computing system. Moreover, the ability to transfer information in an intrinsically secure way over long distances, exploiting the laws of quantum mechanics through photons, offers unprecedented flexibility and security in the field of telecommunications.

However, deploying an entanglement distribution in the real world presents major physical and engineering challenges. In particular, photons are highly sensitive to birefringence fluctuations in optical fiber. When the system is based on polarization-entangled states, these fluctuations directly degrade the polarization of the photons, risking the irreversible compromise of the transported quantum information. To overcome this limitation, scientific literature has extensively investigated various compensation strategies. One approach introduces a co-propagating classical calibration signal in the same fiber as a quantum signal, multiplexing them through Time Division Multiplexing (TDM) or Wavelength Division Multiplexing (WDM) to provide a direct, real-time measurement of the fiber's perturbations [11–13]. Alternatively, another solution is to monitor the Quantum Error Rate (QBER) in real time and when it becomes too high an iterative algorithm is triggered to apply the necessary polarization correction [14]. In submarine scenarios, however, it has been demonstrated that polarization compensation can be initially performed manually. Thanks to the high thermal and mechanical stability of the seabed, the polarization remains stable over long periods, removing the need for continuous, active compensation systems [15].

Furthermore, transmitting quantum signals over existing fiber infrastructure requires them to co-exist with intense classical traffic, which introduces significant noise. The ultimate challenge, therefore, lies in distributing entanglement without degrading its characterizing parameters. Specifically, the system must mitigate both environmental and classical disturbances to keep the S -value above their critical thresholds (which defines quantum correlation) and ensure a high level of Fidelity. Despite these strict constraints, several urban fiber trials have recently demonstrated remarkable progress. For instance, early field tests validated the distribution of entangled states across a 34 km deployed loop within the GothamQ infrastructure in New York City [11]. Following these developments, significant progress in the coexistence of quantum and classical technologies was achieved in Germany via the *BearlinQ* urban network [12]. In this work the entanglement is distributed in deployed fiber up to 60 km maintaining high fidelity (between 85–99%) and recording an experimental S -value between 2.36 and 2.74.

These successful field demonstrations provide an excellent baseline for scaling entanglement distribution across different distances without destroying the quantum properties. This reliable distribution is the essential requirement for implementing secure and efficient Quantum Key Distribution (QKD). This thesis focuses specifically on the laboratory implementation and analysis of the entanglement-based QKD protocol.

2 | Quantum Cryptography

This chapter explores the transition from classical to quantum cryptography, introducing Quantum Key Distribution (QKD) with a specific focus on entanglement-based systems. In particular, it details the BBM92 protocol and describes all the laboratory hardware required for its experimental implementation: the SPDC source, single-photon detectors, and the time tagger.

2.1. Classical Cryptographic Systems

Ensuring the security of communications is one of the most important challenges in the modern world. In this context, **cryptography** plays a fundamental role. By encrypting information, it guarantees the confidentiality of data during transmission, ensuring that only authorized recipients can access the content and neutralizing any attempts at interception by *eavesdroppers*.

Cryptosystems can be distinguished into two main categories [16]:

1. **Asymmetrical cryptosystems** (*or Public-key*): Each user (Alice or Bob) possesses a pair of keys: a public and a private key. The sender uses the receiver's public key to encrypt messages, while the receiver uses their own private key to decrypt it. The security of this system relies on the mathematical complexity of deriving the private key from the public one.
2. **Symmetrical cryptosystems** (*or Secret-key*): The two users possess the same private key used for both encryption and decryption. This key must be shared through a secure channel to prevent interception by potential eavesdroppers.

A model of **symmetrical cryptosystems** is the *One Time Pad* (OTP), proposed by Gilbert Vernam in 1926. Based on information theory results, the OTP is considered information-theoretically secure, as it satisfies the conditions for **perfect secrecy (or unconditional security)** defined in Shannon's Theorem [17]. Perfect secrecy requires that the ciphertext (c) provides no information about the plaintext (m). Mathematically, this is expressed as:

$$P(m|c) = P(m) \tag{2.1}$$

implying that the posterior probability of the message is identical to its prior probability. In the

OTP framework, this condition holds even against an adversary with unlimited computational capacity. According to Shannon, this condition holds if and only if the key (k) satisfies three fundamental criteria:

1. Length: the key must be at least as long as the message to be encrypted.
2. Randomness: the key must be generated uniformly at random.
3. Uniqueness: each key must be used only once (hence the name "One-Time") and then discarded to prevent any frequency analysis or statistical leakage.

In the OTP model, encryption is achieved through the XOR operator ($\oplus$), which maps binary strings by applying the logical operation between the message and the key:

$$c = m \oplus k \tag{2.2}$$

However, the practical application of the One-Time Pad is limited by the difficulty of distributing keys. The model requires the key to be as long as the message and exchanged in advance over a secure channel. If two parties already had a secure channel capable of safely transmitting an n -bit key, they could use that same channel to send the message in plaintext directly, rendering the entire encryption process useless. To overcome these limitations, modern cryptography relaxes the requirement of perfect secrecy in favor of semantic (or computational) security. This paradigm assumes an adversary with limited computational power, allowing the use of keys significantly shorter than the message. Within the framework of symmetric cryptography, the main approaches are stream ciphers and block ciphers, with the Advanced Encryption Standard (AES) [18] representing the most widespread and trusted standard for block-based encryption.

Nevertheless, symmetric cryptography inherently suffers from the key distribution problem, as it requires Alice and Bob to securely share a common key without any prior relationship. To resolve this challenge, modern security architectures combine symmetric and asymmetric methodologies into a hybrid cryptographic framework, exploiting the advantages of both. In these systems, asymmetric protocols, which rely on mathematically complex one-way functions, such as the integer factorization problem underpinning the RSA scheme [19], are employed exclusively during the initial phase to securely distribute the symmetric session key. Once this key is established, the actual data encryption is handed over to symmetric ciphers like AES, which are orders of magnitude faster and computationally more efficient than their asymmetric counterparts.

However, because this hybrid framework relies on computational security rather than information-theoretic security, its foundations are not absolute. The security of the asymmetric component is severely threatened by the current advancements in quantum computing. In 1994, Peter Shor proposed a quantum algorithm capable of solving the prime factorization and discrete

logarithm problems in polynomial time [20]. Consequently, the development of a sufficiently powerful quantum computer would efficiently break current asymmetric schemes like RSA. By compromising the key distribution phase, a quantum adversary could intercept the symmetric keys, thereby undermining the security of the entire communication channel, regardless of the classical robustness of AES.

2.2. Quantum Key Distribution

The introduction of Shor's algorithm exposed the vulnerability of public-key systems to the growing power of quantum computing, making the secure generation and distribution of keys essential. In response to this issue, two main approaches have been proposed for the development of encryption schemes defined as quantum-safe [21]:

- **Post-Quantum Cryptography (PQC)**: this approach uses cryptography to develop encryption algorithms resistant to both classical and quantum attacks. The only limitation is that security remains computational. Although these algorithms are secure against currently known quantum threats, it cannot be excluded that future advances in computational theory or computing power could compromise their security, rendering them potentially obsolete.
- **Quantum Key Distribution (QKD)**: this method offers promising information-theoretical security based on the fundamental laws of quantum physics. In contrast to PCQ, QKD allows for the generation and distribution of unconditionally secure keys. Its effectiveness is entirely independent of the computational power of an adversary: since security derives from physical properties, it remains immune even to quantum computers with virtually unlimited computational capacity.

The **QKD protocol** enables two authorized users, Alice and Bob, to establish secure communication through the generation and distribution of cryptographic keys. To implement this system, the two parties require two separate channels:

1. **Quantum Channel**: used for the exchange of quantum signals necessary for key generation. This channel is vulnerable to potential manipulation or intrusion by a malicious user, commonly referred to as Eve.
2. **Classical (Public) Channel**: used for the exchange of encrypted messages and for auxiliary operations on the key, such as error correction and privacy amplification. Although this channel can be eavesdropped on by third parties, it must be authenticated to ensure that Alice and Bob correctly identify each other.

Unlike classical cryptography, the security of QKD is based on the principles of quantum mechanics:

- **Measurement disturbance** (*Section 1.1*): any attempt to measure a quantum system inevitably disturbs its state. If Eve intercepts and measures the signals in the quantum channel to identify the key, she introduces noise that allows Alice and Bob to detect the compromise.
- **No-Cloning Theorem**: as published by Wootters and Zurek in 1982 [6], this theorem establishes the impossibility of creating an identical and perfect copy of an unknown quantum state. Therefore, Eve cannot copy the key-related states and send them back through the channel to Bob without revealing her presence.

Quantum states of light are used to transmit the cryptographic key. Photons represent ideal candidates as they interact weakly with matter; this characteristic allows them to travel over long distances without the encoded quantum state undergoing significant decoherence [22]. The main limitation in transmission is represented by channel losses: the sent photon does not always reach its destination. For this reason, the quantum channels used are typically optical fibers or, if Alice and Bob are in line-of-sight, free space (free-space or satellite communications). While maximum theoretical security is achieved by employing ideal single-photon sources, practical implementations often rely on attenuated laser pulses, which occasionally yield multi-photon states. This technological limitation exposes the system to the **Photon Number Splitting (PNS) attack**. In this scenario, an eavesdropper (Eve) can non-destructively determine the photon-number statistics of each incoming pulse. Utilizing this information, she selectively blocks all single-photon pulses while splitting the multi-photon ones, storing her harvested fraction in a quantum memory. Eve then simply waits for Bob to register his portion and for the basis reconciliation phase to occur over the public classical channel. Once the correct measurement bases are disclosed, she measures her stored photons in the exact same bases as Bob. This allows Eve to extract identical key information without introducing any detectable disturbance or revealing her presence.

Depending on the physical property used for encoding, QKD is divided into two main categories:

- **Discrete Variable QKD (DV-QKD)**: information is encoded in discrete properties of the single photon, such as polarization or phase. In this case, bits are mapped onto well-defined and distinct quantum states.
- **Continuous Variable QKD (CV-QKD)**: information is encoded in the amplitude and phase quadratures of the electromagnetic field. Unlike DV-QKD, these variables are continuous and can take any value within a given interval, allowing the use of reception technologies similar to those used in classical telecommunications.

In addition to the nature of the variables used (discrete or continuous), QKD protocols are distinguished by the approach used to establish the secret key. This classification essentially depends on how the quantum states are generated and distributed:

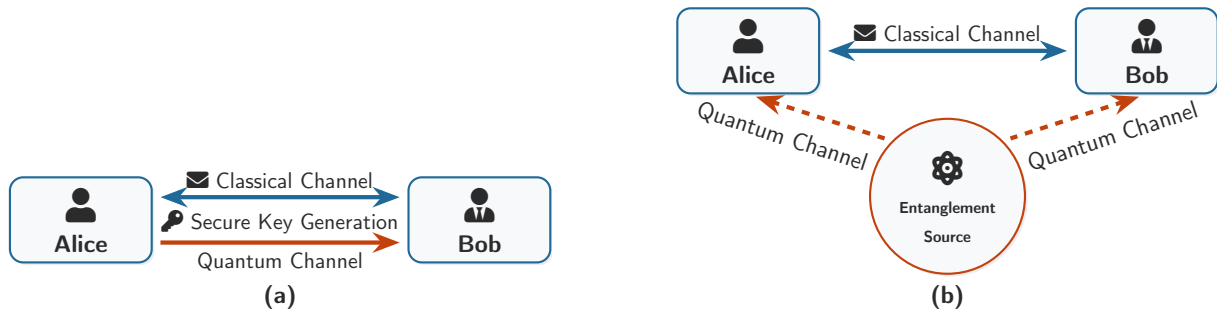


Figure 2.1: Sketch of QKD setups: (a) Prepare and Measure; (b) Entangled-based protocol.

1. **Prepare-and-Measure (P&M)**: the sender (Alice) prepares a sequence of single photons (typically via attenuated laser pulses) and sends them to Bob, who performs a measurement to obtain the key bits. For each photon, Alice randomly chooses a bit value and a coding base (X or Z). In this scheme, security is guaranteed by the non-orthogonality of the bases: if an interceptor (Eve) attempts to decipher the bit using the wrong basis, she inevitably causes the collapse of the quantum state, introducing an error with a non-zero probability, detectable by Bob. Conversely, states within the same basis must be orthogonal to each other; this condition is necessary for Bob to determine the bit value without ambiguity when using the correct basis.
2. **Entanglement-Based (E-B)**: the source generates pairs of photons in an entangled state (usually a Bell state) and distributes one to Alice and one to Bob. In this architecture - the focus of this thesis - the key is not "prepared" in advance but emerges from the quantum correlations between the distributed particles.

Every **E-B protocol** rests on three fundamental pillars:

1. *The Bell State Source*: the system is based on a source that generates pairs of particles in a state of maximum entanglement (such as a Bell State). In this condition, neither particle possesses a definite value before measurement, even though they are linked by a perfect correlation.
2. *Indeterminacy and Correlation*: the key does not physically exist during transit through the channel. Information manifests only when Alice and Bob perform a measurement, causing the system to collapse into a specific state. If both measure using the same basis, quantum mechanics guarantees that they will obtain perfectly correlated results.
3. *Security via Non-Locality (Bell Test)* (Section 1.3.2): security is fundamentally guaranteed by verifying the non-local nature of the correlations. Depending on the specific implementation, this can be achieved by monitoring the Quantum Bit Error Rate (QBER) or, more

robustly, by performing a Bell Test. By employing non-orthogonal measurement bases, Alice and Bob can confirm that the correlations between the photons are purely quantum in nature. If an eavesdropper (Eve) attempts to intercept the signal via an intercept-and-resend attack (measuring the photon and sending a substitute to the destination), she inevitably triggers the collapse of the wavefunction, destroying the entanglement. In this scenario, the system loses its quantum properties and reverts to a local hidden-variable system - effectively becoming classical. Consequently, the S -parameter (used in protocols like E91 to certify security or as a benchmark in BBM92) inevitably drops from the quantum limit of $2\sqrt{2} \approx 2.82$ to the classical threshold of $S \leq 2$. Therefore, any S -value failing to violate Bell's inequality indicates an intrusion or prohibitive environmental noise, leading to the rejection of the key. Conversely, $S > 2$ provides a mathematical guarantee that the correlations are private and exclusive to Alice and Bob.

Finally, while entanglement theory is universal, its practical implementation requires choosing which physical degree of freedom of the photon to use for encoding information. The most popular are:

1. *Polarization*: it is an easy method to implement, but the polarization of the photons is highly sensitive to fluctuations due to the birefringence present in optical fiber, which degrades the entanglement fidelity.
2. *Time-Bin*: The information is encoded in the arrival time of the photons. If two photons are emitted simultaneously and their arrival times cannot be distinguished from one another, they belong to the same *time bin*.
3. *Energy-Time*: this method exploits the correlation between the time of creation and the energy of the photons. The two photons entangled are created simultaneously, which makes it possible to determine the exact moment of their generation. Also, by the principle of conservation of energy, the sum of the energies of the two photons is equal to the energy of the pump photon that generated them. Consequently, measuring the energy of one photon instantly determines that of its partner (complementary energy).
4. *Frequency-Bin*: the information is encoded based on the frequency (and therefore the wavelength) of the emitted photons. The photons have different frequencies and are identified by these. Although the frequency of an individual photon is uncertain, their sum is well defined by the conservation of energy.

Once the quantum distribution phase is complete, the string of data collected by Alice and Bob is not yet ready for cryptographic use. The process then shifts to a classical communication channel (public but authenticated) to transform these raw correlations into a shared, perfectly secret key. This stage, known as Post-Processing, is essential for eliminating environmental

noise and neutralizing any information potentially leaked to an eavesdropper. It consists of three fundamental steps:

1. **Sifting** (Basis Reconciliation): Alice and Bob compare their measurement bases and retain only the bits where their choices coincide, discarding the rest. This process results in the *Raw Key*.
2. **Error Correction** (Information Reconciliation): classical algorithms are employed to identify and correct bit errors caused by channel noise or potential eavesdropping. During this stage, the *Quantum Bit Error Rate* (QBER) is estimated to ensure that the channel's security remains within theoretical limits.
3. **Privacy Amplification** (Universal Hashing): this final step eliminates any partial information that an interceptor may have gained during the quantum transmission or the reconciliation phase. By "distilling" the string, Alice and Bob ensure the generation of a perfectly secure *Final Secret Key*.

2.3. BBM92 Protocol

The first QKD protocol was BB84, proposed by Bennett and Brassard in 1984 [23]. This scheme belongs to the category of prepare-and-measure protocols, where security relies on the deliberate preparation and subsequent measurement of specific quantum states.

The introduction of protocols based on entanglement occurred in 1991, when Artur K. Ekert proposed the **E91 protocol** [24]. As previously discussed, E91 exploits the non-local correlations of a pair of particles in an entangled state (EPR pair) and certifies security through the verification of Bell's inequality violation. If the observed parameter S exceeds the classical threshold, it can be demonstrated that the correlation between the particles is purely quantum and that no information has been leaked to a potential eavesdropper. E91 also represents the first Device-Independent (or quasi-device-independent) protocol. In this approach, the source can potentially be provided by an eavesdropper; however, as long as the violation of Bell's inequalities is verified, the reliability of the source is guaranteed regardless of its manufacturer. This marks a significant departure from BB84, where security strictly depends on the trust placed in Alice's preparation apparatus.

Nevertheless, E91 presents certain operational challenges. The requirement to perform measurements across three different bases to estimate the parameter S significantly increases experimental complexity and reduces the protocol's efficiency, as a larger fraction of data must be discarded during the sifting phase. To overcome these inefficiencies, Bennett, Brassard, and Mermin introduced the **BBM92 protocol** in 1992 [25]. This scheme can be interpreted as an entanglement-based version of BB84. While maintaining an architecture with an external source, BBM92 drastically simplifies the process: Alice and Bob utilize only two measurement

bases, thereby optimizing the key generation rate. Unlike E91, BBM92 does not require the explicit calculation of Bell's inequalities in every session to guarantee secrecy; instead, it relies on the estimation of the Quantum Bit Error Rate (QBER) derived from the intrinsic correlations of the entangled states.

The BBM92 protocol exploits non-local quantum correlations to establish a secure cryptographic key between two users, Alice and Bob. In BBM92, a source (which may be located with Alice, Bob or a third party) emits polarization-entangled photon pairs. Typically, these photons are prepared in the Bell singlet state:

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A |V\rangle_B - |V\rangle_A |H\rangle_B). \quad (2.3)$$

In this state, the polarization assumed by the two photons are perfectly anti-correlated: if Alice measures a horizontal polarization (H), Bob will necessarily measure a vertical polarization (V) if he uses the same basis. To generate keys, the protocol follows three main phases:

1. **Measurement:**

- (a) Alice and Bob each receive one photon from the entangled pair. They independently and randomly choose a measurement basis between two non-orthogonal bases:
 - **Linear (Z):** States $|H\rangle$ (Horizontal, 0°) and $|V\rangle$ (Vertical, 90°)
 - **Diagonal (X):** States $|D\rangle$ (Diagonal, $+45^\circ$) and $|A\rangle$ (Anti-diagonal, -45°)
- (b) For every measurement event, a record is kept of both the chosen basis and the corresponding bit value obtained from the outcome.

Due to quantum entanglement, if Alice and Bob use the same basis, their results are perfectly correlated. If they use different bases, the results are completely random and uncorrelated. Consequently, their initial raw keys consist of a mixture of both correlated and uncorrelated bits.

- 2. **Sifting:** Alice and Bob publicly communicate their chosen bases via a classical channel, keeping the measurement results strictly private. If the bases match, the bits are retained to form the *Sifted Key*. If the bases differ, the data is discarded, which dynamically eliminates roughly half of the raw key string.
- 3. **Error Determination and Key Generation:** Alice and Bob publicly compare a random subset of the bits in their Sifted Key to calculate the Quantum Bit Error Rate (QBER):

$$QBER = \frac{\text{Erroneous Bits}}{\text{Total Bits Compared}} \quad (2.4)$$

Any eavesdropping attempt by Eve destroys the entanglement due to the no-cloning theorem and wave-function collapse, inevitably introducing errors.

- **If QBER $\leq 11\%$:** The channel is secure. Alice and Bob proceed with error correction and privacy amplification, distilling a shorter but completely secure, identical **Secret Key**.
- **If QBER $> 11\%$:** The potential information leaked to Eve is too high; the key is discarded, and the process restarts.

To clarify this workflow, Table 2.1 illustrates a step-by-step execution example over 12 quantum rounds.

Because the source emits the anti-correlated Bell singlet state $|\psi^-\rangle$, matching bases (rounds 1, 3, 4, 6, 8, 10, 11, and 12) yields opposite states (e.g., in round 1, Alice measures $|H\rangle \rightarrow 0$ and Bob measures $|V\rangle \rightarrow 1$). Bob reconciles this systematic anti-correlation by performing a deterministic bit inversion ($B \rightarrow \bar{B}$) during the public discussion phase. Rounds with mismatched bases are discarded (—), while sampled rounds (11 and 12) are sacrificed for the QBER check.

Pulse (Photon Pair)	1	2	3	4	5	6	7	8	9	10	11	12
QUANTUM TRANSMISSION												
Alice's Random Basis	Z	X	X	Z	Z	X	X	Z	Z	X	Z	X
Bob's Random Basis	Z	Z	X	Z	X	X	Z	Z	X	X	Z	X
Alice's Measured State	$ H\rangle$	$ D\rangle$	$ D\rangle$	$ V\rangle$	$ H\rangle$	$ A\rangle$	$ D\rangle$	$ H\rangle$	$ V\rangle$	$ A\rangle$	$ H\rangle$	$ A\rangle$
Bob's Measured State	$ V\rangle$	$ V\rangle$	$ A\rangle$	$ H\rangle$	$ A\rangle$	$ D\rangle$	$ H\rangle$	$ V\rangle$	$ D\rangle$	$ D\rangle$	$ V\rangle$	$ D\rangle$
Alice's Raw Bit (A)	0	0	0	1	0	1	0	0	1	1	0	1
Bob's Raw Bit (B)	1	1	1	0	1	0	1	1	0	0	1	0
PUBLIC DISCUSSION												
Alice Checks Bases	OK		OK	OK		OK		OK		OK	OK	OK
Bob Inverts Bits ($B \rightarrow \bar{B}$)	0		0	1		1		0		1	0	1
QBER Check (Samples)											Check	Check
OUTCOME												
Final Secret Key	0	—	0	1	—	1	—	0	—	1	—	—

Table 2.1: Example of the BBM92 QKD protocol execution using the anti-correlated Bell singlet state $|\psi^-\rangle$ and Z/X bases. The table illustrates the measurements, the sifting process, and how Bob inverts his anti-correlated bits to match Alice's key.

2.4. Physical Implementation

In order to implement the protocol described above, specific hardware components are required. This section analyzes the principal components normally present in practical realization of entanglement-based QKD systems, focusing on the source, the single-photon detectors, and the time-tagger.

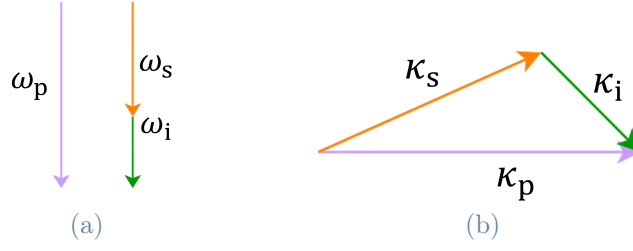


Figure 2.2: Principles of type II phase matching: (a) energy conservation; (b) momentum conservation

2.4.1. SPDC Source

Entangled photon pairs are generated based on different physical principles, which are primarily implemented through three technological approaches: semiconductors [26], atomic ensembles [27], and Spontaneous Parametric Down-Conversion (SPDC) [28].

Among these methodologies, SPDC is considered in this thesis. It is a non-linear optical process in which a single high-energy photon, called the pump photon, interacts with a material and is spontaneously converted into two lower-energy photons, respectively called the signal and idler. For this conversion to take place, the high-intensity pump photon must pass through a material with specific second-order non-linear properties ($\chi^{(2)}$). Under these conditions, the polarization P has a non-linear dependence on the electric field E :

$$\mathbf{P}^{(2)} = \varepsilon_0 \chi^{(2)} \mathbf{E}^2 \quad (2.5)$$

The process is governed by two fundamental laws that ensure conversion efficiency both illustrated in Figure 2.2:

1. Conservation of Energy: The energy of the pump photon must be equal to the sum of the energies of the two produced photons:

$$\omega_p = \omega_s + \omega_i. \quad (2.6)$$

2. Momentum Conservation (Phase Matching): The wave vectors (indicating the direction of propagation) must be conserved according to the relation:

$$\vec{k}_p = \vec{k}_s + \vec{k}_i \quad (2.7)$$

In Type-II SPDC, the generated photons (signal and idler) have mutually orthogonal polarizations (one horizontal and one vertical). Due to momentum conservation conditions, the photons are not emitted in a single direction, but rather along the surfaces of two light cones as showed

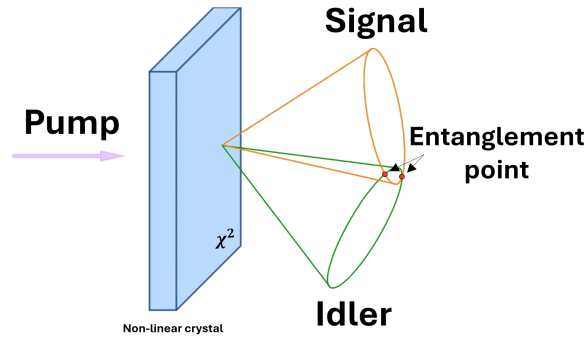


Figure 2.3: Two emitted cones from the non-linear crystal

in Figure 2.3. At each intersection point, a photon can be either horizontally or vertically polarized with equal probability. However, once the polarization of a photon at one intersection point is measured, the polarization of the photon at the other intersection point is immediately determined to be orthogonal. Thus, the two photons corresponding to the intersection points are entangled and form a Bell-state pair.

2.4.2. Single Photon Detectors

Photon detection is the fundamental process of converting the energy of an incident photon into a measurable electrical signal. This process generates a discrete pulse, commonly referred to as a ‘click’, enabling the precise registration of the particle’s arrival time.

In applications such as QKD, it is essential to employ detectors that maximize detection probability while minimizing dark count rate. The performance of these devices is characterized by four fundamental parameters:

- **Quantum Efficiency (η):** the ability of the detector to convert an incident photon into an electrical signal. Specifically, it is defined as the ratio of detected photons to the total number of incident photons.
- **Dark Count Rate (DCR):** it represents the rate of counts registered in the absence of incident photons. These false positives are caused by thermal noise or quantum fluctuations within the material. A low DCR is crucial for ensuring the integrity of data transmission.
- **Temporal Response (Jitter):** it refers to the statistical fluctuation in the time interval between the photon absorption and the generation of the output electrical pulse. Low jitter is essential for accurate temporal synchronization.
- **Recovery Time:** the time interval required for the detector to return to its operational state following a detection event. It includes the quenching phase, the final voltage reset

and time of hold-off.

In quantum key distribution setups, single-photon detection is primarily implemented using two main technologies: Superconducting Nanowire Single-Photon Detectors (**SNSPDs**) and semiconductor-based Single-Photon Avalanche Diodes (**SPADs**). SNSPDs offer outstanding performance, including near-unity detection efficiency, ultra-low jitter, and minimal dark counts. However, they rely on cryogenic cooling systems to operate at temperatures below 2 K, which significantly increases cost.

To avoid these requirements, this work focuses on semiconductor **SPADs**. While they exhibit lower detection efficiencies and higher timing jitter compared to superconducting devices, SPADs provide a much more cost-effective solution.

A SPAD is a p-n junction operating in Geiger Mode. In this configuration, the junction is reverse-biased with a voltage ($V_a = V_{ex} + V_{BD}$) exceeding the breakdown voltage (V_{BD}). In this state of unstable equilibrium, the electric field within the active region is so high that the absorption of a single photon can trigger an electron transition from the valence band to the conduction band, generating an electron-hole pair. The carriers generated in this process are accelerated by the electric field (electrons toward the anode, holes toward the cathode), colliding with the crystal lattice and liberating additional carriers. This impact ionization process triggers a self-sustaining avalanche, which produces a measurable macroscopic current. To interrupt conduction and prepare the device for a new event, an external quenching circuit is required. The operational cycle of a SPAD is easily understood by analyzing the temporal evolution of its electrical signals. When an incident photon triggers the avalanche, the current (I) rises rapidly, as shown in figure 2.4. Then, the Active Quenching Circuit (AQC) detects this rise and rapidly lowers the bias voltage (V_a) below the breakdown threshold, resetting the system for the next detection.

SPADs can operate in two primary modes:

- **Free-running mode:** the detector is constantly active and becomes ready for detection immediately following each quenching and reset cycle.
- **Gated mode:** the detector is enabled only during specific time windows (with a determined Gate Width). This mode is widely used in QKD to synchronize the receiver with the expected arrival of photons, drastically reducing the probability of registering out-of-band photons and dark counts.

Despite the use of AQC, performance is further limited by **afterpulsing**. This occurs when charges trapped during an avalanche are released later, triggering a new avalanche uncorrelated with a new photon. As shown in Figure 2.4, the introduction of a $T_{hold-off}$ is the primary method to mitigate this effect. During this interval, the SPAD bias voltage is kept forcibly

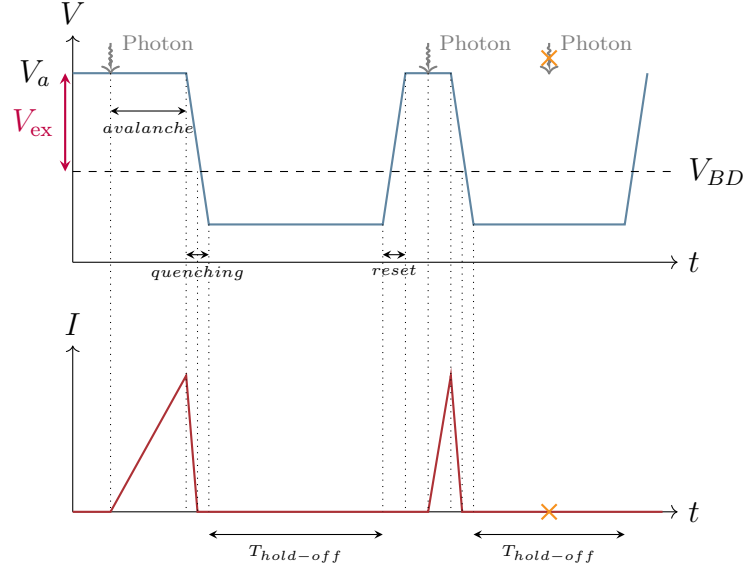


Figure 2.4: Time diagram of SPAD current and voltage during photon detection.

below the breakdown voltage (V_{BD}), making the device temporarily insensitive. Consequently, if a photon strikes the detector during the hold-off period, or during the subsequent reset phase before the voltage has recovered a value sufficiently above V_{BD} , it will fail to trigger an avalanche and will not be counted. In this condition, the photon is lost due to the sensor's dead time, which sets the upper limit for the system's maximum counting rate.

The presence of noise, which is also influenced by ambient light and temperature, increases the probability of error in the system (QBER), making it essential to optimize the balance between efficiency and hold-off time. So, the best trade-off of the measurement parameters should be found for each measurement setup. As shown in Figure 2.5a, the detection efficiency η increases monotonically with the excess voltage V_{ex} , as a higher voltage generates a stronger electric field that raises the probability of triggering an avalanche. This operational behavior is directly related to the noise performance illustrated in Figure 2.5b. At short hold-off times, the counts experience an exponential increase due to the afterpulsing effect of trapped carriers. At longer hold-off times, the curves flatten out toward the pure thermal Dark Count Rate (DCR), the baseline of which inherently increases at higher excess voltages. Consequently, selecting the operating parameters requires solving an interconnected optimization problem. The specific experimental trade-offs and the selection of these values for our BBM92 setup will be fully discussed and justified in Chapter 5.

2.4.3. Time Tagger

A highly effective approach for measuring the arrival time of a photon detected by single-photon detectors consists in using a Time-to-Digital Converter (TDC), such as the Time Tagger Ultra

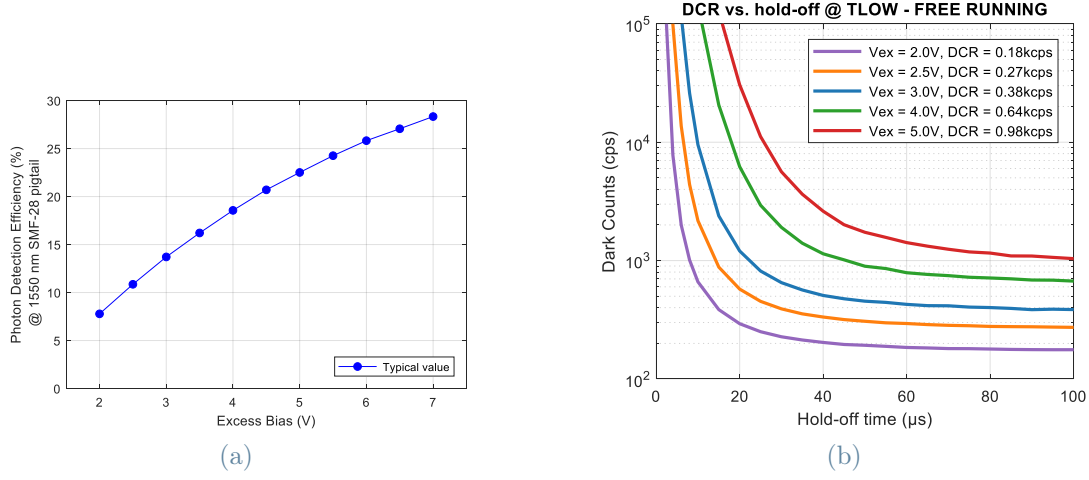


Figure 2.5: (a) PDE VS Excess Bias; (b) Dark counts vs Hold Off (Ref.[29])

manufactured by Swabian Instruments. The Time Tagger features multiple input channels to which the SPADs are connected. When a photon is detected, the SPAD generates an electrical pulse that is sent to one of the Time Tagger input channels. Inside the device, a comparator circuit continuously compares the incoming signal with a user-defined trigger threshold voltage. This discrimination threshold plays a crucial role in isolating genuine photodetection events from electronic background noise and from the intrinsic dark counts of the SPAD. As soon as the rising edge of the electrical pulse crosses the predefined threshold, the corresponding temporal value (**timestamp**) is recorded. Each detection event, commonly referred to as a "click", is therefore identified by the channel–timestamp pair. This process occurs continuously, generating a chronological stream of time-tagged events with high temporal resolution.

In the present thesis work, the Time Tagger was specifically employed to identify and count temporal coincidences generated by entangled photon pairs. Since quantum-correlated photons are emitted simultaneously by the source, the corresponding electrical pulses must reach the Time Tagger channels within an extremely narrow temporal window in order to be recognized as coincident events. The performance and precision of the Time Tagger Ultra are mainly limited by an intrinsic temporal uncertainty known as timing jitter. This parameter describes the temporal spread associated with the measurement of an ideally instantaneous event and is characterized by two main quantities: an RMS (Root Mean Square) jitter of approximately 42 ps and a Full Width at Half Maximum (FWHM) of the Gaussian error distribution of approximately 100 ps. The low RMS jitter enables the definition of very narrow coincidence windows, significantly reducing the number of accidental coincidences caused by noise. The FWHM parameter, on the other hand, characterizes the overall temporal response of the acquisition system and describes the temporal distribution of the detected events. In this way, each Time Tagger channel can register with high precision the photon counts detected by the corresponding SPAD.

3 | System Parametrization and BBM92 Numerical Modeling

3.1. Source Characterization

This chapter outlines the steps toward the BBM92 protocol realization, beginning with a comprehensive characterization of the source in terms of optical power emission and spectral properties. Next, the identification of coincidence peaks is addressed to analyze the entangled photon pairs, providing the foundation for the numerical optimization model of the polarization-encoded BBM92 system.

3.1.1. The Sagnac Interferometer Source

The source used in this thesis (HEPS-1000-3A3A-1550-9/125, OZ OPTICS) produces polarization-entangled photon pairs by exploiting the process of Spontaneous Parametric Down-Conversion (SPDC) inside a Periodically Poled Lithium Niobate (PPLN) crystal. The crystal is placed inside a Sagnac interferometer configured in a closed loop of polarization-maintaining (PM) optical fiber.

The pairs of photons are generated along two opposite propagation paths. A high-energy linearly polarized pump laser at a wavelength of 775 nm is directed toward the Sagnac interferometer. At the input of the loop, a Polarization Beam Splitter (PBS) splits the pump light into its two orthogonal components: the p -polarized (horizontal) component and the s -polarized (vertical) component. Since the PM fiber circuit is configured in a closed loop, the two components propagate in opposite directions: one travels clockwise (CW) and the other counter-clockwise (CCW). The PPLN crystal, positioned at the center of the loop, is thus traversed in opposite directions by the two pump beams. Inside the crystal, the Type-II SPDC process occurs (Section: 2.4.1). In the clockwise direction, the pump photon generates a photon pair with halved energy and a wavelength centered at 1550 nm, in a polarization state $|HV\rangle$ (one horizontal photon and one vertical photon). In the counter-clockwise direction, the process occurs in a specular manner but with inverted polarizations, generating a photon pair in the state $|VH\rangle$. The generated photons continue to traverse the loop until they reach the initial PBS once again. Because the two optical paths are geometrically identical and intrinsically indistinguishable, the two quan-

tum alternatives coexist in a coherent superposition. At the output of the PBS, the photons are projected into a maximally polarization-entangled state (Bell state).

The state of polarization of the pump photon before entering the Sagnac loop is controlled by a rotatable half-wave plate (HWP) capable of covering an angular range from 0° to 360° . Adjusting the HWP rotates the polarization angle of the pump laser, which determines how the light splits into the two directions and changes the output state. In particular under optimal operating conditions, the HWP is set to 45° . In this configuration, the plate perfectly balances the pump laser power, distributing it equally between horizontal and vertical polarization. Consequently, the beam is split by the PBS into 50% clockwise and 50% counter-clockwise components; this perfect balance ensures the absolute indistinguishability of the two optical paths, allowing the creation of a perfectly entangled photon pair, as illustrated in Figure 3.1c). The quantum regime can be interrupted by rotating the plate to 90° or 0° , which is useful during the calibration phase. By rotating the HWP to 90° , the pump polarization becomes orthogonal to the x -axis (s -polarization), which causes the PBS to reflect the entire light intensity exclusively along the clockwise path (Fig. 3.1a)). In this way, the crystal generates a non-entangled photon pair in the $|V\rangle, |H\rangle$ state which, once returned to the PBS, separates by routing the two photons to Output 2 and Output 1, respectively. Conversely, setting the plate to 0° aligns the pump polarization parallel to the x -axis (p -polarization) (Fig. 3.1b)). In this case, the light passes linearly through the PBS, entering only the counter-clockwise path; the SPDC process then generates a pair with inverted polarizations compared to the previous case, and the PBS separates it by directing the photons to Output 1 and Output 2, respectively.

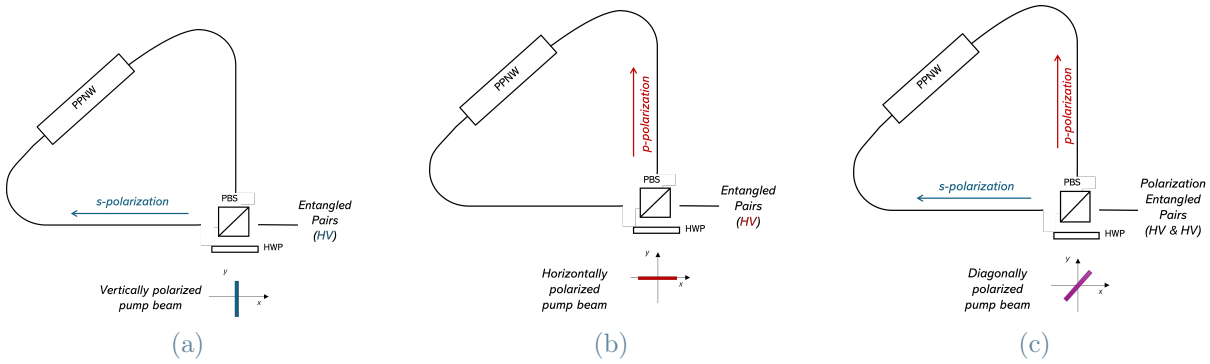


Figure 3.1: Schematic representation of the Sagnac interferometer configuration under different pump polarization states: (a) vertical (s -polarized) pump, (b) horizontal (p -polarized) pump, and (c) diagonal pump creating polarization-entangled photon pairs.

3.1.2. Optical Power Characterization

The system consists of the SPDC source, which features two optical output channels labeled Channel A and Channel B. To characterize the emission, a single SPAD is employed sequentially:

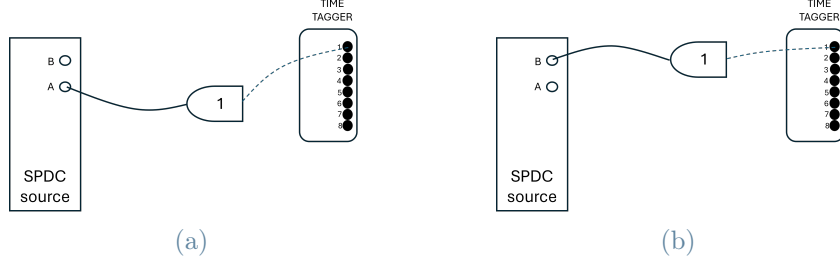


Figure 3.2: Setups used to measure power at output A (a) and (b) B

it is first coupled to Channel A 3.2a and subsequently moved to Channel B 3.2b. The electrical transistor-transistor logic (TTL) pulses generated by the SPAD upon each photon detection event are sent to the Time Tagger. To estimate the count rate, the Time Tagger's integrated *counter* and *countrate* analytical functions are used. While the *counter* mode allows the discretization of detections into temporal intervals (bins), providing a time-resolved distribution, the *countrate* function directly yields the total average rate over the entire acquisition period ($t_{\text{acq}} = 60$ s).

Initially, a preliminary calibration is performed to characterize the SPAD dark counts by carrying out an acquisition with the source turned off (Figure 3.3). Subsequently, the SPDC source is activated, and the steady-state photon-counting acquisition is recorded for each channel independently. The net count rate is then calculated by subtracting the background noise from the raw data.

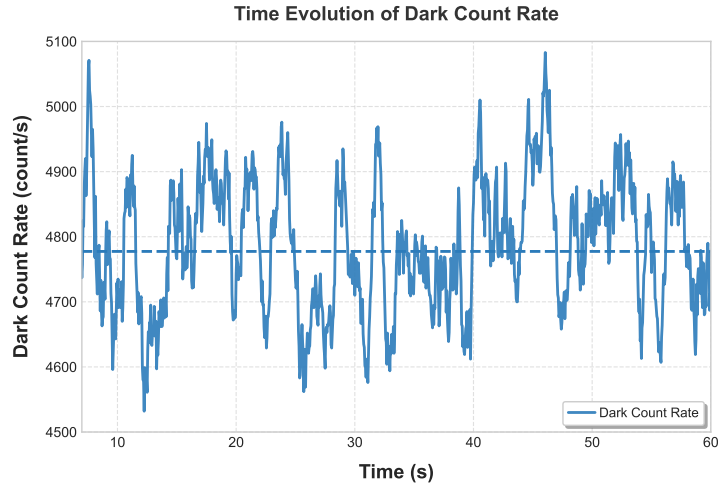


Figure 3.3: SPAD Dark Count characterization

The data obtained are presented in Table 3.1.

	Detector	Raw (counts/s)	Dark Counts (counts/s)	Net (counts/s)
Config. A	SPAD 1 (output A)	9603.53	~ 4777.41	4826, 12
Config. B	SPAD 1 (output B)	8950, 14	$\sim 4777, 41$	4172, 72

Table 3.1: Summary of raw, background, and net count rates for the configurations showed in Figure (3.2)

So, the stable background-subtracted rates (CR_A and CR_B) is determined (Fig. 3.4):

$$CR_A = 4826.12 \text{ counts/s} \quad (3.1)$$

$$CR_B = 4172.72 \text{ counts/s} \quad (3.2)$$

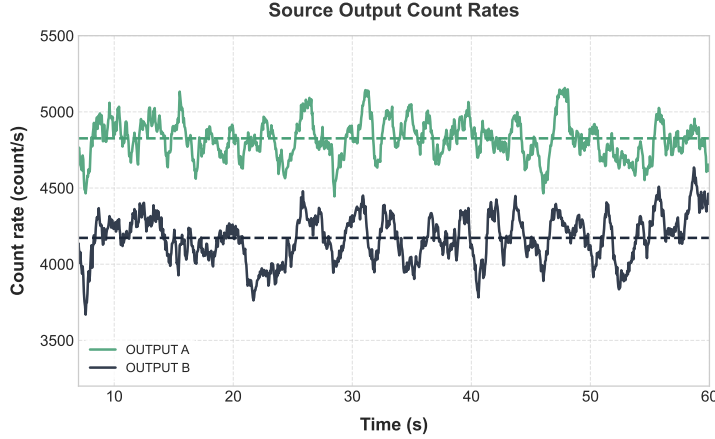


Figure 3.4: Count Rates in function of time.

The system performance is affected by the photon detection efficiency (PDE) and the hold-off time (T_{HO}) of the SPAD. This temporary blind time allows for the recovery of the device bias voltage and prevents false counts from afterpulsing, as discussed in Section 2.4.2. Therefore, to reconstruct the true photon emission rate of the source ($CR_{corrected}$) from the measured rate ($CR_{measured}$), both parameters must be taken into account:

$$CR_{corrected} = \frac{1}{PDE} \cdot \frac{CR_{measured}}{1 - CR_{measured} \cdot T_{HO}} \quad (3.3)$$

The optical output power from the two channels of the SPDC source can then be determined by analyzing the corrected photon emission rate. Specifically, the power P_i for each channel

$i \in \{A, B\}$ is evaluated as:

$$P_i = \frac{E_{\text{tot},i}}{t_{\text{acq}}} = CR_i \cdot E_{\text{ph}} \quad (3.4)$$

where CR_i represents the photon emission rate (expressed in photons per second) associated with output i , and E_{ph} is the energy of a single photon at the specific operating wavelength λ , defined as:

$$E_{\text{ph}} = h\nu = \frac{hc}{\lambda n} \quad (3.5)$$

with h being the Planck constant, c the speed of light in vacuum, and n the refraction index.

By setting an excess bias voltage $V_{\text{EX}} = 5$ V, appropriate temperature and T_{HO} values were selected. Higher temperatures lead to increased dark counts, whereas a longer T_{HO} is required to mitigate afterpulsing. The optimal trade-off for this measurement session was found to be:

$$T_{\text{HO}} = 40 \text{ } \mu\text{s}$$

$$T = T_{\text{lower}}$$

The operational PDE value was determined based on the characteristic curves extracted from the manufacturer's datasheet, shown in Figure 2.5a). Consequently, the following operating parameters are considered for the correction:

$$\text{PDE} = 0.225$$

$$T_{\text{HO}} = 40 \text{ } \mu\text{s}$$

Substituting these values into Eq. (3.3) yields the corrected counting rate for Channel A:

$$\text{CR}_{\text{corrected A}} = \frac{1}{0.225} \cdot \frac{4826.12}{1 - 4826.12 \cdot 40 \cdot 10^{-6}} = 26580.69 \frac{\text{counts}}{\text{s}} \quad (3.6)$$

and the corrected counting rate for Channel B:

$$\text{CR}_{\text{corrected B}} = \frac{1}{0.225} \cdot \frac{4172.72}{1 - 4172.72 \cdot 40 \cdot 10^{-6}} = 22261.01 \frac{\text{counts}}{\text{s}} \quad (3.7)$$

From Eq. (3.4), the equivalent optical powers for both channels are calculated and showed in Figure 3.5, considering $\lambda = 1550 \text{ nm}$:

$$P_A = 2.27 \text{ fW} = -116.43 \text{ dBm}$$

$$P_B = 1.90 \text{ fW} = -117.20 \text{ dBm}$$

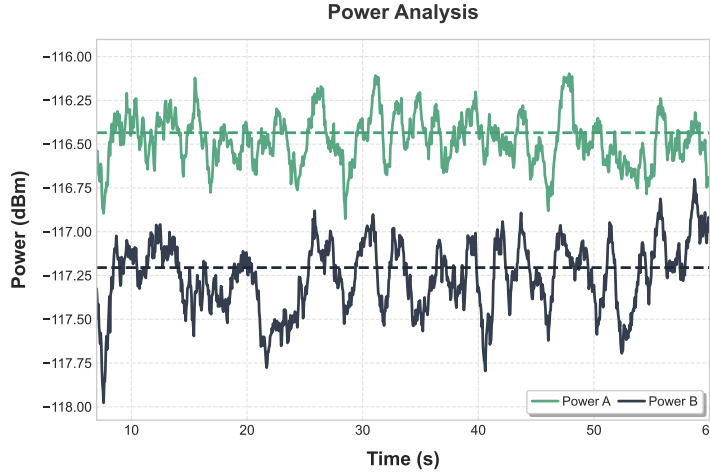


Figure 3.5: Output power SPDC source

3.1.3. Spectral Analysis

In order to determine the emission spectrum of the SPDC source, specifically identifying the wavelengths of the photons emitted from both outputs, the following experimental setup is implemented. One output of the source is connected via optical fiber directly to a SPAD detector, while the other output is sent to a programmable optical filter (**II-VI Waveshaper 4000 A**) before reaching its respective detector, as shown in Figure 3.6. The Waveshaper is an optical signal manipulation and processing device based on Liquid Crystal on Silicon (LCoS) technology, which allows for controlling the phase of light to direct the optical beam. In this experiment, the Waveshaper is used as a dynamically tunable spectral filter, capable of focusing and transmitting the wavelength of interest while blocking the unselected spectral components [30].

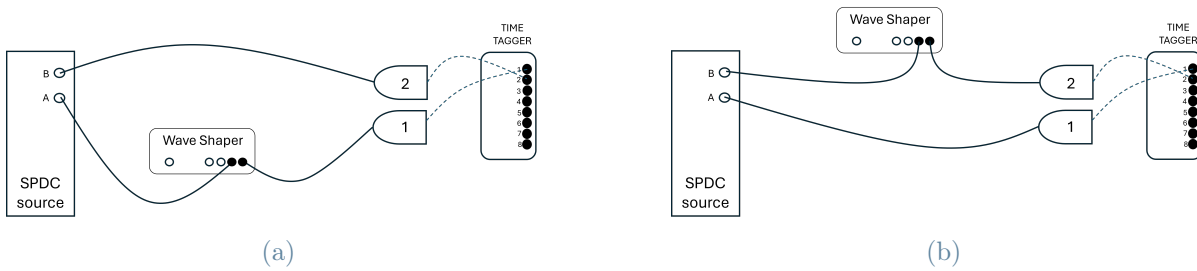


Figure 3.6: Sketch of experimental setup to determine spectrum of the source (a) normal setup configuration for output A; (b) inverted setup configuration for output B

In this framework, the spectrum is reconstructed by measuring the time coincidences between the two detectors while sweeping the filter window across the Waveshaper's bandwidth. Since the source generates entangled photon pairs, a detection event occurs simultaneously on both arms only when one photon successfully passes through the Waveshaper filter and the other

photon is registered by the opposite SPAD. Conversely, if a photon is blocked by the filter, the coincidence is lost because the remaining photon arrives without its partner. This method allows counting only the photon pairs that pass through the filter. Therefore, a higher number of coincidences at a specific wavelength means a higher probability of emission from the source at that frequency.

For the quantitative analysis, the peak value of the second-order correlation function $g^{(2)}(\tau)$ is calculated instead of using raw counts, providing a normalized coincidence measurement. The operational parameters of the detectors are configured as follows:

- Excess voltage: $V_{ex} = 2.5$ V
- Cooling system: "LOW" temperature
- Mode: FREE-RUNNING MODE
- Hold-off Time: $10 \mu s$.

The Waveshaper features an operational frequency range of $[191.1; 196.46]$ THz, corresponding to $[1526; 1568.7]$ nm, with a typical insertion loss of 4 dB. The acquisition time for the correlation calculation is fixed at 5 s, which is the minimum time required to clearly resolve the coincidence peak; a waiting time of 2 s is added to allow for filter reconfiguration and sweeping. A custom Python script was developed to interface with the Waveshaper, fully automating both data acquisition and analysis. The correlation is computed using the dedicated Time Tagger function, configured with 100 bins and a bin width of 1000 ps.

Data acquisition is first performed by analyzing the configuration shown in Figure 3.6a) to identify the wavelength of Output A. The measurement is then repeated by moving the Waveshaper to the opposite arm of the source, resulting in a specular configuration that allowed for the determination of the corresponding wavelength of Output B, Fig. 3.6b). Initially, the filter window is set to a bandwidth of $\Delta f = 0.1$ THz = 100 GHz. The sweep covered the interval $[191.2; 196.3]$ THz ($[1527; 1568]$ nm) centered at 193.79 THz (1547 nm). As shown in Figure 3.7a, a distinct coincidence peak is observed at 194.55 THz. To further investigate the spectral region of interest, the sweep range was narrowed to $[194; 195]$ THz ($[1539; 1546.8]$ nm), while halving the filter bandwidth to $\Delta f = 50$ GHz. This configuration enabled a more precise isolation of an emission peak centered around 1541.43 nm (194.63 THz) (Fig. 3.8). On the other hand, by reducing the filter bandwidth down to 10 GHz, the coincidence count rate dropped significantly: the signal-to-noise ratio became too low to statistically identify the peak, preventing from performing a correct characterization of the source spectrum. Therefore, considering the 50 GHz filter as the optimal choice to clearly visualize the peak, a Gaussian fit was performed on the corresponding experimental data points.

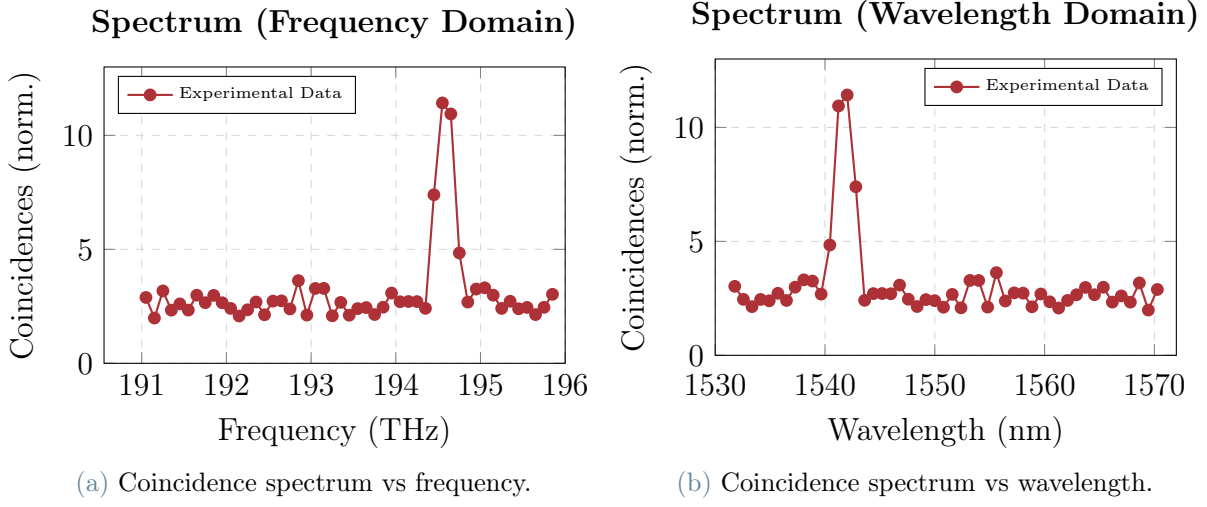


Figure 3.7: Experimental second-order correlation measurements for Output A using a filter bandwidth of 0.1 THz. The coincidence peak is plotted as a function of (a) Frequency and (b) Wavelength

For **Output A**, the fit localized the peak center at:

$$f_A = 194.5858 \pm 0.0087 \text{ THz} \quad (\lambda_A = 1541.7387 \pm 0.0685 \text{ nm}) \quad (3.8)$$

By moving the Waveshaper to the opposite arm (Fig.3.6b)), the complementary spectrum was observed. Using first the 100 GHz filter (in the [191;196] THz range) (Fig. 3.9) and then the 50 GHz filter (in the [191.5;193] THz range), a peak centered at 1560 nm was identified (Fig. 3.10). In particular, by applying a Gaussian fit to the experimental data points of this optimal configuration, the central emission frequency for **Output B** was precisely determined to be:

$$f_B = 192.3251 \pm 0.0047 \text{ THz} \quad (\lambda_B = 1558.78 \pm 0.04 \text{ nm}) \quad (3.9)$$

The two experimentally obtained emission frequencies are in perfect agreement with the SPDC process on which the source relies for generating the idler and signal photon pairs. As discussed in section 2.4.1, for this phenomenon to occur both the principle of energy conservation and the phase-matching condition must be strictly satisfied.

Indeed, the data analysis highlights a spectral symmetry between the two output channels. By summing the central frequencies extracted from the respective fits, the energy of the initial pump photon can be reconstructed with high precision.

$$f_{\text{pump}} = 194.5858 \text{ THz} + 192.3251 \text{ THz} = 386.9109 \text{ THz} \quad (3.10)$$

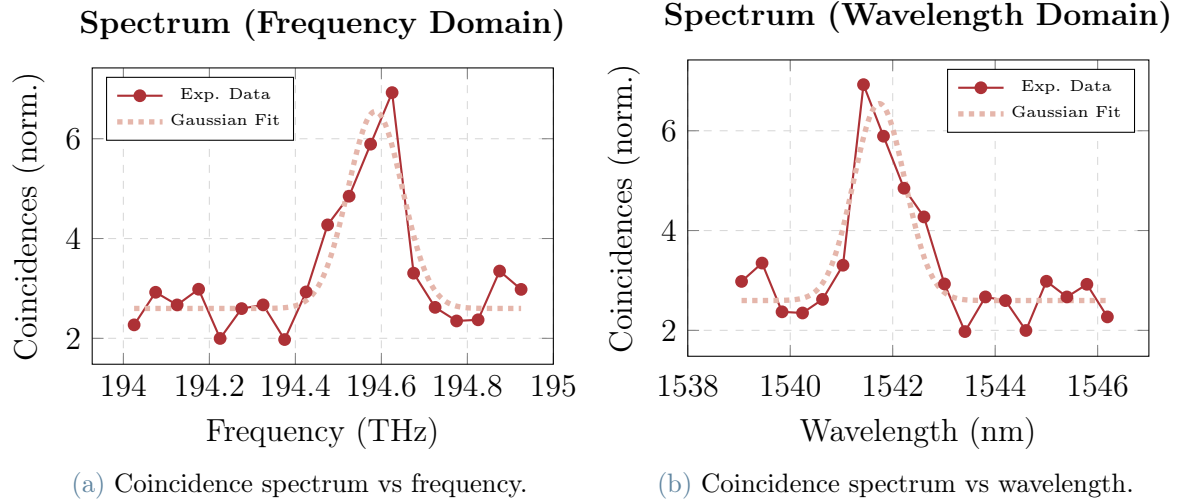


Figure 3.8: Experimental second-order correlation measurements using a filter bandwidth of 50 GHz for Output A, as a function of (a) Frequency and (b) Wavelength. The experimental data (points) are fitted with a Gaussian model (dashed line) to determine the central emission parameters.

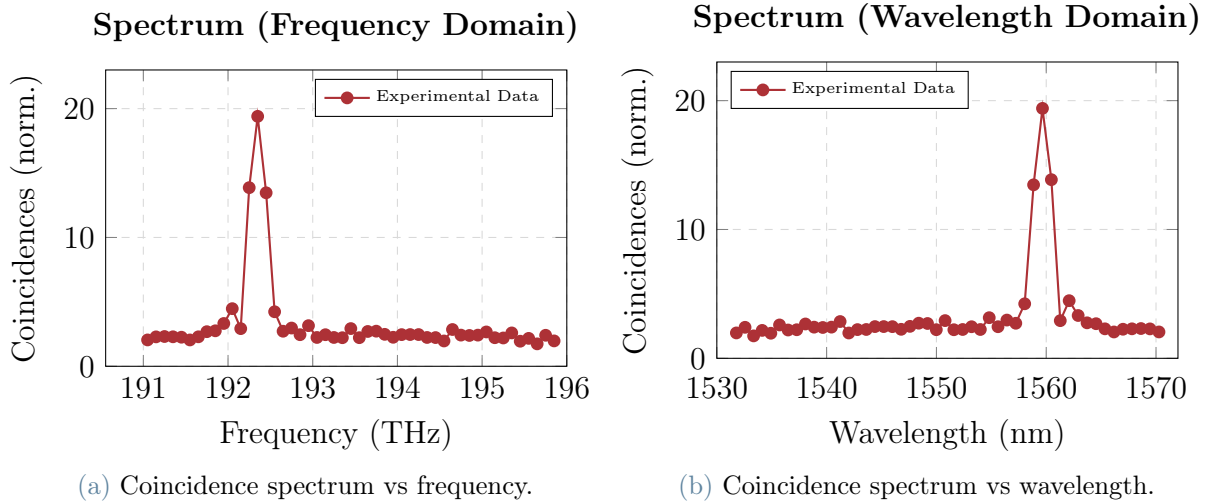


Figure 3.9: Experimental second-order correlation measurements for Output B using a filter bandwidth of 100 GHz. The coincidence peak is plotted as a function of (a) Frequency and (b) Wavelength.

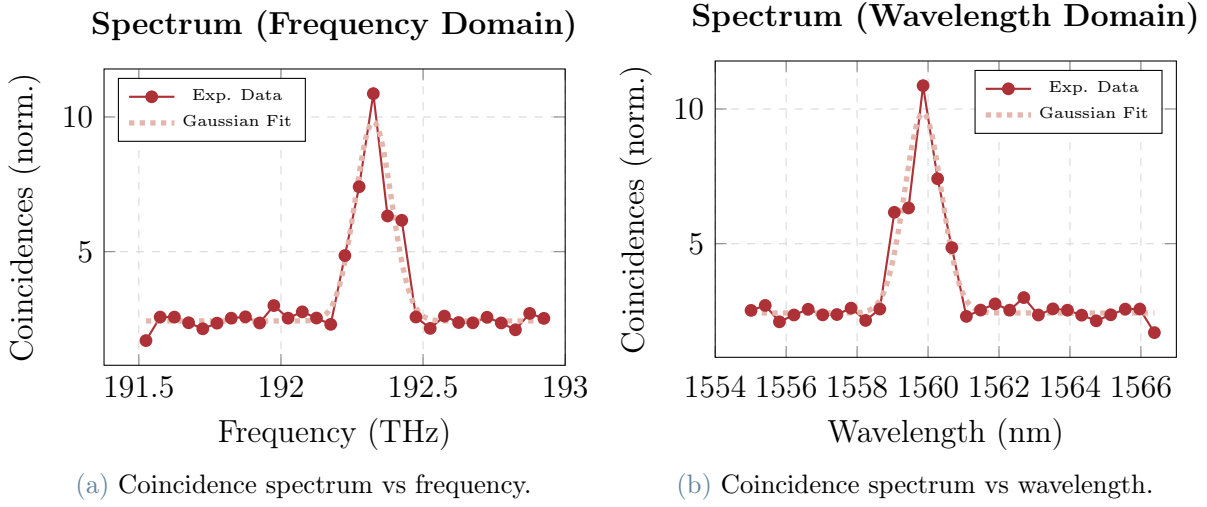


Figure 3.10: Experimental second-order correlation measurements using a filter bandwidth of 50 GHz for Output B, as a function of (a) Frequency and (b) Wavelength. The experimental data (points) are fitted with a Gaussian model (dashed line) to determine the central emission parameters.

which closely corresponds to a calculated pump wavelength of:

$$\lambda_{\text{pump}} = 774.835 \text{ nm} \approx 775 \text{ nm} \quad (3.11)$$

matching the effective operating wavelength of the experimental pump laser.

3.1.4. Coincidence Peak Identification

Before analyzing the experiments in the next chapters, it is essential to establish the methodology for characterizing the coincidence peak and determining the optimal time window.

In the used experimental setup, each output of the SPDC source was connected via optical fiber to a SPAD detector and then to a channel of Time Tagger. Time Tagger records the arrival timestamps on both channels and computes, for each pair, the time difference $dt = t_1 - t_2$.

As shown in the Figure 3.11, to construct the coincidence histogram, the time delay axis was discretized into different intervals called bins, whose width (bin width) can be configured by the user, depending on the desired resolution. Each time the calculated time difference between two events falls within the range defined by a specific bin, the counter associated with that bin is incremented by one. By accumulating these differences over the acquisition period, a histogram is generated that maps the temporal distribution of the arriving pairs.

There are two possible approaches to analyze this histogram with the Correlation class provided

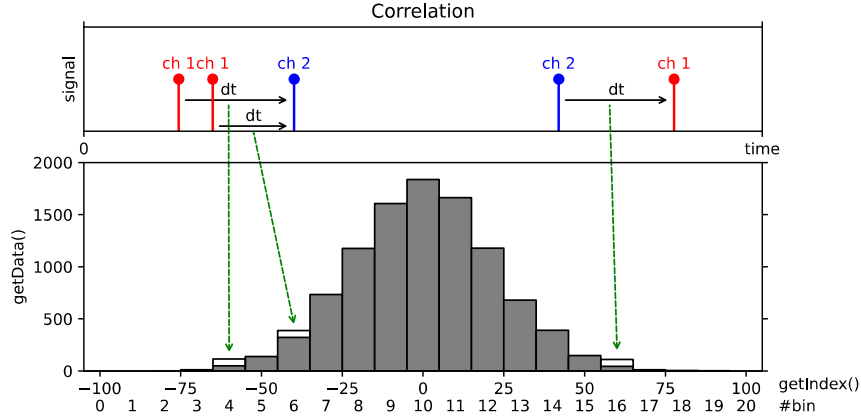


Figure 3.11: Schematic representation of the coincidence histogram construction by the Time Tagger [31].

by the Time Tagger: using raw counts or using the normalized $g^{(2)}(\tau)$ function.

The first approach relies on cumulative raw counts retrieved via the "getData()" method. This representation simply shows the actual number of accumulated events, which scales linearly and increases over the acquisition time. To overcome the dependence on the acquisition time, the second approach utilizes the second-order correlation function, $g^{(2)}(\tau)$, which measures the probability of detecting a true photon pair compared to the background noise. Since the number of counts registered by a photon-counting detector is proportional to the incident intensity, this function can be expressed directly in terms of detection events as [28]:

$$g^{(2)}(\tau) = \frac{\langle n_1(t), n_2(t + \tau) \rangle}{\langle n_1(t) \rangle, \langle n_2(t + \tau) \rangle} \quad (3.12)$$

where $n_i(t)$ is the number of counts registered by detector i at time t , and τ is the temporal delay between the two detection channels.

Physically, $g^{(2)}(\tau)$ represents the normalized conditional probability of detecting a photon on channel 2 at time $t + \tau$ given a detection on channel 1 at time t . This makes this approach robust to the specific measurement duration. The Time Tagger calculates a digital approximation of the function (3.12) using the "getDataNormalized()" method in [31].

The temporal position of the correlation peak is strictly determined by the optical path lengths traveled by the photons, which are directly related to the length of the fibers connecting the source to each detector.

By defining $\Delta s = s_1 - s_2$ as the difference between the lengths of the two fiber arms, the position of the correlation peak depends strictly on the symmetry of the optical paths. In the case of symmetric paths, where both fibers have identical lengths ($\Delta s = 0$), the pairs of photons reach the detectors simultaneously, centering the correlation peak exactly at $\tau = 0$. Conversely, in the

presence of asymmetric path lengths ($\Delta s \neq 0$), the length mismatch introduces a relative arrival delay that shifts the peak along the time axis ($\tau \neq 0$).

So, the precise position of the peak directly reflects the temporal delay between the two photons, and measuring this shift allows us to backward-calculate the exact spatial mismatch Δs between the two links.

Assuming a maximum of 100 km of optical fiber on each link, the scenario leading to the highest spatial mismatch corresponds to the two fiber lengths being equal to 100 km and 0 km. This condition establishes a maximum spatial interval equal to:

$$[-\Delta s_{\max}, +\Delta s_{\max}] = [-100 \text{ km}, +100 \text{ km}] \quad (3.13)$$

To determine the corresponding temporal search window, we consider the propagation velocity of light within a standard single-mode fiber with a refractive index of $n = 1.5$:

$$v = \frac{c}{n} = \frac{3 \cdot 10^8 \text{ m/s}}{1.5} = 2 \cdot 10^8 \text{ m/s} \quad (3.14)$$

Under these conditions, a maximum path difference of 100 km translates into a maximum expected temporal delay of:

$$\Delta t_{\max} = \frac{100 \text{ km}}{2 \cdot 10^8 \text{ m/s}} = 500 \text{ } \mu\text{s} \quad (3.15)$$

To guarantee that the correlation peak is captured regardless of any fiber mismatch, the Time Tagger's histogram must scan the entire symmetric interval of $[-500 \text{ } \mu\text{s}, +500 \text{ } \mu\text{s}]$. This defines a total expected delay range of:

$$T = 2 \cdot \Delta t_{\max} = 1 \text{ ms} \quad (3.16)$$

To initialize the Correlation class of the Time Tagger for this preliminary search, we must define the histogram's bin width (BW) and calculate the required number of discrete bins (N_{bin}). To preserve sufficient resolution, the bin width is chosen to match the characteristic timing jitter of the SPAD detectors:

$$\text{BW} = 100 \text{ ps} \quad (3.17)$$

The total number of discrete bins required to sample the full 1 ms window is therefore given by:

$$N_{\text{bin}} = \frac{T}{\text{BW}} = \frac{1 \text{ ms}}{100 \text{ ps}} = 10^7 \quad (3.18)$$

By passing these parameters to the Time Tagger software and considered ad acquisition time of $T_{\text{acq}} = 60 \text{ s}$, the system generates the $g^{(2)}(\tau)$ histogram shown in Fig. 3.12.

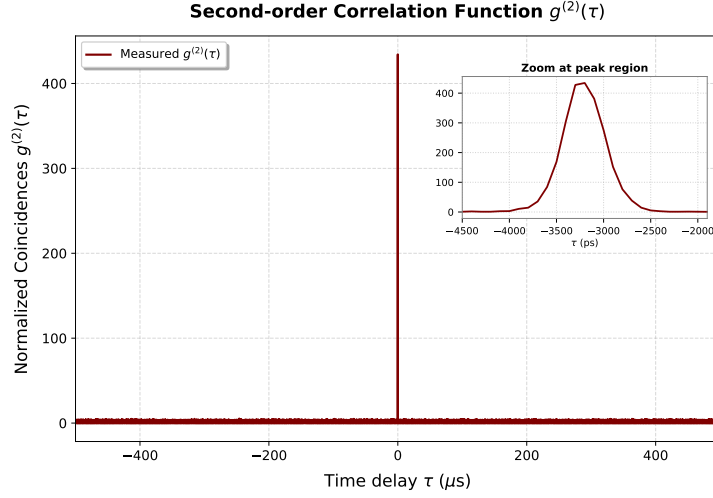


Figure 3.12: Second-order correlation function $g^{(2)}(\tau)$ as a function of the time delay τ , measured over a 1 ms wide time range $[-500 \mu\text{s} \text{ to } 500 \mu\text{s}]$ to locate the coincidence peak.

Inset: A zoom of the peak region revealing its peak position shifted to $\tau \approx -3200$ ps

The resulting $g^{(2)}(\tau)$ function exhibits a coincidence peak centered at a temporal delay of:

$$\Delta t_{\text{peak}} = -3200 \text{ ps} \quad (3.19)$$

From this measured value, the precise physical path mismatch between the two optical fibers can be extracted as follows:

$$|\Delta s| = |\Delta t_{\text{peak}} \cdot v| = 64 \text{ cm} \quad (3.20)$$

To obtain a higher-resolution zoom on this coincidence peak, a second correlation measurement must be performed with finer temporal resolution. However, prior to this step, the histogram must be centered at $\Delta t = 0$. This is achieved by introducing a software delay that is equal and opposite to the measured offset on one of the two detection channels.

Given the delay definition $\tau = t_1 - t_2$, the sign of the shift determines which optical path is longer:

$\tau > 0$ Photons on ch 1 arrive after photons on ch 2, indicating a longer fiber on Ch 1.

$\tau < 0$ Photons on ch 1 arrive before photon on ch 2, indicating a longer fiber on Ch 2.

In this configuration, the cross-correlation is evaluated between ch_1 and ch_2 . Since the observed delay is negative ($\tau < 0$), it implies that channel 2 photon arrives before channel 1 photon. This directly indicates that channel 1 features a longer fiber link, requiring the compensating delay to be applied to channel 2. To determine the optimal coincidence window, a Gaussian fit was performed on the high-resolution $g^{(2)}(\tau)$ dataset, yielding a Full Width at Half Maximum

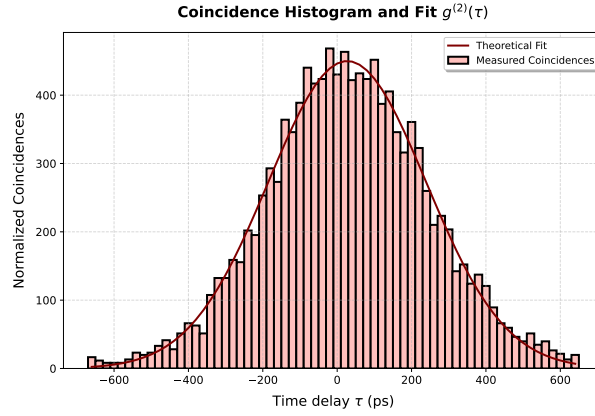


Figure 3.13: Coincidence peak centered at zero delay after delay compensation

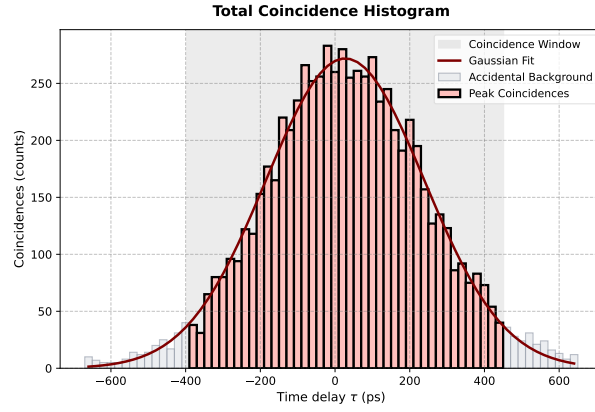


Figure 3.14: Coincidence histogram acquired over a 60 s integration time, illustrating the optimal integration window. The time bins are colored to differentiate signal from noise

(FWHM) of 499.77 ps. This parameter directly reflects the overall timing jitter of the detection system, which sets the fundamental limit on the temporal resolution of the measurement.

To ensure that the entire coincidence peak is reliably captured while minimizing the inclusion of accidental noise, the optimal coincidence window is defined as $\pm 2\sigma$ around the peak center. From a statistical perspective, this interval encompasses approximately 95.4% of the true photon pair events described by the Gaussian distribution.

$$t_{cc,final} = 4 \cdot \sigma = 848.48 \text{ ps} \quad (3.21)$$

Once this optimal window was established, it was applied directly to the true coincidence peak, as shown in Figure 3.14. By integrating all the discrete time bins contained within this localized window, the total number of accumulated coincidence counts was determined to be $N_{tot} = 6862$.

3.2. Optimization Model of BBM92

In literature, the analysis and implementation of the BBM92 protocol under realistic conditions are generally investigated using a dual approach: experimental testing and numerical simulations.

On the practical size, the challenges over long distance is addressed by Neumann et al. [14]. Their work demonstrated the feasibility of secure key generation over a 248 km terrestrial link using an already deployed telecommunication infrastructure. Connecting three distinct cities across two countries (Vienna and St. Pölten in Austria and Bratislava in Slovakia), the system achieved an average asymptotic secure key rate (SKR) of 1.4 bit/s while maintaining an entangled state visibility of 86% through an active stabilization system. Later, laboratory experiments reached even longer distances. In particular, Huang et al. reached a distance of 343 km on fiber spools by implementing a multi-wavelength single-photon-level calibration technique [13], while the Zhuang and Pan group established the current laboratory record of 404 km utilizing an ultra-bright entangled photon source [32]. On the other hand, a simulation model to predict system performance has been developed by Chalupnik et al. [33].

Behind these experiments and simulations, the choice of the source changes how the photon detection events are processed and synchronized. There are two modes of creating the pairs of entangled photons, by pumping with a pulsed laser or by continuous wave laser (CW). In pulsed regime, photons are confined within discrete, synchronized time intervals governed by a system clock [34]. Consequently, the position of each bit is defined a priori by the pump pulses, allowing for a computation of the Quantum Bit Error Rate (QBER) as described previously in Equation (2.4). On the other hand, when the entangled photon pairs are generated via spontaneous parametric down-conversion (SPDC), normally continuous-wave pump lasers are exploited. Instead of being emitted in predefined time intervals, the photons are generated stochastically. In such systems, the emission probability of a photon pair is uniformly distributed over time. Consequently, high-resolution time-tagging electronics are required to record the exact arrival times (t_A and t_B) of individual photons. Coincident pairs are subsequently identified by checking if two photons arrived within a narrow temporal tolerance limit, denoted as the coincidence window t_{CC} . So, two events are registered as a valid entangled pair if they satisfy the condition:

$$|t_A - t_B| \leq \frac{t_{CC}}{2} \quad (3.22)$$

While this asynchronous approach eliminates the need for active synchronization, it requires finding the optimal coincidence window size t_{CC} to balance noise inclusion against pair loss. The ideal size of the coincidence window t_{cc} is directly related to the Full Width at Half Maximum (FWHM) of the coincidence peak t_{Δ} , which is determined by the timing jitter of the detectors and time-tagging electronics.

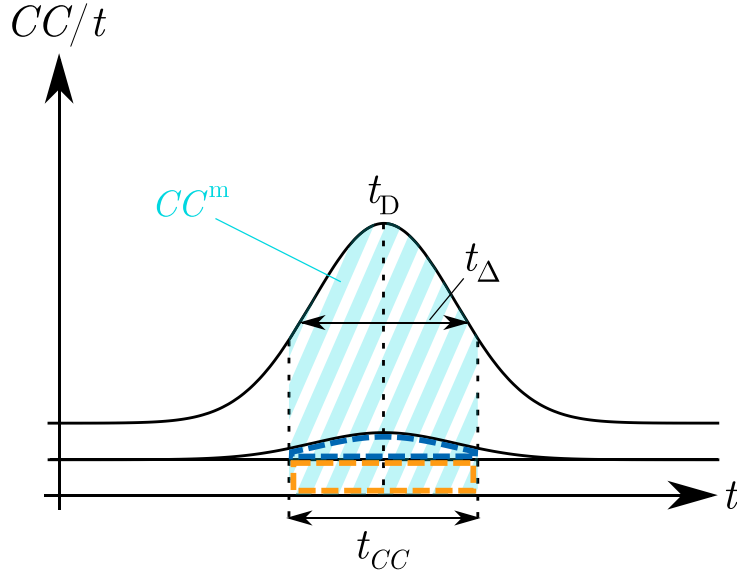


Figure 3.15: Geometrical interpretation of the coincidence counts per time unit for different relative measurement times, adapted from [35].

To quantitatively evaluate the behavior of a realistic system and optimize its performance, the mathematical model developed by [35] is adopted in the following. This framework allows for the expansion of the macroscopic QBER definition introduced in Equation (2.4), expressing it as a function of physical parameters that can be directly measured in the laboratory. In particular, in a continuous-wave (CW) architecture, the QBER can be expressed as the ratio between the coincidence rates recorded within a specific coincidence window t_{cc} :

$$QBER(t_{cc}) = \frac{CC^{err}(t_{cc})}{CC^m(t_{cc})} \quad (3.23)$$

The total measured coincidence rate $CC^m(t_{cc})$ is physically decomposed into the sum of two distinct contributions:

$$CC^m = \eta_r CC^t + CC_{acc}. \quad (3.24)$$

A geometrical interpretation of this physical decomposition is illustrated in Figure 3.15, which shows how the total counts are distributed over time. Visually, the model distinguishes between a localized Gaussian peak, representing the true pairs, and a flat background, which corresponds to the noise. To map the mathematical terms of the model directly onto this geometric representation, each contribution is defined below:

- True coincidence (CC^t): this term represents the ideal rate of entangled photon pairs generated by the SPDC source that successfully reach the detectors.
- Window Detection Efficiency (η_r): in a practical implementation, the true coincidences

are not registered instantly due to the timing uncertainties of the system. Therefore, $\eta_r(t_{cc})$ represents the amount of true coincidences that successfully fall within the chosen temporal window t_{cc} , with:

$$\eta_r = \text{erf}\left(\sqrt{\ln(2)} \frac{t_{cc}}{t_{\Delta}}\right) \quad (3.25)$$

The product $\eta_r(t_{cc})CC^t$ physically defines the effectively detected true coincidences, which correspond to the integrated area under the Gaussian peak (light blue region) in Figure 3.15.

- Accidental Coincidences (CC_{acc}): these are false coincidences caused by dark counts, background noise or the accidental registration of uncorrelated photon pairs as entangled. Since these noise events occur stochastically and uniformly over time, they are visually represented by the flat orange background in Figure 3.15 and their rate is heavily dependent on the width of t_{cc} .

The total error rate $CC^{err}(t_{cc})$ accounts for all coincidences that yield discordant results within identical bases. It is composed of two independent error contributions:

$$CC^{err} = CC_{opt,err} + CC_{acc,err} \quad (3.26)$$

It is composed by the sum of:

- Optical Errors Coincidence $CC_{opt,err}$ (blue region in Fig. 3.15): rate of all those correlated photons which are measured erroneously due to imperfections in the optical setup. It is mathematically modeled as the product of the effectively detected true coincidences and an intrinsic optical error parameter e_{pol} :

$$CC_{opt,err}(t_{cc}) = e_{pol}\eta_r(t_{cc}) CC^t \quad (3.27)$$

The parameter e_{pol} indicate the error that depends on two main factors: the extinction ratio of the optical components in the setup, and the alignment of the polarization bases between Alice and Bob.

- Accidental Errors Coincidence $CC_{acc,err}$: it is exactly half of the total accidentals coincidence. Since these are independent and completely uncorrelated events, they have an equal probability of generating either a correct bit or a key error.

By substituting the explicit physical definitions of $CC_{opt,err}$ and $CC_{acc,err}$ into the initial equation, we obtain the first comprehensive expansion of the QBER as a function of the time window t_{cc} :

$$QBER(t_{cc}) = \frac{e_{\text{pol}}\eta_r(t_{cc})CC^t + \frac{1}{2}CC_{\text{acc}}(t_{cc})}{\eta_r(t_{cc})CC^t + CC_{\text{acc}}(t_{cc})} \quad (3.28)$$

In a continuous-wave (CW) regime, the absence of a system clock renders photon emissions and noise events completely stochastic. Thus, spurious clicks on Alice's and Bob's detector are statistically independent and governed by a Poisson distribution. To derive the false positive rate, the model starts from the total measured single counts per second (S_A^m, S_B^m), which represent the individual photon detection rates recorded at Alice's and Bob's detectors, respectively. The mean number of clicks within a single coincidence window t_{cc} is defined as:

$$\mu_{S_A} = S_A^m t_{cc} \quad \text{and} \quad \mu_{S_B} = S_B^m t_{cc} \quad (3.29)$$

Assuming Poisson statistics, the joint probability P_{acc} of registering at least one simultaneous click at both terminals is given by:

$$P_{\text{acc}} = (1 - e^{-\mu_{S_A}}) (1 - e^{-\mu_{S_B}}) \quad (3.30)$$

In high-loss regimes (where $\mu_{S_i} \ll 1$), this relationship simplifies via first-order Taylor approximation ($1 - e^{-x} \approx x$), becoming:

$$P_{\text{acc}} \approx \mu_{S_A} \mu_{S_B} = (S_A^m t_{cc}) (S_B^m t_{cc}) \quad (3.31)$$

Dividing this joint probability by the fundamental temporal sampling unit t_{cc} yields the accidental coincidence rate per second:

$$CC^{\text{acc}}(t_{cc}) = \frac{P_{\text{acc}}}{t_{cc}} \approx S_A^m S_B^m t_{cc} \quad (3.32)$$

The total registered single counts (S_A^m and S_B^m) do not consist only of useful photons, but are rather the linear combination of two microscopic physical contributions:

$$S_i^m = S_i^t + DC_i \quad \text{where } i \in A, B \quad (3.33)$$

- Dark Counts DC_i : Intrinsic thermal and electronic noise of detectors.
- Single Counts S_i^t : Average local photon detection rate of Alice and Bob, excluding noise.

To relate the model to measurable parameters, the single counts and the true coincidence rate must be expressed as functions of the source dynamics and the transmission channels. The two

primary physical parameters are the source brightness B , which represents the total number of entangled photon pairs generated per second within the non-linear SPDC crystal, and the total channel efficiencies $(\eta_{A,tot}, \eta_{B,tot})$. The rate of useful photons contributing to the single counts at Alice and Bob scales linearly with the source brightness and the efficiency of that specific optical path:

$$S_A^{(t)} = B \eta_{A,tot} \quad \text{and} \quad S_B^{(t)} = B \eta_{B,tot}. \quad (3.34)$$

Similarly, the ideal true coincidence rate CC^t represents the case where both photons from a given pair survive transmission and trigger their respective detectors. This joint probability is governed by the product of both channel efficiencies:

$$CC^t = B \eta_{A,tot} \eta_{B,tot} \quad (3.35)$$

To account for the realistic physical impairments during measurement operations, the total efficiency $\eta_{i,tot}$ ($i \in \{A, B\}$) is defined using three distinct contributions:

$$\eta_{i,tot} = \eta_D \eta_{dt,i} \eta_i \quad (3.36)$$

where:

- η_D is the intrinsic detection efficiency of the single-photon detector at the operating wavelength.
- $\eta_{dt,i}$ is the efficiency contribution due to the dead time of detector i . This term models the probability that a detector is active, accounting for the temporary blindness of the device immediately following a detection event. It is expressed as:

$$\eta_{dt,i} = \frac{1}{1 + (B\eta_i\eta_D + DC_i)\frac{t_d}{n_{d,i}}} \quad (3.37)$$

where t_d is the dead time of detectors and $n_{d,i}$ the number of detectors associated at receiver i .

- η_i is the transmission efficiency that represent all optical losses along the link between the entangled photon source and detector i , excluding those explicitly caused by the detector's maximum efficiency, dead time, and temporal resolution.

Once all the physical parameters are defined, the ultimate performance of the continuous-wave BBM92 protocol can be evaluated through the determination of the *Secure Key Rate* (SKR). In

the asymptotic limit, the rate of secure bits generated per second is defined as:

$$SKR(t_{cc}) = q \cdot CC^m(t_{cc}) [1 - f(E_{bit}) H_2(E_{bit}(t_{cc})) - H_2(E_{ph}(t_{cc}))] \quad (3.38)$$

where q is the sifting efficiency factor, representing the joint probability that Alice and Bob choose identical measurement bases. For a standard, symmetric passive or active basis choice between the X and Z bases, we obtain $q = \frac{1}{2}$. In this expression, the security penalty depends on:

- $E_{bit}(t_{cc})$ represents the error rate measured on the shared bits after the sifting phase, which directly determines how much data must be removed during error correction.
- $E_{ph}(t_{cc})$ represents the phase error rate, which sets the maximum amount of information that an eavesdropper (Eve) could have intercepted.
- $f \geq 1$ is the error correction inefficiency factor that depends on the $E_{bit}(t_{cc})$. It accounts for the extra information that algorithms (such as Cascade or LDPC) must require to fix errors compared to a theoretically perfect system ($f = 1$)
- $H_2(x)$ is the binary Shannon entropy function, mathematically defined as:

$$H_2(x) = -x \log_2(x) - (1 - x) \log_2(1 - x) \quad (3.39)$$

Physically, the term $f H_2(E_{bit})$ quantifies the fraction of information leaked over the public classical channel to reconcile discrepancies within the raw key. Concurrently, the phase entropy term $H_2(E_{ph})$ determines the compression factor necessary during the privacy amplification phase to completely decouple Eve from the final, unconditionally secure cryptographic key.

Assuming a generalized model where the polarization misalignment and optical imperfections affect both bases symmetrically (i.e., where $QBER_{ph} = QBER_{bit} = QBER(t_{cc})$), the Secure Key Rate simplifies to its compact form:

$$SKR(t_{cc}) = q \cdot CC^m(t_{cc}) [1 - (1 + f(QBER(t_{cc})) H_2(QBER(t_{cc})))] \quad (3.40)$$

By combining the explicit formulations of $CC^m(t_{cc})$ and $QBER(t_{cc})$ derived in Eq. (3.28), the final comprehensive equation for the Secure Key Rate as an explicit function of the coincidence window size is obtained:

$$SKR(t_{cc}) = q \cdot CC^m(t_{cc}) \left[1 - (1 + f(QBER(t_{cc}))) H_2 \left(\frac{e_{pol}\eta_r(t_{cc})CC^t + \frac{1}{2}CC_{acc}(t_{cc})}{CC^m(t_{cc})} \right) \right] \quad (3.41)$$

However, when the two $QBER$ values in their respective bases are different, the traditional symmetric model is no longer accurate. In this case, the user can independently choose which basis to use for key generation and which one to assign for the security check. Specifically, the $QBER$ in the basis chosen to extract the key bits determines the bit error rate (E_{bit}), used for classical error correction. Concurrently, the error in the opposite basis is used as the phase error rate (E_{ph}) to detect if an eavesdropper (Eve) has intercepted the photons or if a loss of quantum coherence has occurred.

In this situation, since both channels can be processed independently, the total Secure Key Rate (SKR_{total}) is calculated as the sum of the key rates extracted from each individual basis:

$$SKR_Z(t_{cc}) = p^2 CC^m(t_{cc}) [1 - H_2(QBER_X(t_{cc})) - f(QBER_Z(t_{cc})) \cdot H_2(QBER_Z(t_{cc}))] \quad (3.42)$$

$$SKR_X(t_{cc}) = (1 - p)^2 CC^m(t_{cc}) [1 - H_2(QBER_Z(t_{cc})) - f(QBER_X(t_{cc})) \cdot H_2(QBER_X(t_{cc}))] \quad (3.43)$$

$$SKR_{tot}(t_{cc}) = SKR_Z(t_{cc}) + SKR_X(t_{cc}) \quad (3.44)$$

where p and $(1 - p)$ represent the probabilities of Alice and Bob selecting the Z and X bases, respectively.

To get the best secure key rate, it is crucial to choose the right coincidence window size. In fact, changing t_{cc} leads to two opposite results:

- A very narrow window t_{cc} reduces accidental coincidence but severely penalizes the true coincidence. Due to timing jitter, many real photon pairs are scattered and cut out of the window, driving the SKR to zero.
- An excessively wide window t_{cc} captures all available true pairs but causes the accidental coincidence increase and consequently also the $QBER$.

4 | Interface Tool for Experimental Setup Management

To transform the BBM92 quantum key distribution (QKD) protocol from a theoretical model into a real laboratory experiment, the development of dedicated control software is essential. This software was designed as a key enabling tool to orchestrate three main experimental tasks: hardware management, polarization basis alignment and automated data acquisition. Specifically, the application streamlines instrument configuration while providing a real-time graphical visualization of both single-photon counts and the coincidences generated by the entangled source. This live data stream allows to precisely align the polarization bases, a crucial step for correctly extracting the correlations required by the BBM92 protocol.

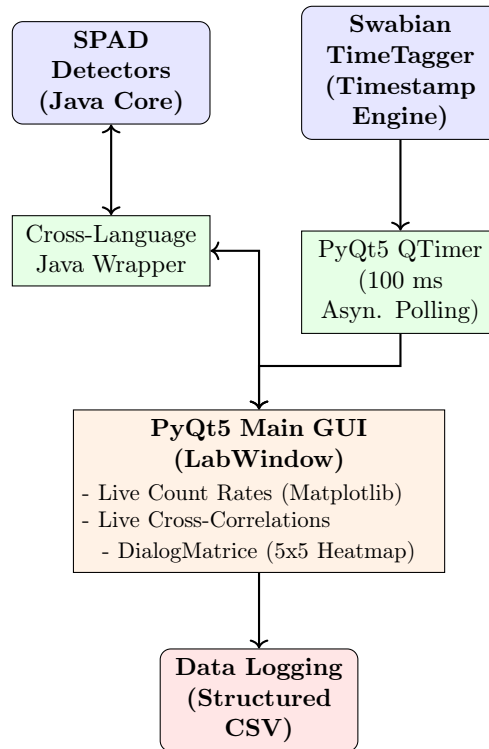


Figure 4.1: Block diagram of the developed software architecture.

4.1. Software Architecture

The software architecture was entirely developed in a *Python* environment, leveraging the *PyQt5* framework and structured according to the block diagram shown in Figure 4.1. It has been engineered following the **Model-View-Controller (MVC)** design pattern, which allows for a clear separation between the hardware management logic, data processing, and the user interface. The infrastructure can be divided into three macro-functional areas:

1. **Model:** this component manages the physical instruments and provides the necessary functions to communicate with them. In this application, the *Model* integrates two distinct hardware interfaces:
 - the native Python SDK of the *Time Tagger* (provided by Swabian Instruments), used to trigger measurements and collect the physical experimental data on the field through specific functions such as Counter and Correlation.
 - a custom interface for controlling the SPAD modules, entirely developed in Python starting from the manufacturer's native Java SDK, enabling the integration and control of both measurement instruments within the same management script.
2. **View:** this constitutes the Graphical User Interface (GUI), developed using the *PyQt5* framework and visually designed through the Qt Designer tool. It plays a purely passive and peripheral control role: it does not generate scientific data, but restricts itself to capturing the configuration settings chosen by the user (such as channel activation via checkboxes, measurement durations, or trigger voltage levels via *spinboxes*) and displaying the final results.
3. **Controller:** this acts as the actual logical engine of the application, orchestrating a circular and continuous communication flow between the operator and the instrument. The moment the user interacts with the *View* by modifying a checkbox or a parameter, the *Controller* immediately intercepts the change and invokes the specific functions provided by the *Model* to configure the hardware. Once the Time Tagger and the SPADs begin collecting data on the field, the *Controller* periodically queries the *Model* to retrieve the accumulated raw data. At this stage, it handles the entire computational logic of the system to clean and process the data, and finally sends it back to the *View*, whose sole task is to display it on the screen by updating the live plots.

To ensure real-time (*live*) plot generation, the software leverages the *QTimer* class from the Qt framework to manage system asynchrony. Specifically, the application implements two distinct timing logics:

- **Cyclic Timer for Data Polling:** a periodic timer is activated and configured to query the hardware every 100 ms. This interval allows the system to retrieve the most updated

data from the device buffers and immediately redraw the plots on the screen. This approach is essential to prevent the freezing or hanging of the graphical user interface, a critical issue that would inevitably occur due to the overhead generated by the massive and uncontrolled arrival of high-frequency timestamps.

- **SingleShot Timer for Measurement Control:** in parallel, a *singleShot* timer is employed to manage time-limited acquisitions. The duration of this timer is not fixed, but is directly configured by the user through the numerical inputs provided by the graphical interface. Upon clicking the start button, the timer begins counting in the background; at the exact expiration of the configured time, it triggers a single event that automatically stops the hardware measurement and initiates the routine for saving and serializing the recorded data into a structured CSV file.

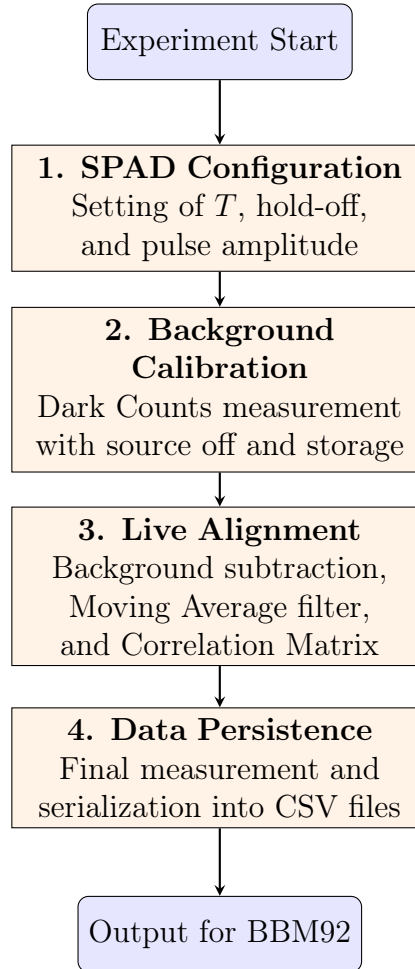


Figure 4.2: Sequential flowchart of the macro-functional logical phases managed by the software application.

The software architecture is designed to support the execution of the BBM92 protocol. The data workflow, shown in Figure 4.2, consists of four sequential phases:

1. **Initial Configuration of SPAD Detectors:** the first step involves setting up the single-photon detectors. The user can configure critical parameters for each channel, such as the excess voltage, hold-off time, and operating temperature. This phase ensures that the hardware is stable before any data collection starts.
2. **Instrumental Characterization and Background Subtraction (Dark Counts):** once the detectors are ready, a calibration phase is performed with the light source turned off to measure the background noise (*Dark Counts*). The software records the dark count rate for each channel during an acquisition of a few seconds. When the source is turned on, the system displays the total counts. The software offers two visualization modes: gross counts and net counts (calculated by subtracting the *Dark Counts*). This net count visualization is essential because it allows to align the optical system within the correct measurement bases without noise interference.
3. **Signal Stabilization and Live Alignment:** single-photon counts naturally exhibit high statistical variability. To solve this, the software includes a moving average algorithm that acts as a digital low-pass filter. This filter reduces quick fluctuations and creates a more stable, readable graph for the user. At the same time, the software displays a live correlation matrix to help with the optical alignment. This matrix maps the coincidences between channels in real time, allowing to monitor both the second-order correlation function ($g^{(2)}$) and the cumulative coincidence counts.
4. **Data Saving and CSV Export:** after completing the optical alignment and optimizing the measurement bases, the user can run the final experiment. Once the measurement is complete, the software automatically handles data storage. It saves both the single-channel detection streams and the coincidence curves into structured CSV files. These files are the final output of the application and provide the clean baseline data needed for the subsequent BBM92 post-processing analysis.

4.2. Graphical User Interface (GUI)

While the software architecture defines the robustness and logical flow of the simulation, the Graphical User Interface (GUI) determines its usability and practical effectiveness. The interface is organized into a layout with 3 main tabs that directly implement the chronological sequence of the experimental phases.

4.2.1. Tab 1: SPAD Parameters Configuration

The first GUI Tab, named *SPAD configuration*, implements the functionalities described in Phase 1 (Initial Configuration of SPAD Detectors) of the previous section 4.1. This Tab allows the user to set up and stabilize the single-photon detection hardware before starting the measurement

acquisition.

As shown in Figure 4.3, the top of the screen shows the global “Refresh SPAD” button, which allows the user to instantly update the SPAD status, identifying which modules are currently connected via USB. Immediately below this is a structured central table in which each row corresponds to a specific detection module, identified by its “Serial Number” and a “Name” that maps the respective experimental channel (for example, 1AH or 2AV, associated with the polarization states of Alice or Bob). By refreshing the page, the user can check the “Connected” column, which indicates in real time (with the status ‘Yes’ or ‘No’) which physical modules are actually connected. This check is essential because, even if the USB cable is disconnected, the devices retain the previously set parameters in memory; consequently, the column allows you to immediately identify which modules require configuration or have been disconnected.

To define the operating parameters for each active channel, the user interacts with the specific fields in the row: numerical input boxes (QSpinBox) for the pulse amplitude ("Amplitude", in mV) and hold-off time ("Hold Off", in ns), alongside drop-down menus (QComboBox) for selecting the temperature ("Temperature") and gate mode ("Gate Shape").

The entered values can be transmitted in two ways: individually, using the "Apply" button at the far right of each row, or simultaneously for all connected channels, using the global "Apply Setting" button located at the top of the screen. Once the controller successfully transmits the settings to the module, the software automatically checks the box in the "Configured" column. This double-confirmation mechanism ensures that the user knows with absolute certainty both which channels are physically connected via USB (Connected) and which have been correctly programmed (Configured) before proceeding with the measurements.

4.2.2. Tab 2: Dark Counts and Background Calibration

The second Tab (Fig. 4.4), named Dark Count, implements the functionalities described in Phase 2 (Instrumental Characterization and Background Subtraction) of the previous section 4.1. The purpose of this section is to quantify and record the background noise of the SPAD modules before activating the quantum source. First, the channels to be calibrated are selected via a dedicated row of checkboxes, while the corresponding voltage threshold, expressed in Volts, is defined using the "Trigger level" numerical boxes (setting, for example, a value of 0.50 V). Within the same area, the measurement duration is specified in the "Acquisition time" field before launching the process by clicking the "START" button. During acquisition, an integrated Matplotlib graph ("Count rate") displays in real time the number of counts per second (Events/s) versus time (s). Upon expiration of the set time, or following a manual press of the "STOP" command, the system calculates the mathematical average of the counts for each module and sends the results to the upper section of the screen. There, a summary table maps the hardware by associating the "Serial Number" and "Name" of each SPAD with its respective

Serial Number	Name	Configured	Connected	Amplitude	Temperature	Gate Shape	Hold Off	
183	1AH	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply
186	2AV	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply
190	3AD	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply
188	4AA	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply
189	5BH	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply
185	6BV	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply
159	7BD	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply
187	8BA	☐	No	5000 mV	TLOW	FREERUNNIN	10000 ns	Apply

Figure 4.3: TAB 1: SPAD Parameters Configuration

logical channel (from 1 to 8), instantly updating the final column, labeled "Dark Count Rate", with the calculated noise values. These baselines are then stored in the application memory to enable the subsequent "Net Counts" mode, which is essential for isolating the actual signal from noise during the following optical alignment procedures.

4.2.3. Tab 3: Live Counts, Coincidences, and Data Export

The third tab of the interface, named "Time Tagger", represents the operational core of the entire system and implements the functionalities described in Phases 3 and 4 of the previous section 4.1, coordinating live alignment operations and the export of correlated data, Fig. 4.5.

For each channel, the interface allows the user to adjust both the discrimination threshold in Volts and the respective "Input delay" expressed in picoseconds (ps), an essential parameter to compensate for geometric propagation delays introduced by coaxial cables or optical fibers. Furthermore, the user can dynamically customize the graphical response by toggling specific mode checkboxes. The "Counter" option enables photon counting, while the check "Delete dark count" allows the software to automatically subtract the noise values previously stored in the "Dark Count" tab from the data stream, ensuring the isolation of the actual signal. Meanwhile, the "Moving Average" checkbox applies a moving average filter that dampens statistical-temporal fluctuations in the count rate to stabilize the plots. Once acquisition is launched via the controls on the lower panel, the central section of the GUI updates two stacked Matplotlib-integrated graphs in real time. The upper plot ("Count rate") monitors the stability of the single-photon

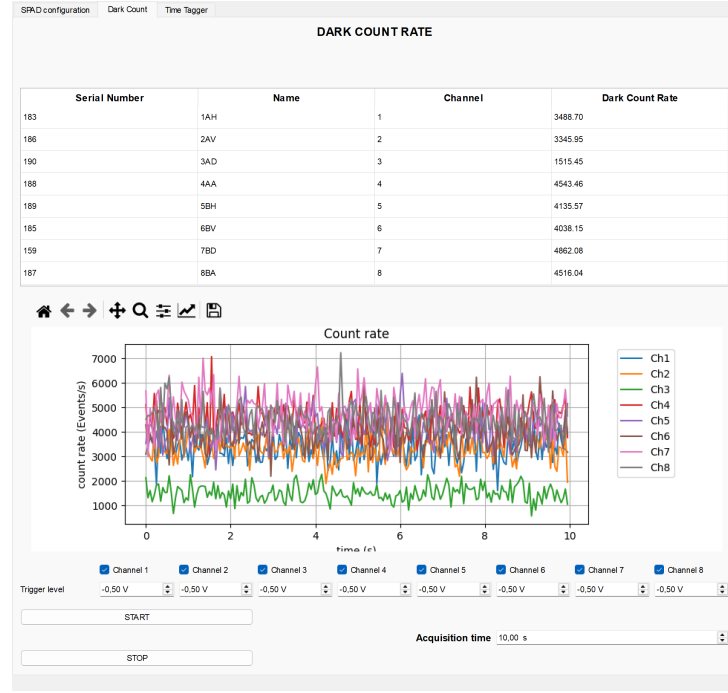


Figure 4.4: Tab 2: Dark Counts and Background Calibration

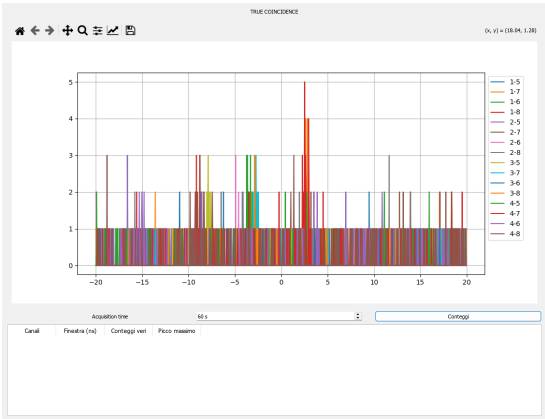
streams expressed in Events/s, providing the flexibility to selectively show or hide the trace of each channel at any moment by checking the corresponding checkboxes at the very top of the tab. Conversely, the lower plot ("normalized correlation") displays the second-order correlation function, $g^{(2)}(\tau)$, relative to the time delay (expressed in nanoseconds). Within this section, a dedicated checkbox grid located directly below the graph allows the user to select and activate exclusively the coincidences between the specific pairs of channels relevant to the experimental protocol. In the lower section of the screen, the total acquisition duration can be defined using the "Start acquisition for" field, while the measurement flow and reset are controlled via the available stream commands. Since the time correlation function implemented in the software operates in an inherently cumulative mode, the recorded counts tend to accumulate progressively over time. Consequently, the user utilizes the "CLEAR" button to flush the memory buffers and reset the history of accumulated events; this allows the graph traces to restart from zero, accurately isolating the new acquisition window. Upon expiration of the set timer, the system automatically stops the process and triggers the archiving routine, saving both the single-count data for each channel and the traces of the $g^{(2)}(\tau)$ function in structured CSV files, ready for subsequent offline analysis.

The bottom panel includes dedicated buttons to open two auxiliary dialog windows designed for detailed data analysis and setup calibration. The first of these windows, named Dialog-Matrice and recalled via the "MATRIX" or "CAPTURE MATRIX" commands, displays a $5 \cdot 5$ matrix grid, Fig.4.6b. Graphically, the core $4 \cdot 4$ sub-matrix maps the temporal correlation peaks calculated for each channel pair, where the matrix axes geometrically cross Alice's



Figure 4.5: Tab 3: Live Counts, Coincidences, and Data Export

bases along the rows (Alice H, Alice V, Alice D, Alice A) with Bob's bases along the columns (Bob H, Bob V, Bob D, Bob A). Rather than displaying raw cumulative counts, each of these sixteen cells dynamically shows the calculated numerical value of the temporal correlation peak for that specific channel pair. The additional fifth row and fifth column act as marginal indicators, displaying the mathematical sum of the counts recorded across their respective columns and rows within the matrix. To simplify the visual interpretation of the data, the background of each cell is linked to a heatmap that updates in real time, where the intensity of the blue color varies in proportion to the recorded peak height. Operationally, this layout serves as a fundamental tool during the optical alignment phase. Instead of using the graph, the user can evaluate protocol efficacy and cumulative channel performance observing the chromatic distribution across the grid. When launched in continuous mode via the "MATRIX" command, the heatmap changes in real time following manual optical adjustments; conversely, under the timed "CAPTURE MATRIX" command, the numerical values and color intensities stabilize progressively throughout the measurement, freezing the final configuration at the end of the countdown for accurate validation prior to CSV export. The second window is recalled via the "**TRUE COINCIDENCE**" button located in Tab 3. This interface embeds a live plot that visualizes the raw, unnormalized coincidence peaks as they progressively accumulate over the integration time (Fig.4.6a). While this visualization shares its mathematical operational framework with the normalized second-order correlation function ($g^{(2)}$), it fundamentally diverges by mapping the absolute, effective coincidence counts rather than a relative statistical distribution. Through this interface, it is possible to set the desired time interval in the dedicated field and start the



(a) True coincidence

	Bob H	Bob V	Bob D	Bob A	Tot
Alice H	2.00	3.00	5.00	5.00	15.00
Alice V	3.00	3.00	3.00	3.00	12.00
Alice D	3.00	2.00	3.00	3.00	11.00
Alice A	2.00	4.00	4.00	3.00	13.00
Tot	10.00	12.00	15.00	14.00	0.00

Refresh Time:
1000 ms

(b) Matrix

Figure 4.6: Dialog Tab: (a) True coincidence; (b) Matrix.

measurement by pressing the count acquisition button. Once the preset time has elapsed, the system allows the data to be exported into a structured CSV file. This file synchronously saves both the normalized trace of the correlation function $g^{(2)}$ and the absolute values of the actual counts, providing a complete statistical and quantitative overview.

5 | BBM92 Implementation and Results

This chapter describes the experimental implementation of the BBM92 quantum key distribution protocol. A detailed analysis of the entire process is presented, starting from the configuration of the experimental setup and data acquisition, and leading to the determination of coincidences for the calculation of the Quantum Bit Error Rate and the Secure Key Rate. Furthermore, the development of a numerical simulator designed to analyze the system's performance and estimate parameters that cannot be directly measured is illustrated. Finally, the results of these simulations are presented and discussed, serving to evaluate potential optimizations of the setup, such as the integration of a different type of detectors.

5.1. Experimental Setup

The setup used for the implementation of the protocol is illustrated in Figure 5.1. As described in detail in Chapter 3.2, an SPDC source (HEPS-1000-3A3A-1550-9/125, OZ OPTICS) operates as the polarization-entangled photon pairs at 1550 nm, leveraging a PPLN crystal driven by a 775 nm pump laser.

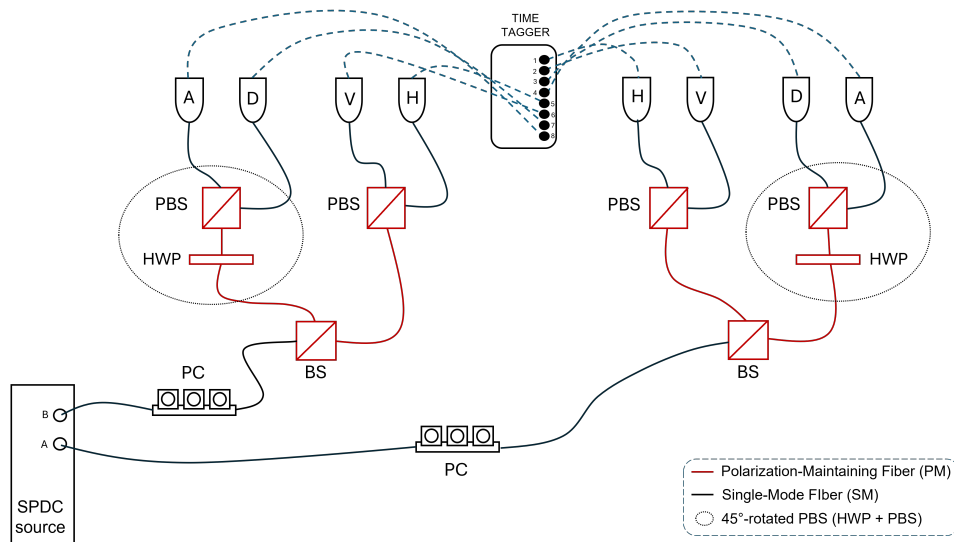
Specifically, the Bell state generated by the source is given by:

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|HV\rangle + |VH\rangle) \quad (5.1)$$

where $|H\rangle$ and $|V\rangle$ represent the orthonormal states of linear polarization, horizontal and vertical respectively. This specific state exhibits perfect anti-correlation in the Z basis (horizontal and vertical) and perfect correlation in the X basis (diagonal and anti-diagonal). Then, the entangled photons are coupled into two separate Single-Mode Fiber branches to reach the communication partners, Alice and Bob. Propagation within single-mode optical fibers inevitably introduces birefringence along with mechanical or thermal stress, which alters the polarization state of the photons along the path. To overcome this issue and correctly align the reference frames, a 3-paddle fiber polarization controller (Thorlabs 3-Paddle Fiber Polarization Controller, PC) is integrated into each branch. This device requires manually moving and orienting the three



(a)



(b)

Figure 5.1: Experimental setup for the BBM92 protocol: (a) photo of actual set-up; (b) Sketch

paddles, which physically act as wave retarders (quarter-wave and half-wave plates) by exploiting stress-induced birefringence. Their adjustment allows for the active compensation of polarization drift until the desired initial state is restored. After the polarization controllers, in each branch the entangled photons encounter a 50:50 fiber beam splitter (BS). This component acts as a passive switch, randomly routing the photon either to the Z measurement basis or to the X basis, thereby ensuring the random basis choice required by the BBM92 protocol. The outputs of the 50:50 beam splitter are connected to two fiber polarization beam splitters (PBSs): in the branch dedicated to the basis Z , the PBS features a 0° rotation and allows the separation of $|H\rangle$ and $|V\rangle$ states; on the other hand, in the branch dedicated to the basis X , the PBS is physically rotated by 45° , enabling the decomposition and measurement of diagonal $|D\rangle$ and anti-diagonal $|A\rangle$ states. While the initial polarization controllers operate on standard single-mode (SM) fiber, polarization-maintaining (PM) fiber is exclusively used from the 50:50 beam splitter up to the detectors. This choice is fundamental to preserve the alignment of the previously calibrated polarization states throughout the entire internal measurement path of Alice and Bob.

A total of 4 single-photon avalanche diodes (SPADs) are connected to each of the PBS outputs (for both Alice's and Bob's arms), dedicated to counting the incoming photons for each specific basis. The signals from the SPADs are sent to a Time Tagger, which records the timestamp of each event to enable the execution of the desired measurements.

5.2. Experimental Procedure and Measurement Phases

Following the physical description of the setup, this section focuses on the experimental implementation of the protocol. The practical workflow is structured into three consecutive macro-phases, fully controlled via the Python software application introduced in Chapter 4.

5.2.1. Initial Hardware Configuration

The first phase of the experiment involves initializing and stabilizing the SPAD modules to optimize their performance and reduce noise. For this experiment, the operating parameters were set as follows:

- Excess voltage: $V_{ex} = 2.5$ V
- Cooling system: "LOW" temperature
- Mode: FREE-RUNNING MODE
- Hold-off Time: $10 \mu s$.

Each of these choices is based on a precise practical trade-off linked to the characteristic curves discussed in Chapter 2. Selecting an Excess voltage of 2.5 V provides an optimal balance between

the detector quantum efficiency and the generated dark count rates (see Figure 2.5). A higher voltage would indeed increase the PDE and improve the timing resolution, but at the cost of a severe increase in dark counts. This increase would inevitably generate more false counts, resulting in a higher QBER that degrades the overall performance of the BBM92 protocol. To prevent this, the excess voltage is paired with a "LOW" temperature configuration to ensure that the baseline dark count rate remains minimal, which effectively stops random thermal counts from ruining the live correlation matrix. This stabilization of voltage and temperature is closely related to the choice of a hold-off time of $10\ \mu\text{s}$, an interval specifically calibrated because it must be long enough to block any potential afterpulsing, but short enough to avoid excessive dead time. An unnecessarily long dead time would cause the system to miss valid photons arriving during that window, thereby lowering the total detection count and directly reducing the final secure Key Rate.

Ultimately, this entire balance is sustained by operating in free-running mode, which is mandatory since the SPDC source generates photons continuously and asynchronously. In this mode, the SPADs remain constantly ready to capture incoming photons.

5.2.2. Noise Calibration and Polarization Alignment

Before proceeding with polarization alignment, a preliminary calibration of dark counts is strictly required. This step is essential because each SPAD module naturally exhibits a slight difference in quantum efficiency, directly resulting in distinct baseline levels for dark counts. To achieve precise alignment, all channels must share a common reference baseline, effectively canceling out the noise offset at the software level. This normalization ensures that, during polarization adjustment, any variation in count rates is due solely to optical alignment rather than hardware asymmetries. Minimizing these background counts across the channels is crucial for maximizing the polarization extinction ratio, a parameter that directly affects the final security of the quantum channel.

Once the noise calibration is complete, the polarization alignment procedure can be performed. As explained in 3.1, the Sagnac interferometer inside our source allows for three different operating configurations by physically rotating an internal half-wave plate (HWP). This feature is directly exploited to align the system through **three sequential steps**:

1. Set the **source's HWP to 0°** (which corresponds to an actual physical offset of -1.5°). In this specific configuration, the source emits a purely horizontal state ($|H\rangle$) on Output 1 and a purely vertical state ($|V\rangle$) on Output 2. By monitoring the software interface in real time from Alice's side (connected to Output 1), alignment is performed by rotating the paddles of the polarization controllers to maximize counts on the channel mapped to the $|H\rangle$ state, while simultaneously reducing counts on the $|V\rangle$ channel to zero. In parallel, the opposite correction is applied to the other output, Bob's side, maximizing counts on

the $|V\rangle$ channel and minimizing those on the $|H\rangle$ channel.

2. As a verification phase, the **source's HWP is rotated by 45°** . This rotation completely inverts the quantum states: Output 1 now produces a state $|V\rangle$ and Output 2 a state $|H\rangle$. If the initial alignment was successful, the count distribution on the software interface should be completely reversed, showing that what was previously maximized has now dropped to the background noise level, and vice versa.
3. Set the system to true entanglement mode by rotating the **source's HWP by 22.5°** . At this point, the source ideally generates the Bell state $|\Psi^+\rangle$, characterized by specific non-local correlations. Consequently, optimal alignment is verified by monitoring both the real-time coincidence matrix and the coincidence peak plot. The goal is to confirm that the coincidences clearly emerge from the background noise and are concentrated exclusively within the expected correlated pairs, specifically HV and VH for the computational basis, and DD and AA for the diagonal basis.

It is important to note, however, that while aligning the computational Z basis ($|H\rangle / |V\rangle$) is relatively simple and highly precise thanks to the use of the PM fiber, the alignment of the diagonal X basis turns out to be significantly more complicated. The theoretical justification for this asymmetry between the two bases lies in the effect that environmental fluctuations have on the relative phase of the quantum state. To formally understand how the phase exclusively affects the diagonal basis, it is necessary to analyze the mathematical evolution of the state.

The two-photon state generated by the source and propagating along the fiber channel can be described by introducing a **relative phase** shift ϕ :

$$|\Psi^+\rangle = \frac{|HV\rangle + e^{i\phi} |VH\rangle}{\sqrt{2}} \quad (5.2)$$

When performing a measurement in the rectilinear (Z) basis, the probability of registering a coincidence on a specific pair of channels (for example, the $|HV\rangle$ pair) is determined by the squared modulus of the transition amplitude, in accordance with Born's rule introduced in the previous chapter (see Equation (1.3)):

$$P(HV) = |\langle HV | \Psi^+ \rangle|^2 \quad (5.3)$$

By developing the scalar product in vector form, where the single-particle states are defined as $|H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, the joint polarization state and its corresponding projection bra

take the following configurations within the tensor space:

$$|VH\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |HV\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (5.4)$$

$$\langle HV| = \begin{pmatrix} 0 & 1 & 0 & 0 \end{pmatrix} \quad (5.5)$$

Calculating the mutual orthogonal projections yields:

$$\langle HV|HV\rangle = 1, \quad \langle HV|VH\rangle = 0 \quad (5.6)$$

By substituting these values into the probability amplitude, the phase term vanishes due to the orthogonality of the basis vectors. Consequently, when computing the measurement probability via the squared modulus, no relative phase term survives:

$$\langle HV|\Psi^+\rangle = \frac{1}{\sqrt{2}}(\langle HV|HV\rangle + e^{i\phi}\langle HV|VH\rangle) = \frac{1}{\sqrt{2}} \quad (5.7)$$

$$P(HV) = |\langle HV|\Psi^+\rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (5.8)$$

The situation changes radically when projecting the system onto the states of the X basis, defined by the linear combinations:

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \quad |A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}} \quad (5.9)$$

Taking the parallel coincidence $|DD\rangle$ as a reference:

$$P(DD) = |\langle DD|\Psi^+\rangle|^2 \quad (5.10)$$

The projective vector in the two-particle space corresponds to:

$$\begin{aligned}
|DD\rangle &= |D\rangle \otimes |D\rangle = \left(\frac{|H\rangle + |V\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|H\rangle + |V\rangle}{\sqrt{2}} \right) \\
&= \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] \otimes \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}
\end{aligned} \tag{5.11}$$

With the matching bra:

$$\langle DD| = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} \tag{5.12}$$

Calculating the transition amplitude of the state $|\Psi^+\rangle$ onto this basis reveals the direct interference of the two contributions:

$$\begin{aligned}
\langle DD|\Psi^+\rangle &= \frac{1}{\sqrt{2}} \left[\frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + e^{i\phi} \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right] \\
&= \frac{1}{2\sqrt{2}} (1 + e^{i\phi})
\end{aligned} \tag{5.13}$$

Applying Born's rule and expanding the complex exponential via Euler's identity ($e^{i\phi} = \cos \phi + i \sin \phi$), the coincidence probability is derived as:

$$\begin{aligned}
P(DD) &= \left| \frac{1}{2\sqrt{2}} (1 + e^{i\phi}) \right|^2 = \frac{1}{8} |(1 + \cos \phi) + i \sin \phi|^2 \\
&= \frac{1}{8} [(1 + \cos \phi)^2 + \sin^2 \phi] = \frac{1}{8} [1 + 2 \cos \phi + \cos^2 \phi + \sin^2 \phi]
\end{aligned} \tag{5.14}$$

Leveraging the fundamental trigonometric identity $\cos^2 \phi + \sin^2 \phi = 1$, the expression reduces to:

$$P(DD) = \frac{1}{8} (2 + 2 \cos \phi) = \frac{1}{4} (1 + \cos \phi) \tag{5.15}$$

By analogously extending the calculation to all combinations within the diagonal basis, the complete set of probabilities is obtained:

$$P(DD) = P(AA) = \frac{1}{4} (1 + \cos \phi) \tag{5.16}$$

$$P(AD) = P(DA) = \frac{1}{4}(1 - \cos \phi) \quad (5.17)$$

If the phase is perfectly compensated ($\phi = 0$), we obtain $P(DD) = P(AA) = \frac{1}{2}$ and $P(AD) = P(DA) = 0$, ensuring maximum correlations. Conversely, if the phase drifts toward $\phi = \pi$, the state decays into the orthogonal channel, completely inverting the coincidence counts (such as shifting from the $|\Psi^+\rangle$ to the $|\Psi^-\rangle$). This demonstrates that the relative phase ϕ directly controls the state, fundamentally determining how the coincidence counts are distributed between the channels of the X basis.

In the setup, a polarization-maintaining (PM) fiber is utilized, which is characterized by a controlled and intentional birefringence. From a physical standpoint, this structural asymmetry generates two propagation axes for the light, commonly referred to as the slow axis and the fast axis. The difference between their respective refractive indices ($\Delta n = n_V - n_H$) causes the horizontally and vertically polarized components to travel at different velocities. This gap introduces a time-of-flight delay, which translates directly into a relative phase ϕ linked to the fiber length L according to the relation:

$$\phi = \frac{2\pi\Delta nL}{\lambda} \quad (5.18)$$

This phase shift has a clearly asymmetric impact on the two measurement bases. For the linear basis (Z), the alignment remains completely unaffected because, as demonstrated by the previous calculations, the measurements on the $|H\rangle$ and $|V\rangle$ channels are mathematically independent of the relative phase ϕ , ensuring the perfect stability of the computational basis. Conversely, significant alignment issues are encountered for the diagonal basis (X), which strictly requires the two polarization components to be perfectly synchronized to generate constructive interference. However, even minimal variations in ambient temperature alter both the physical length of the fiber and its intrinsic birefringence. This phenomenon continuously modifies the accumulated delay, causing the phase ϕ to drift over time and introducing measurement errors.

To address this issue, a possible solution could be active control of the phase via a **mechanical fiber squeezer**, the schematic operating principle of which is illustrated in Figure 5.2. By applying physical pressure to the PM fiber, it would be possible to induce a controlled variation in the birefringence profile, allowing for the tuning of the phase ϕ at will to counteract the thermal drift.

Consequently, the next section provides a detailed analysis of the finest dataset collected during the experiment, highlighting the optimal results achieved through the native stability and standard optimization of the setup.

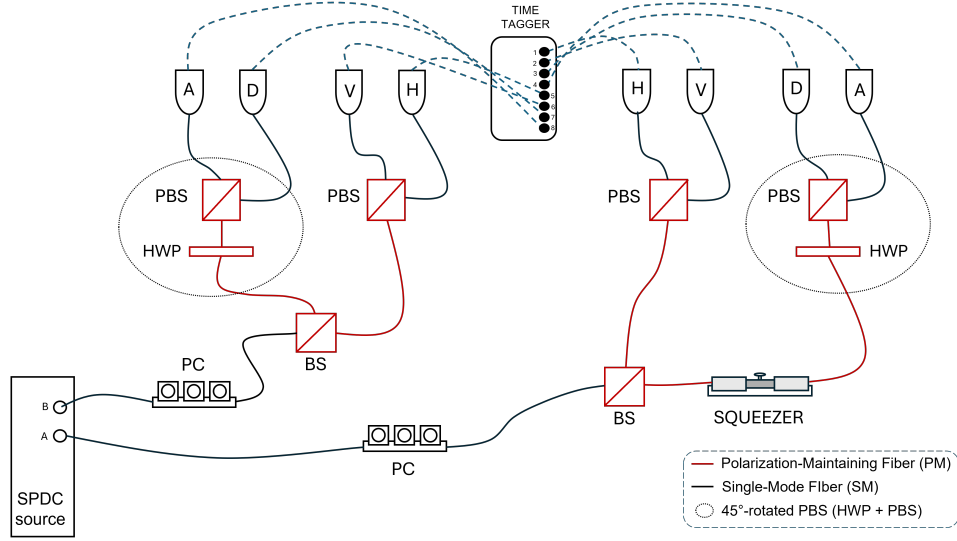


Figure 5.2: Sketch of the experimental setup for the BBM92 protocol with the squeezer source.

5.2.3. Data Analysis and Key Rate Evaluation

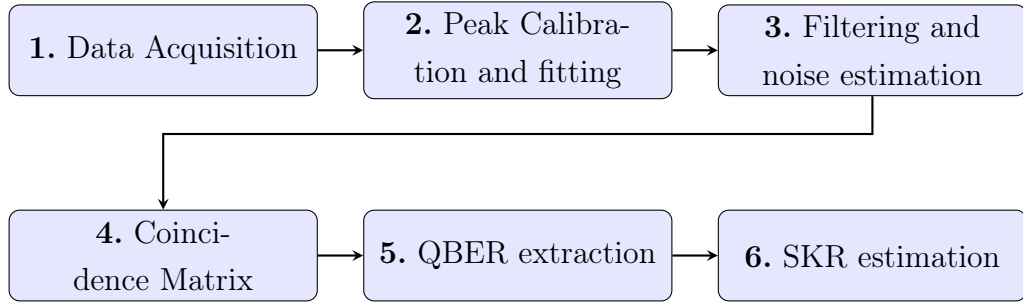


Figure 5.3: Block diagram of the BBM92 data analysis pipeline.

The analysis of the experimental data is structured following the sequential pipeline outlined in the block diagram of Figure 5.3, aimed at extracting the quantum signal and evaluating the SKR with BBM92 protocol [13, 14]. Once the alignment phase is successfully completed, the procedure proceeds with the data acquisition phase, where photon counts are recorded directly via the control software. The measurement is performed with an integration time T_{aq} set to 120 seconds, providing both the raw cumulative coincidence data and the coincidence measurements needed to determine the second-order correlation function ($g^{(2)}$).

To isolate the events relevant to the protocol, the detection channels are classified based on the theoretical correlations of the target Bell state Ψ^+ . This state establishes a clear distinction between two types of registered events. On one hand, **correct coincidences** match the expected correlations HV and VH in the computational Z -basis, and DD and AA in the diagonal X -

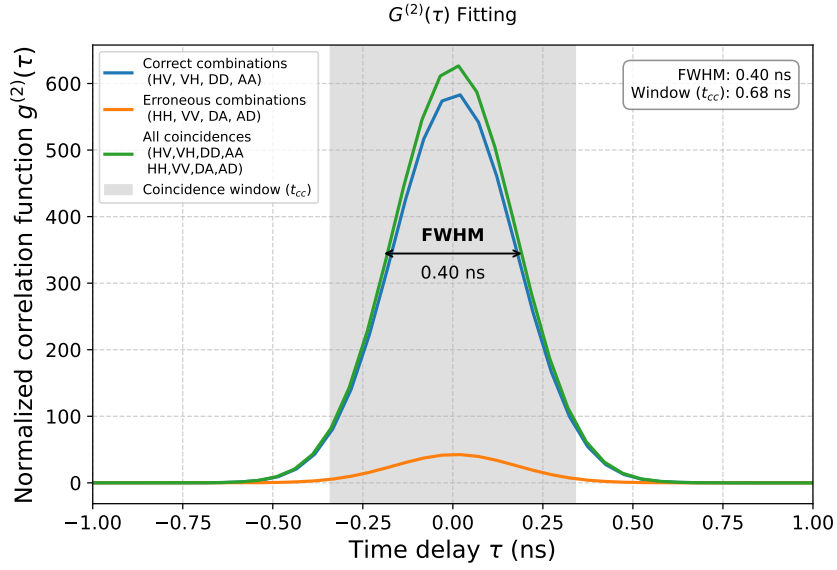


Figure 5.4: Gaussian fit of the overall second-order correlation function $g^{(2)}(\tau)$ (green curve) along with the identification of the partial contributions (blue and orange curves). The gray area defines the optimal coincidence window.

basis. These events directly confirm the proper generation of entanglement. On the other hand, **erroneous coincidences** consist of the mismatched cross-combinations (HH, VV, DA, AD). Each pair of channels generating a coincidence is characterized by a specific temporal peak in the corresponding histogram. As established by the calibration procedure described in Section 3.1.4, identifying the optimal time window is necessary to extract the real counts, isolating them from the accidental events caused by detector background noise or uncorrelated photons. To determine the width of this window, a Gaussian fit is performed on the overall $g^{(2)}(\tau)$ correlation peak. This overall peak is obtained by summing both correct and erroneous coincidence contributions, whose relative propagation delays were preventively compensated to ensure accurate alignment at $\Delta t = 0$. As shown in Figure 5.4, this fitting procedure allows for the extraction of both the standard deviation (σ) and the Full Width at Half Maximum (FWHM) of the distribution:

$$\sigma = 0.17 \text{ ns} \quad (5.19)$$

$$\text{FWHM} = 2\sqrt{2 \ln 2} \sigma \approx 0.40 \text{ ns} \quad (5.20)$$

From these parameters, the optimal coincidence window t_{cc} is defined by selecting a symmetric temporal interval centered around zero, with width $[-2\sigma; +2\sigma]$:

$$t_{cc} = 4\sigma = 0.68 \text{ ns}. \quad (5.21)$$

Once the coincidence window has been determined, the next step is to identify the effective

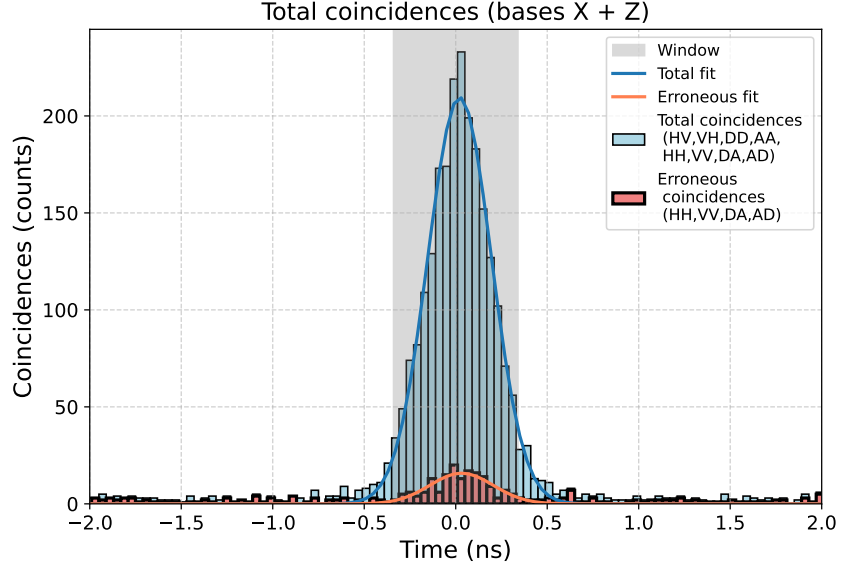


Figure 5.5: Coincidence peak histogram showing total (light blue) and erroneous (red) events for the X and Z bases. The respective Gaussian fits (blue and orange lines) and the time coincidence window (grey region) centered around $\tau = 0$ ns are reported

coincidences, which represent the data that will be subsequently used for the practical implementation of the protocol. As shown in Figure 5.5, the analysis considers the total coincidences recorded in both the X and Z measurement bases. The graph shows a prominent peak corresponding to the overall coincidences, while the underlying curve traces the temporal distribution of the erroneous coincidences.

By integrating the total counts within the selected temporal interval t_{cc} , the resulting values and their corresponding temporal rates, normalized over the total acquisition time $T_{aq} = 120$ s, can be determined. To determine the accidental counts, the coincidence window is shifted away from the main peak (specifically around $\tau \approx -2$ ns), as shown in Figure 5.6.

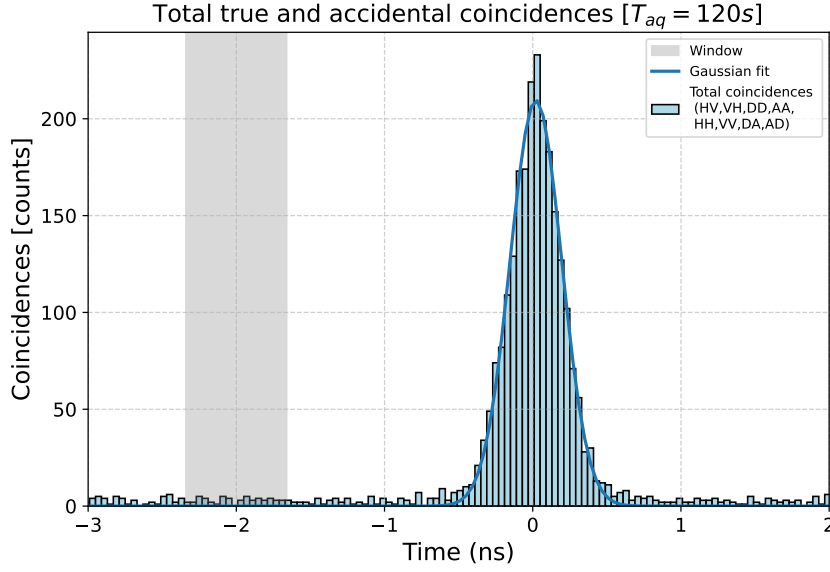


Figure 5.6: Coincidence peak histogram and accidental coincidence background estimation. The blue line represents the Gaussian fit of the true coincidence peak. The grey region highlights the off-peak coincidence window used to isolate and quantify the accidental coincidences.

The results of a 120-s long measurement are summarized in Table 5.1.

To evidence a possible dependence of the measurement basis on the performance in terms of total and error counts, an individual analysis has been carried out separately on the X and Z bases. The two measured peaks are shown in Figure 5.7, while the data obtained by integrating the counts within the coincidence window are reported in Table 5.2.

Coincidence type	Counts (N)	Rate (counts/s) (CC)
Total (tot)	2166	18.05
Erroneous (err)	161	1.34
Accidental (acc)	46	0.38

Table 5.1: Summary of Counts and Rate within the selected temporal window $t_{cc} = 0.68$ ns.

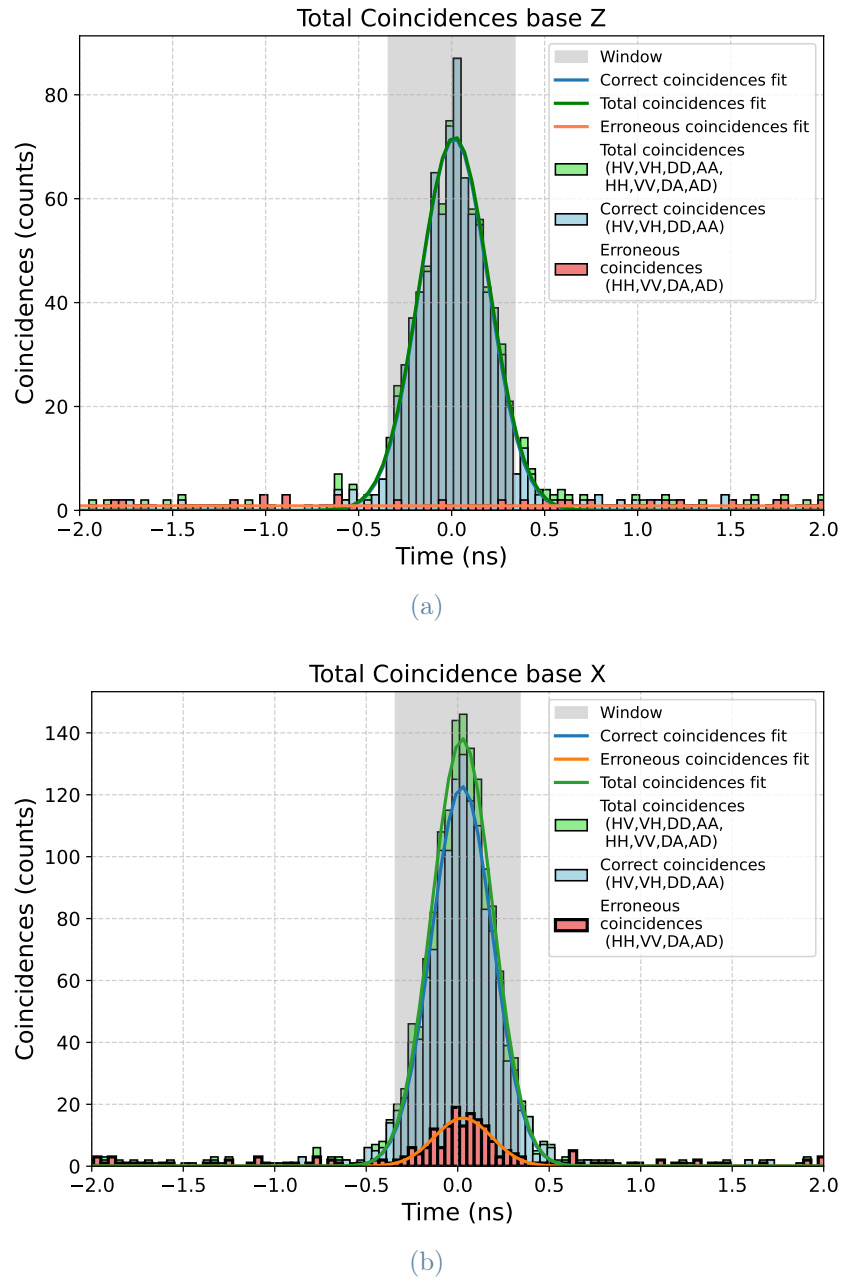


Figure 5.7: Coincidence peak recorded separately for the two measurement bases: (a) Z-basis; (b) X-bases

	Coincidence type	Counts (N)	Rate (counts/s) (CC)
BASE Z	Total (tot)	791	6.59
	Erroneous (err)	12	0.1
	Accidental (acc)	11	0.1
BASE X	Total (tot)	1226	11.45
	Erroneous (err)	149	1.24
	Accidental (acc)	13	0.1

Table 5.2: Summary of Counts and Rate for each bases within the selected temporal window $t_{cc} = 0.68$ ns.

Fig. 5.7 experimentally confirms the polarization predictions discussed in the previous Section 5.2.2. Specifically, the area under the erroneous fit curve represents the alignment errors, referred to as polarization error (e_{pol}). This error peak is significantly more pronounced in the X -basis than in the Z -basis. This discrepancy highlights that the overall polarization error is a combination of both device limitations and polarization alignment. Specifically, for the Z -basis, which was finely aligned, the observed error rate corresponds directly to the intrinsic extinction ratio of the employed optical components measured in the setup. Conversely, the higher error counts observed in the diagonal X -basis not only incorporates this hardware baseline but is predominantly driven by the misalignment of the polarization states between Alice and Bob. Consequently, it can be concluded that the excess erroneous coincidences observed in the overall data are mainly caused by this polarization misalignment affecting the diagonal measurements.

In order to give a complete overview of the coincidence combinations across Alice and Bob's detection channels, the data from the previous temporal plots have been organized into two distinct matrices shown in Figure 5.8. The first matrix shows the coincidence counts collected during the 120 s acquisition time, while the second matrix displays the corresponding measured probabilities. In these representations, each of the four main blocks corresponds to a specific basis combination (Z/Z , X/X , X/Z , and Z/X). Ideally, the correlations are expected within the **same basis blocks** (Z/Z , X/X), with the total absence of correlation between the **cross basis blocks** (X/Z , Z/X). This behavior is essential to guarantee the security of the BBM92 protocol, which strictly requires the choice of measurement bases that are mutually orthogonal. Specifically, in the case of mismatched bases (X/Z , and Z/X) the absence of correlations should ideally result in a uniform 0.25 probability across all internal cells. However, as can be observed from the matrix shown in Fig. 5.8b), this condition is not fully satisfied, resulting in a non-uniform probability distribution across these cells. This discrepancy is directly attributed to the non-optimal alignment of the X basis discussed previously. In contrast, the two same basis blocks show a correlation distribution that properly reflects the expected behavior of the Bell

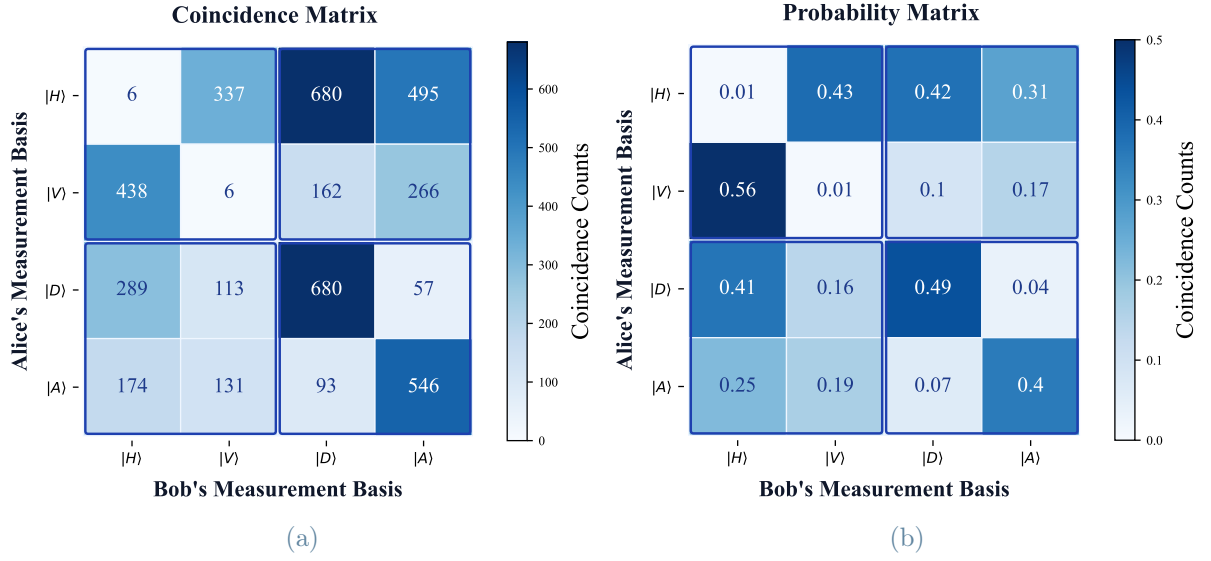


Figure 5.8: Experimental data represented in matrix form: (a) coincidence counts accumulated over an acquisition time of $T_{aq} = 120$ s; (b) normalized correlation matrix, where the conditional probabilities are evaluated independently within each of the four basis combinations (Z/Z , X/X , X/Z , and Z/X)

state generated by the source (Ψ^+): a higher number of coincidence counts is concentrated along the anti-diagonal of the Z/Z block and the main diagonal of the X/X block, while the other cells represent the erroneous coincidences. As expected from the coincidence peak analysis, the error counts within the X/X block are visibly higher than those in the Z/Z block due to polarization alignment issues.

The effect of these polarization imperfections affects the Quantum Bit Error Rate ($QBER$). In this case, the general $QBER$ formula (Eq. 2.4) has been adapted to express the $QBER$ as a function of the coincidences:

$$\begin{aligned}
 QBER &= \frac{N_{erroneous}}{N_{corrected} + N_{erroneous}} = \\
 &= \frac{N_{HH} + N_{VV} + N_{DA} + N_{AD}}{(N_{HV} + N_{VH} + N_{DD} + N_{AA}) + (N_{HH} + N_{VV} + N_{DA} + N_{AD})} = 7.5\%
 \end{aligned} \tag{5.22}$$

where N_{ij} (with $i, j \in \{H, V, D, A\}$) denotes the coincidence counts recorded for each state combination, corresponding to the data reported in Tables 5.1 and 5.2. Splitting the $QBER$ evaluation into its Z and X bases provides a clear understanding of how each basis contributes to the overall error rate:

$$QBER_Z = \frac{N_{erroneous,z}}{N_{corrected,z} + N_{erroneous,z}} = \frac{N_{HH} + N_{VV}}{N_{HH} + N_{VV} + N_{HV} + N_{VH}} = 1.58\% \tag{5.23}$$

$$QBER_X = \frac{N_{erroneous,x}}{N_{corrected,x} + N_{erroneous,x}} = \frac{N_{DA} + N_{AD}}{N_{DD} + N_{DA} + N_{AD} + N_{AA}} = 10.90 \% \quad (5.24)$$

From the reported individual evaluation on the two bases, it appears that $QBER_X$ is significantly higher than $QBER_Z$, i.e. the diagonal basis constitutes the main source of the overall system error as expected from the evaluation of the coincidence curves reported before. However, the $QBER_X$ is below the threshold of 11%, which is strictly required to generate a secure key. This specific asymmetry provides definitive evidence that the performance degradation in the diagonal basis is driven solely by current experimental limitations in polarization alignment.

As discussed in Section 3.2, when different $QBER$ performance are achieved on the two bases the evaluation of the bit error rate (E_{bit}) and the phase error rate (E_{ph}) depends on the basis selected for key generation. If the key is extracted from Z -base $E_{bit} = QBER_Z$ and $E_{ph} = QBER_X$. Conversely, the parameters are symmetrically inverted, setting $E_{bit} = QBER_X$ and $E_{ph} = QBER_Z$.

Assuming a situation where the bases are selected independently with an equal probability of $p = 1/2$ and the $f = 1.2$ the two secure key rates are evaluated. By substituting the experimental values into the general model derived in Eqs. (3.42) and (3.43), the secure key rates are respectively given by:

$$SKR_Z(t_{cc}) = 1.68 \text{ bits/s} \quad (5.25)$$

$$SKR_X(t_{cc}) = 1.51 \text{ bits/s} \quad (5.26)$$

The total Secure Key Rate (SKR_{tot}) of the system is subsequently evaluated as the sum of these two independent contributions:

$$SKR_{tot} = SKR_X + SKR_Z = 3.2 \text{ bits/s} \quad (5.27)$$

The similarity between $SKR_Z(t_{cc})$ and $SKR_X(t_{cc})$ is a direct mathematical consequence of the nearly identical value of the overall information that must be discarded for error correction and privacy amplification. In the secure key rate framework, this total reduction is represented by the combined entropy contribution:

$$\Delta = f(E_{bit}) \cdot H_2(E_{bit})(t_{cc}) + H_2(E_{ph}(t_{cc})) \quad (5.28)$$

In the case of $SKR_Z(t_{cc})$, the bit error rate is very low ($E_{bit} = QBER_Z$), meaning that almost all raw bits exchanged between Alice and Bob are correct and require minimal exchange of public information for classical reconciliation. However, the high phase error ($E_{ph} = QBER_X$) forces the system to assume a strong presence of Eve, imposing a drastic reduction of bits during

privacy amplification and resulting in a total entropy contribution of $\Delta = 0.636$. Conversely, the opposite occurs for $SKR_X(t_{cc})$, where the bit error rate is very high ($E_{\text{bit}} = QBER_X$) and the phase error ($E_{\text{ph}} = QBER_Z$) is minimal, leading to a total entropy contribution of $\Delta = 0.712$. In conclusion, the two key rates converge to very close values due to a distinct trade-off: a demand for a more intensive compression of the secure key is systematically offset by the minimization of the classical error correction overhead, and vice versa.

5.3. Simulation Framework and Comparative Analysis

This section describes the implementation and validation of a numerical simulation tool designed to model the performance of the BBM92 protocol, based on the framework described in Section 3.2. Subsequently, the tool is then utilized for a predictive analysis to evaluate the system's performance under alternative configurations, specifically focusing on the integration of SNSPD detectors.

5.3.1. Parameter Estimation and Model Validation

As a preliminary step, validating the simulation tool required calculating the input parameters directly from the experimental data. In particular, this involved determining the polarization error, the brightness (B), and the overall losses of the setup.

The intrinsic **polarization error** (e^{pol}) is calculated by removing the impact of the background noise from the measured $QBER$. This isolates the optical misalignment degree of each basis:

$$e_x^{\text{pol}} = \frac{QBER_X \cdot (\eta_r CC_x^t + CC_x^{\text{acc}}) - \frac{1}{2} CC_x^{\text{acc}}}{\eta_r CC_x^t} = 0.10 \quad (5.29)$$

$$e_z^{\text{pol}} = \frac{QBER_Z \cdot (\eta_r CC_z^t + CC_z^{\text{acc}}) - \frac{1}{2} CC_z^{\text{acc}}}{\eta_r CC_z^t} = 0.008 \quad (5.30)$$

The calculated value for e_z^{pol} closely matches the extinction ratio of the real standard components (approximately -22dB), while e_x^{pol} shows a significantly higher value due to the alignment issues related to the X bases.

The **source brightness** represents the average number of entangled photon pairs generated per unit of time, and it's expressed in counts per second prior to coupling losses. The brightness of the entanglement source cannot be measured directly, but it can be analytically calculated from the measurement results as:

$$B(t_{cc}) \approx \frac{(S_A - D_A)(S_B - D_B)}{CC_t(t_{cc})} \quad (5.31)$$

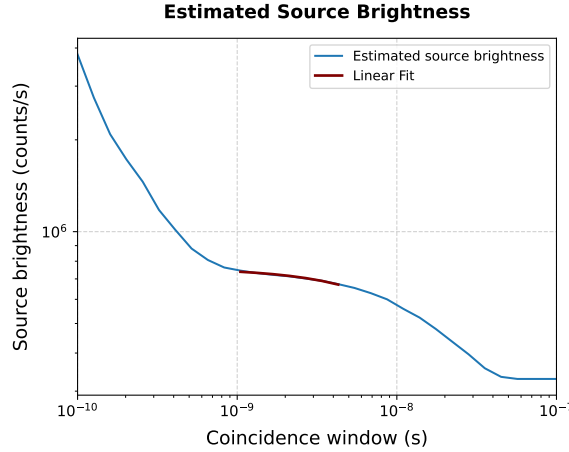


Figure 5.9: Estimated source brightness B as a function of the temporal coincidence window size t_{cc} .

where S_A and S_B are the single counts registered at Alice's and Bob's detectors, while D_A and D_B represent their respective dark counts. As defined by Eq. (5.31), the calculated brightness directly depends on the width of the temporal coincidence window t_{cc} , which determines the total number of recorded coincidence counts. To obtain the intrinsic value of B , the experimental coincidence counts were evaluated over several window sizes; for each window width, the coincidence values were extracted from the experimental data by integrating the correlation peak (as in Figure 5.5) within the corresponding window. Then, the experimental data can be processed through Eq. (5.31) to obtain the dependence of the calculated B versus the coincidence window, as illustrated in Figure 5.9. For narrow windows, the estimated brightness diverges toward higher values because the coincidence window excludes a portion of the coincidence peak, i.e. a part of the emitted entangled photon pairs. So, this loss of data results in an overestimation of the brightness. In contrast, for excessively large windows, the curve reaches a lower plateau as the coincidence counts become saturated and entirely dominated by accidental noise. Therefore, to isolate the true, physical brightness independent of these boundary effects, a linear fit (represented by the dark red line in Figure 5.9) was performed in the stable transition region around 1 ns to 2 ns, where accidental coincidences are still negligible compared to real ones. Furthermore, this result was cross-verified by applying Eq. 5.31 directly at the previously optimized coincidence window of $t_{cc} = 0.68$ ns, which was shown to maximize SKR . By substituting the single counts (cleared of dark counts) (see Tab. 5.3) and the corresponding total coincidence counts within this optimal window (reported in Tab. 5.1), the calculation yields a value in perfect agreement with the intercept of the linear fit, providing a final estimated brightness of $B \approx 7.7 \cdot 10^5$ counts/s. This convergence fully validates the consistency and accuracy of the extraction method.

Once the intrinsic value of the brightness is determined, it is possible to quantitatively estimate the total and individual losses characterizing the two optical channels of the experimental setup

Parameter	Alice (A) (counts/s)	Bob (B) (counts/s)
Raw Single Counts (S_i)	29371.65	40289.09
Dark Count Rate (D_i)	24468.93	37363.00
True Single Counts ($S_i^t = S_i - D_i$)	4902.72	2926.09

Table 5.3: Experimental single counts and dark counts for Alice’s and Bob’s detection channels, measured at $V_{\text{ex}} = 2.5$ V. Values denote the aggregate rates obtained by summing the contributions of the four individual detectors at each station.

(Alice and Bob). Starting from Eq. 3.34 and inserting the experimental values summarized in Tab. 5.3, the total efficiency of each channel, denoted as $\eta_{A,\text{TOT}}$ and $\eta_{B,\text{TOT}}$ respectively, is determined by the ratio between the single counts (cleared of dark counts) and the source brightness:

$$\eta_{A,\text{TOT}} = \frac{S_A^t}{B} \approx 0.0063, \quad \eta_{B,\text{TOT}} = \frac{S_B^t}{B} \approx 0.0038 \quad (5.32)$$

To isolate the net losses of the optical link (η_A and η_B), which include all attenuations introduced by alignment, optical components, and the fiber, while excluding the quantum efficiency of the detectors and saturation effects, the total efficiency was decomposed according to Equation 3.36:

$$\eta_{i,\text{TOT}} = \eta_i \eta_D \eta_{\text{dt},i} \quad (\text{with } i = A, B) \quad (5.33)$$

where η_D is the detection efficiency and $\eta_{\text{dt},i}$ is the efficiency due to the detector dead time at Alice (A) or Bob (B).

The nominal efficiency of the detectors at the set operating voltage ($V_{\text{ex}} = 2.5$ V) is $\eta_D = 0.11$. The dead-time efficiency, on the other hand, was evaluated starting from Eq. (3.37) and manipulated to derive the explicit formulation used for the direct calculation of $\eta_{\text{dt},i}$:

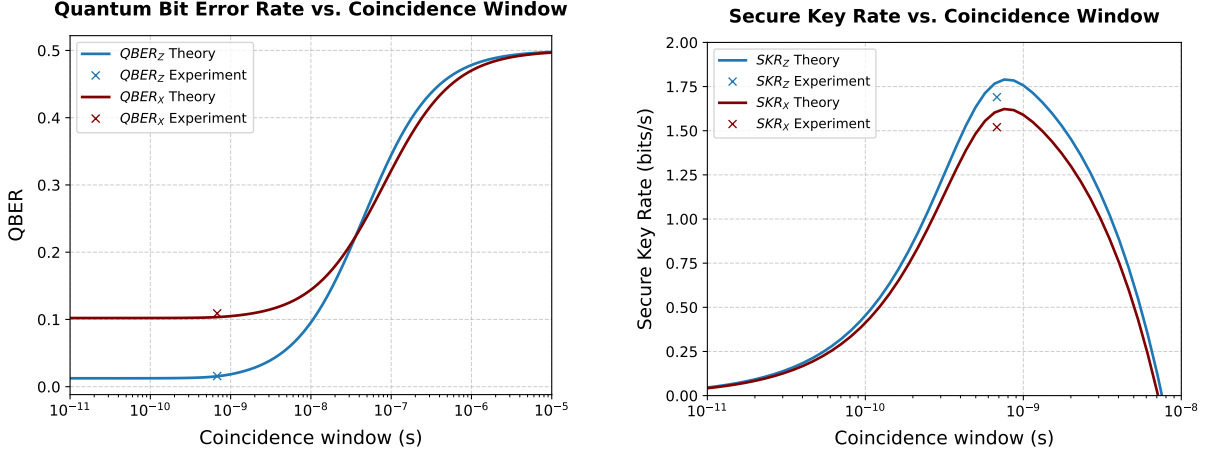
$$\eta_{\text{dt},i} = \frac{1 - \eta_{\text{dt},i} \eta_i \frac{t_d}{n_d} B \eta_D}{1 + D_i \frac{t_d}{n_d}} \quad (5.34)$$

yielding the values $\eta_{\text{dt},A} = 0.93$ and $\eta_{\text{dt},B} = 0.90$.

By removing the calculated detection-related efficiencies, the intrinsic setup losses are obtained, expressed in both linear terms and dB:

$$\eta_A = 0.06 \approx -12.07 \text{ dB} \quad (5.35)$$

$$\eta_B = 0.037 \approx -14.20 \text{ dB} \quad (5.36)$$



(a) Quantum Bit Error Rate (QBER) as a function of the coincidence window.

(b) Secure Key Rate (SKR) as a function of the coincidence window.

Figure 5.10: Experimental behavior of the system parameters as a function of the temporal coincidence window: (a) trend of the QBER, and (b) corresponding impact on the Secure Key Rate (SKR).

These values indicate that Bob's channel exhibits a marginally higher attenuation compared to Alice's, with a discrepancy of approximately 2.13 dB. This mismatch can be attributed to differing insertion losses between the two arms.

Once the physical parameters were determined, the numerical tool could be validated. By providing the extracted values of error of polarization, B , η_A , η_B , and detector specifications into the model, the system simulated the theoretical curves of the SKR . Under these experimental conditions, the simulated results demonstrated perfect consistency with the measured data points. As illustrated in Figure 5.10, the experimental data points, marked as 'X', are in good agreement with the simulated curves. The slight differences between them are entirely due to experimental errors and statistical fluctuations. Specifically, as the width of the temporal window t_{cc} increases, the QBER rises until it reaches 50%, i.e. where there is an equal probability of obtaining correct or erroneous bits and the result is completely random. This behavior occurs because, by widening the window, accidental coincidences grow linearly, and thus the background noise distributes equally between correct and incorrect coincidences. However, the QBER curves for the X and Z bases exhibit different starting points due to their respective optical misalignments. When the window is extremely narrow, the amount of accidental coincidences drops to zero, meaning that the only detected error is the intrinsic optical misalignment of the system. Consequently, the X basis shows a very high baseline QBER due to its prominent misalignment, whereas the Z basis, characterized by a misalignment close to the extinction ratio of the employed optical components, starts at a value near zero.

The behavior of the QBER directly determines the evolution of the SKR in function of the

window size. With wide temporal windows, the SKR drops to zero because the extreme number of accidental counts limits the equivalent signal-to-noise ratio of the system, not allowing any key generation process. Conversely, if the window is too small, its width becomes narrower than the jitter of the detector itself. Consequently, the coincidence peak is partially truncated, preventing the system from recording all the true coincidences. In conclusion, the numerical model demonstrates that the window used in the experiment ($t_{cc} = 0.68$ ns) represents the global optimum for the system, as this operating point sits at the peak of the SKR curve, achieving the optimal compromise between maximizing useful counts and minimizing background noise.

5.3.2. Performance Evaluation of a BBM92 System

The source brightness and link losses extracted from experimental data can be used as baseline input parameters of the numerical model described in Section 3.2 to simulate the performance of a generic BBM92 system with upgraded components. This analysis will permit to underline the ultimate capabilities of the proposed QKD system in terms of overall key rate. In particular, this section focuses on simulating the use of **Superconducting Nanowire Single-Photon Detectors (SNSPDs)** [36] as a replacement for the SPADs employed in the actual measurement setup. These devices present a higher detection efficiency and a superior temporal resolution compared to the SPADs, together with a very limited presence of dark counts (close to 1 dark count per second). To provide a realistic and high-fidelity benchmark for this predictive framework, the hardware parameters configured in the simulator reflect the nominal specifications of a state-of-the-art commercial devices [37]. The specific characteristics of this system, in contrast to the SPAD parameters, are summarized in Table 5.4.

By integrating the parameters calculated from the experimental measurements -including the source brightness (B), link losses, and the different polarization errors between the two bases ($e_Z = 0.008, e_X = 0.1$)- the optimal coincidence window can be determined also considering the new detectors. In particular, regarding the time jitter of the SNSPD detector, with a FWHM of 15 ps, and assuming that the temporal uncertainty of the detectors follows a Gaussian

Hardware Parameter	SPAD	SNSPD
Detector Efficiency (η_D)	11%	90%
Dark Count Rate (D_i)	~ 7 kcps	~ 1 cps
Dead Time (τ_{dead})	10 μs	40 ns
Timing Jitter (FWHM)	~ 200 ps	15 ps

Table 5.4: Hardware parameters comparison between the experimental baseline (SPAD) and the predictive optimization model.

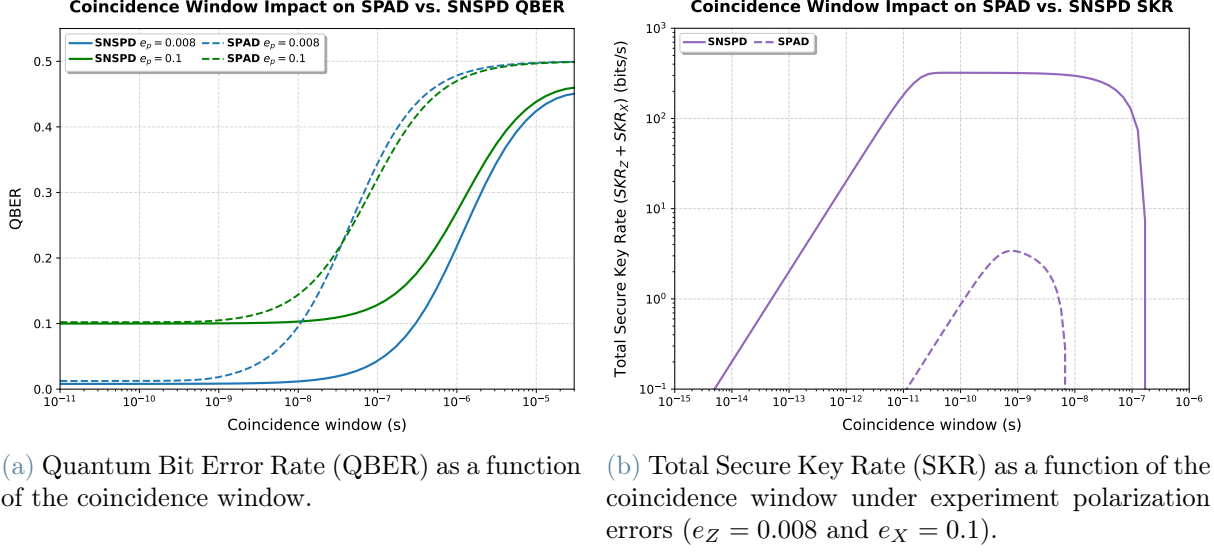


Figure 5.11: Comparative analysis of (a) QBER and (b) SKR between the experimental SPAD detectors and simulated SNSPDs.

distribution, the following relation is obtained:

$$\sigma = \frac{\text{FWHM}}{2.355} \approx 6.37 \text{ ps} \quad (5.37)$$

Consequently, a coincidence window of $\pm 2\sigma$ was selected to capture nearly all authentic coincidences:

$$t_{cc} = 25.47 \text{ ps} \quad (5.38)$$

As illustrated in Fig. 5.11, the calculated window aligns with the simulated performance. While a BBM92 system using SPAD detectors yields a peak Secure Key Rate (SKR) of only 3.2 bits/s, the upgrading to SNSPDs increases the SKR to an impressive 602.66 bits/s under identical polarization errors ($e_Z = 0.008$ and $e_X = 0.1$). Furthermore, the SNSPD SKR remains stable within a wide temporal interval, from approximately 10⁻¹¹ s (10 ps) to 10⁻⁸ s (10 ns). This behavior is in perfect agreement with the corresponding QBER evolution for SNSPD-based system, which shows the minimum achievable value within this same temporal interval. Specifically, the two bases exhibit distinct error floors—approximately 0.8% for the Z basis and 10% for the X basis—directly reflecting the different intrinsic polarization errors ($e_Z = 0.008$ and $e_X = 0.1$) integrated into the simulation parameters. On the other hand, the SKR obtained via the SPAD exhibits a significantly reduced maximum value achievable in the optimal conditions, meaning that the temporal interval suitable for selecting an optimal coincidence window is narrow. It can be observed that the QBER of SPAD remains stable only for a shorter interval compared to the SNSPD configuration, starting a rapid growth precisely at the specific window width where the SKR begins to decline. This limitation is mainly due to the large differences in timing

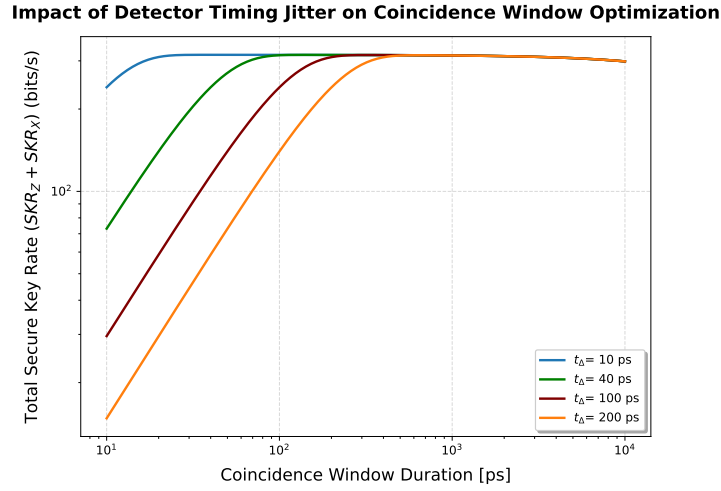


Figure 5.12: Total Secure Key Rate (SKR) versus coincidence window duration, simulated for various detector timing jitter values (t_Δ).

jitter and Dark Count Rate (DCR) between the two detectors. As demonstrated in the jitter impact analysis (Fig. 5.12), a less precise detector (with higher timing jitter) requires a wider coincidence window to collect all valid pairs of photons and reach a high SKR. However, a large window increases the possibility to capture accidental noise coincidences, which trigger a rapid increase in the QBER and a subsequent reduction of the SKR.

Another advantage of using an SNSPD is the very limited dark count rate (~ 1 cps); this permits to significantly widen the measurement window without introducing any penalty due to the presence of background noise. Unlike in case of SPADs, which are penalized by a DCR of 7 kcps, the SNSPD system is not limited by the restrictive trade-off between noise and timing uncertainty. Furthermore, SPAD performance is characterized by a long dead time (tens of microseconds), during which the detector remains blind. This causes a saturation effect at high count rates, further lowering the maximum achievable SKR. In contrast, SNSPDs feature an ultra-short dead time (of the order of few nanoseconds) due to the rapid recovery of the superconducting nanowires, allowing them to sustain a much higher photon counting rate.

By using the measured optical losses for the receiver end (corresponding to the manual polarization controllers, the passive 50/50 splitter and the PBSs), it is possible to calculate the impact of the presence of optical fiber propagation on the BBM92 performance. The result is shown in Figure 5.13 both for SPAD- and SNSPD-based receivers. In the first case, a very rapid drop of the SKR after just 5 dB propagation losses is present. Under standard telecommunication conditions, assuming a fiber attenuation coefficient of 0.2 dB/km at 1550 nm, this low tolerance limits the maximum transmission range to 25 km at most. On the other side, the SNSPD-based receiver allows correct system operations up to a loss of approximately 63 dB, corresponding to a fiber distance of 315 km. It should be noted that the BBM92 protocol can be exploited on fully

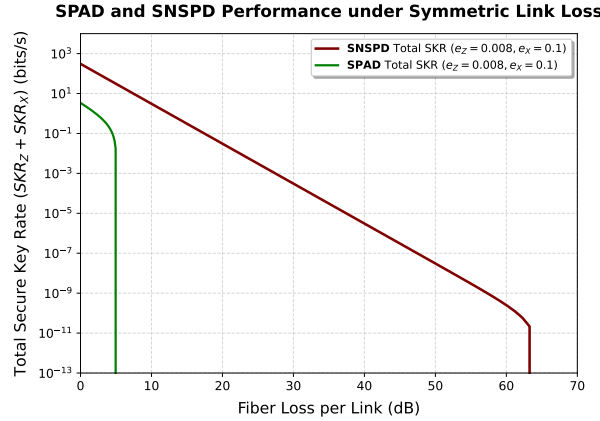


Figure 5.13: Secure Key Rate (SKR) as a function of symmetric link loss for experimental SPADs and simulated SNSPDs.

symmetric links: since the central source splits the photons into two distinct arms, the maximum distance limits should be applied independently to each individual optical link. Consequently, the total communication range separating Alice and Bob can be effectively doubled compared to a standard point-to-point configuration, allowing the system to bridge a total distance of 50 km and 630 km for SPAD- and SNSPD-based receivers, respectively. The performance difference between the two detectors configurations is driven by the degradation of the Signal-to-Noise Ratio (SNR) as fiber length increases. This behavior occurs because, as channel loss increases, the number of detected photons decreases. For the SPAD setup, its high internal noise quickly suppresses the few remaining signal photons, leading to an unsustainable error rate. In contrast, since the SNSPD features a near-zero dark count rate, there is no background noise impacting the signal, enabling the generation of a secure key rate (SKR) over significantly longer fiber distances.

To further investigate the role of detector noise, Figure 5.15 illustrates the impact of different dark count rates on the SKR performance. As the dark count rate increases from the ideal 0 cps up to 1000 cps, the maximum supported attenuation shifts progressively toward lower values. At low loss levels, the different curves overlap because the high photon flux completely dominates the noise. However, as attenuation increases, higher dark count rates dominate over the weak signal, leading to a strong reduction of the maximum tolerated distance.

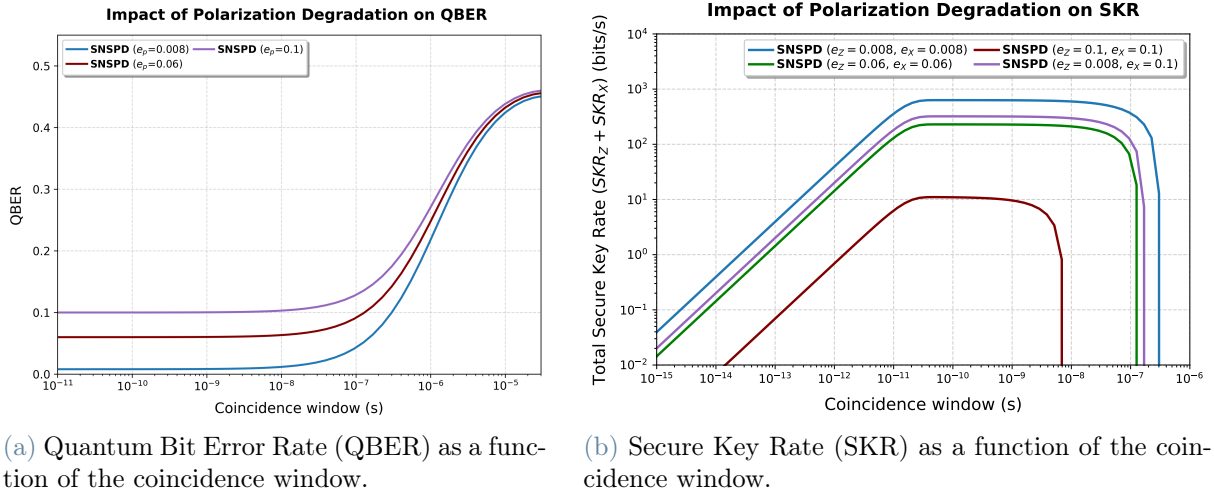


Figure 5.14: Performance optimization as a function of the coincidence window under different polarization error scenarios: (a) QBER evolution and (b) corresponding SKR behavior.

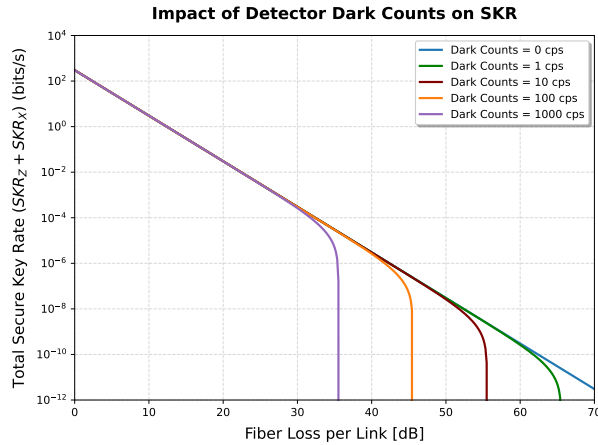


Figure 5.15: Total Secure Key Rate (SKR) as a function of fiber loss per link, simulated for dark count rates ranging from 0 cps to 1000 cps

By varying the polarization error in the system, significant differences in the overall performance can be observed. Assuming operation under an extremely reduced misalignment conditions, i.e. assuming an optimal alignment of the measurement bases with $e_Z = 0.008$ and $e_X = 0.008$, a maximum SKR of 602.67 bit/s can be achieved. Conversely, by introducing more misalignment in the receiver end, the SKR experiences a drastic drop. This behavior is directly correlated with the dependence of the QBER on the polarization error, reported in Figure 5.14a.

The SKR performance is similar in the case of symmetric error configuration with $e_Z = e_X = 0.06$ (green curve) and highly asymmetric one with $e_Z = 0.008, e_X = 0.1$ (purple curve), as shown in

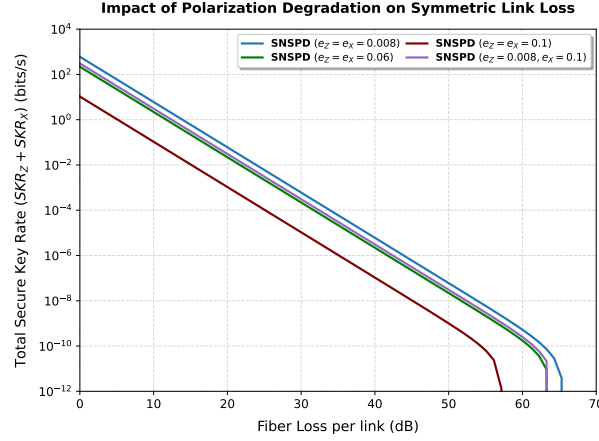


Figure 5.16: SKR as a function of symmetric link loss under different polarization error scenarios.

Fig.5.14b. This proposed asymmetric scenario closely reflects the employed experimental setup, where the alignment of the X -basis proved significantly more challenging than the Z -basis. The overlap of the performance for the two different configurations demonstrates that the system seems robust even against a severe single-basis misalignment.

The total achievable SKR as a function of symmetric link losses, under different polarization errors, is shown in Figure 5.16. Under optimal alignment ($e_Z = e_X = 0.008$, blue curve), the system tolerates up to 65 dB of loss per link, corresponding to a maximum transmission distance of 325 km per link with standard fiber attenuation coefficient of 0.2 dB/km. Conversely, a severe misalignment ($e_Z = e_X = 0.1$, red curve) limits the channel tolerance to ~ 57 dB per link, reducing the maximum reach to 285 km per link and causing a net loss of 40 km of propagation length on each link due to polarization drift. Nevertheless, it is crucial to highlight that even under this worst-case scenario (10% polarization error), the SNSPD architecture sustains secure communication over an impressive distance of nearly 285 km per link. This outstanding range—vastly outperforming the 25 km limit previously observed for SPAD detectors under standard conditions—underlines the technological advantage of SNSPDs. Thanks to a near-zero dark count rate, the system maintains a high SNR over long distances, preventing the SKR from collapsing prematurely even under severe polarization misalignments.

The study on the impact of the source brightness B on the SKR performance has been carried out, maintaining a realistic polarization error equal to 0.008 and by considering the receiver optical losses reported before in Eq. (5.35) and (5.36) in a SNSPD environment. The results are shown in Fig. 5.17. As expected, the SKR initially increases with brightness B . For very low B values, the SKR is close to zero because the source produces a low number of photon pairs. In this regime, the few generated photons are easily hidden by Dark Counts, causing the background noise to dominate over the actual signal.

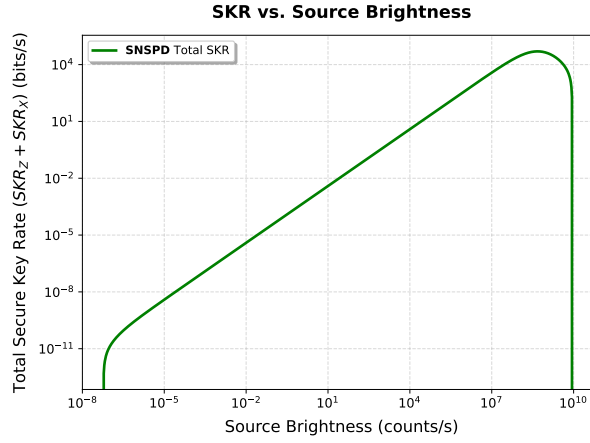


Figure 5.17: Total Secure Key Rate (SKR) as a function of source brightness (B).

As B increases, the SKR grows and improves until it reaches a peak. Beyond this optimal point, however, there is a rapid decrease of the SKR. This drop-off region corresponds to the multi-pair generation regime. In this regime, the measured coincidences could also be generated by photon pairs which are not actually entangled. Therefore, accidental coincidences increase and, consequently, the SKR drops to zero.

In conclusion, this analysis validates the superior performance achievable by state-of-the-art SNSPDs. By simultaneously mitigating time jitter, dead-time saturation, and dark count rates, the simulated superconducting architecture extends the tolerable link loss up to 63 dB, corresponding to an impressive total communication range of 630 km.

6 | Conclusions and Future Developments

This thesis presented the experimental implementation and characterization of an entanglement-based Quantum Key Distribution system exploiting the BBM92 protocol, from the optical characterization of the photon-pair source to the numerical simulation of a generic system using next-generation detector architectures.

The first experimental step focused on the rigorous characterization of the Spontaneous Parametric Down-Conversion (SPDC) source generating the entangled photon pairs. The optical power emitted on each output channel was measured, yielding equivalent single-photon powers of $P_A = 1.07$ fW and $P_B = 0.89$ fW at the signal and idler wavelengths, respectively. Spectral characterization precisely located the two emission peaks at $\lambda_A \approx 1541.7$ nm and $\lambda_B \approx 1558.8$ nm, both within the C-band of optical telecommunications, obtained thanks a 775 nm pump, in full agreement with energy and momentum conservation laws to be fulfilled in the SPDC process. After the spectral characterization, the identification of the coincidence peak and the determination of the optimal integration window has been carried out. In particular, the second-order correlation function $g^{(2)}(\tau)$ was first reconstructed via a wide-range temporal scan. Then, a Gaussian fit was then applied to this profile to determine the exact delay, which shows the fiber path mismatch between the two implemented links. The optimal integration window was then analytically derived from the fitted peak variance. Finally, this optimized window was centered onto the true coincidence peak to selectively isolate and process correlated events.

After a full characterization of the entanglement source, the BBM92 protocol has been experimentally implemented in the laboratory. Its practical realization employed eight SPAD detectors - four assigned to Alice and four to Bob - in a passive configuration, allowing the measurement of photon polarization in both the Z and X bases. Polarization alignment was assisted by a software tool developed on purpose to achieve real-time monitoring of count rates, $g^{(2)}(\tau)$ functions, and coincidence matrix, allowing to dynamically optimize both measurement bases by manually polarization controllers. Using a 120-second acquisition time duration, an optimal coincidence window of $t_{cc} = 0.68$ ns (FWHM ≈ 0.40 ns) was calculated. The measured QBER was $\text{QBER}_Z = 1.58\%$ in the rectilinear basis and $\text{QBER}_X = 10.90\%$ in the diagonal basis, yielding a total Secure Key Rate of $\text{SKR}_{\text{tot}} = 3.2$ bits/s. The asymmetry between the two bases

is consistent with the measured polarization alignments. While the QBER in the Z -basis is independent of the phase ϕ accumulated along the fiber, the X -basis depends directly on it. This makes the X -basis highly sensitive to thermal and mechanical fluctuations, making it the dominant source of error in the system.

To extend the performance evaluation of a BBM92 protocol beyond the current experimental results, a numerical simulator was developed and calibrated on the laboratory data. Considering the measured QBER and coincidence rate curves as a function of the coincidence window, the fundamental physical parameters of the setup - namely the source brightness ($B \approx 7.7 \cdot 10^5$ counts/s), the intrinsic polarization errors of each basis ($e_Z = 0.008$, $e_X = 0.10$), and the link losses for Alice and Bob ($\eta_A \approx -12.07$ dB, $\eta_B \approx -14.20$ dB) - has been obtained. These parameters were then used as input parameters for to numerically simulate the operations of an entanglement-based QKD system with polarization encoding. Specifically, the simulation permitted to evaluate the impact of state-of-the-art Superconducting Nanowire Single-Photon Detectors (SNSPDs) on system performance. Their principal parameters ($\eta_D = 90\%$, DCR ~ 1 cps, dead time 40 ns, jitter FWHM = 15 ps), yields an extreme improvement of the entanglement-based transmission. The better timing resolution of the SNSPDs allows to reduce the optimal coincidence window to $t_{cc} = 25.47$ ps, suppressing accidental coincidences by orders of magnitude, increasing the maximum achievable Secure Key Rate (SKR) from the measured 3.2 bit/s to up to 602.66 bit/s. Furthermore, the employment of the SNSPD technology permits to reduce the QBER at its minimum value, related to the optical alignment of the receiver end. In terms of transmission distance, while a standard SPAD-based receiver can tolerate at most 5 dB of additional fiber loss per link (limiting the propagation distance to 25 km with 0.2 dB/km), the SNSPD technology sustains a positive SKR up to 63 dB of link losses. corresponding to 315 km per link in a symmetrical configuration, enabling a total Alice-Bob distance of 630 km. If optimal basis alignment is achieved ($e_Z = e_X = 0.008$), the system's tolerance scales up to 65 dB of loss per link, pushing the maximum transmission range to 325 km, enabling a total communication distance of 650 km. Concurrently, the near-zero background noise maintains a high Signal-to-Noise Ratio (SNR) over long distances, ensuring that even under a severe 10% polarization misalignment scenario, the SNSPD system prevents a premature SKR collapse and preserves a secure communication range of 285 km per link (570 km total).

To fully realize this potential in real-world environments, future implementations must address the phase instability that currently limits the X -basis performance. An active phase-stabilization mechanism utilizing a mechanical fiber squeezer could be integrated directly into the optical path. By applying a dynamically controlled transverse pressure to the polarization-maintaining (PM) fiber, this device can modify the local birefringence to actively reset the phase accumulated ϕ along the link to 0, thereby restoring and preserving the ideal $|\Psi^+\rangle$ Bell state against thermal and mechanical fluctuations. With phase stability secured, the exploitation of an SNSPD-based detection platform could be the most impactful advancement toward a field-deployable system,

fully unlocking the predicted maximum 650 km communication range. Ultimately, scaling the system with these high-efficiency detectors overcomes the physical constraints of traditional point-to-point QKD. By distributing high-fidelity entanglement from a central source to two remote parties, this architecture establishes a robust, highly resilient, and scalable framework for long-range quantum networks, paving the way for future inter-city nodes and the deployment of a secure global quantum internet.

Bibliography

- [1] Albert Einstein and Max Born. *The Born–Einstein Letters: Correspondence between Albert Einstein and Max Born, 1916–1955*. Macmillan, 1971.
- [2] A. Einstein, B. Podolsky, and N. Rosen. Can quantum-mechanical description of physical reality be considered complete? *Phys. Rev.*, 47:777–780, May 1935.
- [3] P. A. M. Dirac. A new notation for quantum mechanics. *Mathematical Proceedings of the Cambridge Philosophical Society*, 35(3):416–418, 1939.
- [4] Arjun Berera and Luigi Debbio. *Quantum Mechanics*. Cambridge University Press, 2021.
- [5] Sina Jafarpour. Introduction to the world of quantum computers. In Yiyu Yao, Zhongzhi Shi, Yingxu Wang, and Witold Kinsner, editors, *Proceedings of the Firth IEEE International Conference on Cognitive Informatics, ICCI 2006, July 17-19, Beijing, China*, pages 760–764. IEEE, 2006.
- [6] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, 2010.
- [7] J. S. Bell. On the einstein podolsky rosen paradox. *Physics Physique Fizika*, 1:195–200, Nov 1964.
- [8] Alain Aspect, Jean Dalibard, and Gérard Roger. Experimental test of bell’s inequalities using time-varying analyzers. *Phys. Rev. Lett.*, 49:1804–1807, Dec 1982.
- [9] Alain Aspect, Philippe Grangier, and Gérard Roger. Experimental realization of einstein-podolsky-rosen-bohm gedankenexperiment: A new violation of bell’s inequalities. *Phys. Rev. Lett.*, 49:91–94, Jul 1982.
- [10] B.S. Cirel’son (Boris S. Tsirelson). Quantum generalizations of bell’s inequality. *Letters in Mathematical Physics*, 1980.
- [11] Alexander N. Craddock, Anne Lazenby, Gabriel Bello Portmann, Rourke Sekelsky, Mael Flament, and Mehdi Namazi. Automated distribution of polarization-entangled photons using deployed new york city fibers. *PRX Quantum*, 5:030330, Aug 2024.
- [12] Matheus Sena, Mael Flament, Shane Andrews, Ioannis Caltzidis, Niccolò Bigagli, Thomas Rieser, Gabriel Bello Portmann, Rourke Sekelsky, Ralf-Peter Braun, Alexander N. Crad-

- dock, Maximilian Schulz, Klaus D. Jöns, Michaela Ritter, Marc Geitz, Oliver Holschke, and Mehdi Namazi. High-fidelity quantum entanglement distribution in metropolitan fiber networks with co-propagating classical traffic. *J. Opt. Commun. Netw.*, 17(12):1072–1081, Dec 2025.
- [13] Huang, Han Jinqun, Li Hui, Li Jiaao, Peng Jiahao, Yu Tao, Wu Wanrong, Sun Chunqing, Liu Shihai, and Bo. Entanglement-based quantum key distribution via 343 km fiber link with multiple wavelength polarization state compensation. *Science China Physics, Mechanics & Astronomy*, 69(2):220315, 2026.
- [14] Sebastian Philipp Neumann, Alexander Buchner, Lukas Bulla, Martin Bohmann, and Rupert Ursin. Continuous entanglement distribution over a transnational 248 km fiber link. *Nature Communications*, 13(1):6134, 2022.
- [15] Sören Wengerowsky et al. Entanglement distribution over a 96-km-long submarine optical fiber. *Proceedings of the National Academy of Sciences*, 116(14):6684–6688, 2019.
- [16] Nicolas Gisin, Grégoire Ribordy, Wolfgang Tittel, and Hugo Zbinden. Quantum cryptography. *Rev. Mod. Phys.*, 74:145–195, Mar 2002.
- [17] C. E. Shannon. Communication theory of secrecy systems. *Bell System Technical Journal*, 28(4):656–715, 1949.
- [18] National Institute of Standards and Technology. Advanced encryption standard (aes). Technical Report NIST FIPS 197 (Update 1), U.S. Department of Commerce, May 2023.
- [19] R. L. Rivest, A. Shamir, and L. Adleman. A method for obtaining digital signatures and public-key cryptosystems. *Commun. ACM*, 21(2):120–126, February 1978.
- [20] P.W. Shor. Algorithms for quantum computation: discrete logarithms and factoring. In *Proceedings 35th Annual Symposium on Foundations of Computer Science*, pages 124–134, 1994.
- [21] Feihu Xu, Xiongfeng Ma, Qiang Zhang, Hoi-Kwong Lo, and Jian-Wei Pan. Secure quantum key distribution with realistic devices. *Rev. Mod. Phys.*, 92:025002, May 2020.
- [22] Valerio Scarani, Helle Bechmann-Pasquinucci, Nicolas J. Cerf, Miloslav Dušek, Norbert Lütkenhaus, and Momtchil Peev. The security of practical quantum key distribution. *Rev. Mod. Phys.*, 81:1301–1350, Sep 2009.
- [23] Charles H. Bennett and Gilles Brassard. Quantum cryptography: Public key distribution and coin tossing. *Theoretical Computer Science*, 560:7–11, 2014. Theoretical Aspects of Quantum Cryptography – celebrating 30 years of BB84.
- [24] Artur K. Ekert. Quantum cryptography based on bell’s theorem. *Phys. Rev. Lett.*, 67:661–663, Aug 1991.

- [25] Charles H. Bennett, Gilles Brassard, and N. David Mermin. Quantum cryptography without bell’s theorem. *Phys. Rev. Lett.*, 68:557–559, Feb 1992.
- [26] Daniel Huber, Marcus Reindl, Johannes Aberl, Armando Rastelli, and Rinaldo Trotta. Semiconductor quantum dots as an ideal source of polarization-entangled photon pairs on-demand: a review. *Journal of Optics*, 20(7):073002, jun 2018.
- [27] Jiho Park, Jinhyuk Bae, Heonoh Kim, and Han Seb Moon. Direct generation of polarization-entangled photons from warm atomic ensemble. *Applied Physics Letters*, 119(7):074001, 08 2021.
- [28] M. Fox. *Quantum Optics: An Introduction*. Oxford Master Series in Physics. OUP Oxford, 2006.
- [29] Micro Photon Devices Srl. *PDM-IR: InGaAs/InP Single-Photon Avalanche Diode Counting Module*, February 2025. Datasheet, Document version v2.1.
- [30] Coherent Corp. *WaveShaper® A/B-Series User Manual*, 2023.
- [31] Swabian Instruments. *Time Tagger User Manual Release 2.21.2.0*, 2026.
- [32] Shi-Chang Zhuang, Bo Li, Ming-Yang Zheng, Yi-Xi Zeng, Hui-Nan Wu, Guang-Bing Li, Quan Yao, Xiu-Ping Xie, Yu-Huai Li, Hao Qin, Li-Xing You, Feihu Xu, Juan Yin, Yuan Cao, Qiang Zhang, Cheng-Zhi Peng, and Jian-Wei Pan. Ultrabright entanglement based quantum key distribution over a 404 km optical fiber. *Phys. Rev. Lett.*, 134:230801, Jun 2025.
- [33] Michelle Chalupnik, Brian Doolittle, Suparna Seshadri, Eric G. Brown, Keith Kenemer, Daniel Winton, Daniel Sanchez-Rosales, Matthew Skrzypczyk, Cara Alexander, Eric Ostby, and Michael Cubeddu. Realistic quantum network simulation for experimental bbm92 key distribution. *IEEE Transactions on Quantum Engineering*, 7:1–10, 2026.
- [34] Xiongfeng Ma, Chi-Hang Fred Fung, and Hoi-Kwong Lo. Quantum key distribution with entangled photon sources. *Phys. Rev. A*, 76:012307, Jul 2007.
- [35] Sebastian Philipp Neumann, Thomas Scheidl, Mirela Selimovic, Matej Pivoluska, Bo Liu, Martin Bohmann, and Rupert Ursin. Model for optimizing quantum key distribution with continuous-wave pumped entangled-photon sources. *Phys. Rev. A*, 104:022406, Aug 2021.
- [36] Lixing You. Superconducting nanowire single-photon detectors for quantum information. *Nanophotonics*, 9(9):2673–2692, 2020.
- [37] Single Quantum. *Excellence Line: Customized single photon detection systems*, 2026. Accessed: 2026-05-14.

List of Figures

1.1	Representation of measure	6
1.2	Bloch sphere representation of a Qubit adapted from [6].	8
1.3	Quantum correlation coefficient as a function of the relative angle θ . The dots indicate the four geometric configurations used for the maximum violation of the CHSH inequality.	15
1.4	CHSH parameter S as a function of Bob's relative angle θ (with fixed $\alpha = 0^\circ, \alpha' = 45^\circ$). The quantum curve violates the classical local-realistic bounds (dashed lines, ± 2), reaching the Tsirelson limit ($2\sqrt{2}$) at $\theta = 22.5^\circ$	16
2.1	Sketch of QKD setups: (a) Prepare and Measure; (b) Entangled-based protocol. .	27
2.2	Principles of type II phase matching: (a) energy conservation; (b) momentum conservation	32
2.3	Two emitted cones from the non-linear crystal	33
2.4	Time diagram of SPAD current and voltage during photon detection.	35
2.5	(a) PDE VS Excess Bias; (b) Dark counts vs Hold Off (Ref.[29])	36
3.1	Schematic representation of the Sagnac interferometer configuration under different pump polarization states: (a) vertical (s -polarized) pump, (b) horizontal (p -polarized) pump, and (c) diagonal pump creating polarization-entangled photon pairs.	38
3.2	Setups used to measure power at output A (a) and (b) B	39
3.3	SPAD Dark Count characterization	39
3.4	Count Rates in function of time.	40
3.5	Output power SPDC source	42
3.6	Sketch of experimental setup to determine spectrum of the source (a) normal setup configuration for output A; (b) inverted setup configuration for output B .	42
3.7	Experimental second-order correlation measurements for Output A using a filter bandwidth of 0.1 THz. The coincidence peak is plotted as a function of (a) Frequency and (b) Wavelength	44

3.8	Experimental second-order correlation measurements using a filter bandwidth of 50 GHz for Output A, as a function of (a) Frequency and (b) Wavelength. The experimental data (points) are fitted with a Gaussian model (dashed line) to determine the central emission parameters.	45
3.9	Experimental second-order correlation measurements for Output B using a filter bandwidth of 100 GHz. The coincidence peak is plotted as a function of (a) Frequency and (b) Wavelength	45
3.10	Experimental second-order correlation measurements using a filter bandwidth of 50 GHz for Output B, as a function of (a) Frequency and (b) Wavelength. The experimental data (points) are fitted with a Gaussian model (dashed line) to determine the central emission parameters.	46
3.11	Schematic representation of the coincidence histogram construction by the Time Tagger [31].	47
3.12	Second-order correlation function $g^{(2)}(\tau)$ as a function of the time delay τ , measured over a 1 ms wide time range $[-500 \mu\text{s}$ to $500 \mu\text{s}]$ to locate the coincidence peak. Inset: A zoom of the peak region revealing its peak position shifted to $\tau \approx -3200$ ps	49
3.13	Coincidence peak centered at zero delay after delay compensation	50
3.14	Coincidence histogram acquired over a 60 s integration time, illustrating the optimal integration window. The time bins are colored to differentiate signal from noise	50
3.15	Geometrical interpretation of the coincidence counts per time unit for different relative measurement times, adapted from [35].	52
4.1	Block diagram of the developed software architecture.	59
4.2	Sequential flowchart of the macro-functional logical phases managed by the software application.	61
4.3	TAB 1: SPAD Parameters Configuration	64
4.4	Tab 2: Dark Counts and Background Calibration	65
4.5	Tab 3: Live Counts, Coincidences, and Data Export	66
4.6	Dialog Tab: (a) True coincidence; (b) Matrix.	67
5.1	Experimental setup for the BBM92 protocol: (a) photo of actual set-up; (b) Sketch	70
5.2	Sketch of the experimental setup for the BBM92 protocol with the squeezer source.	77
5.3	Block diagram of the BBM92 data analysis pipeline.	77
5.4	Gaussian fit of the overall second-order correlation function $g^{(2)}(\tau)$ (green curve) along with the identification of the partial contributions (blue and orange curves). The gray area defines the optimal coincidence window.	78

5.5	Coincidence peak histogram showing total (light blue) and erroneous (red) events for the X and Z bases. The respective Gaussian fits (blue and orange lines) and the time coincidence window (grey region) centered around $\tau = 0$ ns are reported	79
5.6	Coincidence peak histogram and accidental coincidence background estimation. The blue line represents the Gaussian fit of the true coincidence peak. The grey region highlights the off-peak coincidence window used to isolate and quantify the accidental coincidences.	80
5.7	Coincidence peak recorded separately for the two measurement bases: (a) Z -basis; (b) X -bases	81
5.8	Experimental data represented in matrix form: (a) coincidence counts accumulated over an acquisition time of $T_{aq} = 120$ s; (b) normalized correlation matrix, where the conditional probabilities are evaluated independently within each of the four basis combinations (Z/Z , X/X , X/Z , and Z/X)	83
5.9	Estimated source brightness B as a function of the temporal coincidence window size t_{cc} .	86
5.10	Experimental behavior of the system parameters as a function of the temporal coincidence window: (a) trend of the QBER, and (b) corresponding impact on the Secure Key Rate (SKR).	88
5.11	Comparative analysis of (a) QBER and (b) SKR between the experimental SPAD detectors and simulated SNSPDs.	90
5.12	Total Secure Key Rate (SKR) versus coincidence window duration, simulated for various detector timing jitter values (t_{Δ}).	91
5.13	Secure Key Rate (SKR) as a function of symmetric link loss for experimental SPADs and simulated SNSPDs.	92
5.14	Performance optimization as a function of the coincidence window under different polarization error scenarios: (a) QBER evolution and (b) corresponding SKR behavior.	93
5.15	Total Secure Key Rate (SKR) as a function of fiber loss per link, simulated for dark count rates ranging from 0 cps to 1000 cps	93
5.16	SKR as a function of symmetric link loss under different polarization error scenarios.	94
5.17	Total Secure Key Rate (SKR) as a function of source brightness (B).	95

List of Tables

1.1	Quantum correlation and anti-correlation behavior for the four Bell states in the Z and X bases.	21
2.1	Example of the BBM92 QKD protocol execution using the anti-correlated Bell singlet state $ \psi^-\rangle$ and Z/X bases. The table illustrates the measurements, the sifting process, and how Bob inverts his anti-correlated bits to match Alice's key.	31
3.1	Summary of raw, background, and net count rates for the configurations showed in Figure (3.2)	40
5.1	Summary of Counts and Rate within the selected temporal window $t_{cc} = 0.68$ ns.	80
5.2	Summary of Counts and Rate for each bases within the selected temporal window $t_{cc} = 0.68$ ns.	82
5.3	Experimental single counts and dark counts for Alice's and Bob's detection channels, measured at $V_{ex} = 2.5$ V. Values denote the aggregate rates obtained by summing the contributions of the four individual detectors at each station.	87
5.4	Hardware parameters comparison between the experimental baseline (SPAD) and the predictive optimization model.	89

Acknowledgements

Sono le 2 del 15 luglio e fra 7 ore devo mandare la tesi in stampa (sennò mia madre chi se la sente :)). L'idea era quella di scrivere dei bei ringraziamenti corretti grammaticalmente e commuoventi, ma...misà che non ho tempo quindi vi dovrete accontentare di ringraziamenti sgrammaticati, ma sinceri.

Non credo di riuscire ancora a realizzare che sono passati 8 anni dal momento in cui ho messo piede per la prima volta in un'aula del Poli. Ripensando a quella diciannovenne che ha lasciato il suo paesello per trasferirsi in una città di cui conosceva solo il nome, con una voglia assurda di ottenere quella laurea e ignara di tutte le difficoltà che avrebbe incontrato, provo ammirazione. Ora le vorrei dire che ce l'ha fatta, e che nel bene e nel male è riuscita con la sua determinazione ad ottenere ciò che voleva. Forse questa era la mia paura più grande, quella di dover tornare a testa bassa nel mio paese per aver fallito e non avercela fatta. Invece, tra una settimana discuterò la mia tesi. Assurdo.

Durante questi anni ho conosciuto tante persone, chi più importanti, chi meno, ma credo che ognuno di loro mi abbia dato la possibilità di scoprire qualcosa di me, negativa o positiva che sia. Le vorrei ringraziare tutte, sia quelle di passaggio, sia quelle che sono rimaste. Grazie.

Un ringraziamento speciale va al Prof Gatto. Ahimè, il fatidico 29 Febbraio 2027 è arrivato. Sono stati mesi intensi, mesi in cui ho messo in discussione ogni cosa di me. Lei è stato sempre presente, dicendomi sempre le parole giuste al momento giusto. Ha fatto di tutto per provare a farmi credere in me stessa e nelle mie potenzialità. Grazie per avere speso il suo tempo a farmi quel bellissimo discorso sui divanetti del secondo piano, l'ho impresso nella mente e pian piano sto provando ad iniziare a crederci. Grazie per avermi dato la possibilità di avere un mio banco ottico e la fiducia nell'affidarmi un bel po' di strumentazione costosa. Insomma, grazie per avermi supportata, seguita, aiutata tutto questo tempo.

Grazie al Dott. Gagliano che si è ritrovato a gestire una tesista forse un po' fuori dagli schemi. Alessandro, grazie per avermi sopportato per tutti questi nove mesi, in tutti i miei momenti di sconforto. Ad ogni mia fantastica domanda "E mo che devo fare?", sei riuscito a mantenere la calma e non insultarmi come avresti in realtà voluto, lo so, grazie. Ricordo tutti i tuoi discorsi motivazionali in lab e in ufficio, e anche tutti quelli che hai iniziato e poi guardandomi con rassegnazione hai interrotto esordendo con "E' inutile che inizio tanto non mi credi..". Grazie per averci provato! Grazie per aver passato le tue giornate con me in lab trasmettendomi, la

tua conoscenza e la tua passione. Dai...dal 22 luglio sarai libero finalmente, non avrai più una presenza costante che disturba continuamente il tuo lavoro (e la visione delle partire di curling).

Grazie al Policom dove ho passato intensamente 9 mesi, in cui mi sono disperata tanto, ma dai a tratti qualche piccola soddisfazione la ricordo. Tutti quei mesi passati al buio, su uno sgabello a muovere le lamine di quei benedetti Polarization controller alla ricerca di un'allineamento perfetto che sembrava non arrivasse mai. Vorrei ringraziare la fantastica sorgente entanglement che sembrava mi prendesse in giro, anzi no, mi prendeva proprio in giro, ne sono sicura. E poi i fantastici SPADs, senza di voi sicuramente la mia vita sarebbe stata più semplice. L'unica mia certezza eri tu, Time Tagger sempre lì a costruirmi i fantastici picchi di coincidenze. Mi mancherà il mio banco ottico e credo che un po' mancherò anche a lui.

Vorrei ringraziare i ragazzi del Policom. Grazie Andrea per essere stato sempre lì pronto a cercare di distruggermi il setup, sei finalmente libero di prendere tutti i Polarization beam splitters che vuoi. Grazie a Jacopo che puntalmente non si accorgeva della mia presenza, in quanto sempre al buio, e accendeva la luce rovindandomi le misure. Grazie a Marco per avermi sempre smontato l'entusiasmo ogni volta che avevo un minimo di risultati, trovando sempre un qualcosa che non andava. Grazie a Maddalena, una ventata di aria fresca, sempre pronta a cercarmi un fidanzato. Ovvimente scherzo, siete stati preziosi in questo periodo, sono sicura che un pochino sentirete la mancanza della mia presenza costante.

Grazie Liv, Stefano, Marco (Quantum office + infiltrati) siete stati il mio supporto continuo, siete sopravvissuti alle mie continue perplessità di vita e di tesi e accenni di pianti. Mi sentivo come a casa a stare lì con voi. E poi i Quantum meeting, come farò senza quei momenti di spessore in cui ho scoperto come si fa una palla di fuoco in laboratorio :), mi mancheranno.

Mamma e papà, cosa devo dirvi... penso che sappiate già tutto. Grazie per avermi sempre spinto a non mollare, ad andare avanti e non abbattermi nei momenti in cui forse da un genitore medio mi sarei aspettata solo un "cambia, fai altro, forse non è la tua strada", invece siete sempre rimasti lì, al mio fianco. Grazie per averci creduto più voi rispetto a me. Voi come nessun altro sapete quanto tenga a questa laurea, a questa tesi, a questo traguardo. Voi solo sapete quante lacrime ho versato, quante incazzature e quante delusioni ho vissuto in tutti questi anni. Grazie a voi che avete avuto la forza di lasciare andare una ragazzina di 19 anni in una grande città come Milano, nonostante le perplessità. So quanto vi sia costato e so anche quanto avete sofferto ogni volta che mi disperavo in preda a crisi di pianto isteriche al telefono, oppure ogni volta che mi lasciavate salire su quell'autobus alle 4.45 di mattina. Non smetterò mai di ringraziarvi per avermi permesso di arrivare fino a qui. Non immaginate neanche quanto vi ammiro.

Cristian, so che, nonostante non me lo dici mai, un pochino ti sono mancata per tutti questi anni. Anche perché...vuoi mettere quanto ti annoi senza una sorella che viene lì continuamente a disturbarti e fare dispetti. Grazie per essere stato sempre al mio fianco, anche in silenzio. È impossibile dimenticare la sensazione di quegli abbracci ad ogni mio ritorno in patria. Ti voglio

bene!

Un grazie speciale a nonna Maria e nonno Peppe per aver sempre fatto il tifo per me. Grazie Nonna per avermi sempre coccolato con l'amatriciana, gli arancini e i pranzi infiniti ogni volta che tornavo giù. Nonno, mi spiace che il tuo sogno di vedermi maestra o infermiera è stato fatto in mille pezzi, però ti assicuro che presto alla tua domanda: "ma con questa cosa che stai studiando tu lo trovi lavoro?" ti risponderò di sì e spero tu sia lo stesso fiero di me.

Vorrei ringraziare anche i miei nonni che non ci sono più, nonno Nicolino e nonna Nina. Pagherei ora per avervi qui con me oggi, anche solo per un breve abbraccio. So che sareste contenti, forse non tanto per la laurea, ma per vedermi felice. Mancate giusto un pochino.

Grazie a Zia Irene, Zio Paolo, Jaisa e Alessio, siete sempre stati discreti nel chiedermi degli esami quando sapevate che non stava andando tutto come ci si sarebbe aspettato. Avete capito e rispettato ogni mio momento di silenzio. Grazie per esserci stati quando all'inizio ero un po' spaesata e mi avete presa per mano. Grazie alla piccola Camilla, con le sue risate e la sua voglia di giocare e cantare riesce a far dimenticare qualsiasi cosa. Continua a scoprire il mondo ogni giorno e mi raccomando anche nei momenti di difficoltà non mollare mai, che alla fine ne vale sempre la pena!

Grazie a Zio Lino, zia Concetta, Roberto e Antonello senza di voi probabilmente non sarei qui a Milano. Grazie per avermi ascoltata quando ero in lacrime, supplicando di convincere mio padre a lasciarmi partire. Ma soprattutto grazie per aver spiegato a mio padre quanto fosse importante per me. Avete visto quanto sia stato difficile, e tante volte sentivo che riuscivate a capirmi più di chiunque altro.

Grazie Alessandra, l'amica di una vita, quella persona che ti conosce più di chiunque altro, con cui hai condiviso 20 anni di vita. Quella persona che anche se non senti per mesi, quando poi ti ritrovi sembra come se non fosse passato neanche un secondo. Grazie per esserci sempre stata.

Grazie a tutti i miei amici di Barisciano, ad ogni mio ritorno eravate lì ad aspettarmi. Alessandro, grazie per i discorsi interminabili sul senso della vita. Abbiamo condiviso la stessa città e sappiamo solo noi tutte le difficoltà che abbiamo dovuto affrontare per ambientarci. Massi e Daniele, è sempre bello passare una serata con voi, grazie per avermi riportato un po' di casa qui a Milano.

Grazie a tutte le mie coinquiline che sono passate in questi 8 anni nell'ormai un po' mio appartamento De Gasperis. Di ognuno di voi porto con me un ricordo speciale. In particolare, ci tengo a ringraziare Francesca e Gaia, le veterane. Francesca, siamo entrate insieme in casa, ormai per me è come se fosse un po' casa nostra. Ricordo quando uno dei primi mesi sono venuta a disperarmi sul tuo letto e tu pronta a consolarmi. Grazie di avermi sempre supportata e capita, soprattutto dopo gli esami, argomento tabù in casa. Grazie per avermi raccolta quando mi sono ritrovata un po' sola ad affrontare un qualcosa più grande di me. Gaia, sei una forza

della natura. Non hai idea di quanto io ti stimi. Grazie per esserci sempre stata, per le serate puzzle e quelle Radio Italia, per gli aperitivi improvvisati e le infinite chiacchierate sul divano.

Ok non ho ancora finito, scusate, ma manca di ringraziare la mia famiglia acquisita. Quella con cui condividi la quotidianità, con cui passi 12 ore al giorno tra caffè al bancale e aperitivi in piazza Leo. Quella a cui non puoi nascondere nulla (o quasi), che mi conosce meglio di me. Grazie Giulia, Alessia, Matte, Fresco, Albi, Edo, Andrea, Ema, Fuma. Con voi sono riuscita ad essere me stessa, a non farmi problemi su ciò che dicevo e sul come lo dicevo, e chi mi conoscesse che per me non è una cosa scontata. Sono felice di aver condiviso con voi tanti momenti importanti e li porterò sempre con me nel mio cuore.

Giulia, ricordo quanto volevo venirti a parlare nei primi giorni di lezione, ma la paura che parlassi inglese era troppa. Fortunatamente poi ho sentito delle parole in italiano uscire dalla tua bocca e lì, mi sono fiondata a rivolgerti parola. Grazie per aver scelto me per condividere ogni cosa in questi 3 anni, lezioni, esami, tesi, perplessità, problemi di vita. Sei stata sempre lì, pronta ad ascoltarmi, e hai avuto una pazienza infinita in tutti quei momenti in cui mi rinchiudevo nella mia bolla solitaria. Non mi hai mai lasciata sola, eri sempre lì ad aspettarmi e a rispettare i miei tempi, grazie.

Alessia, grazie per riuscire ogni volta a trasmettermi la gioia della vita, la tua solarità è contagiosa. Grazie per avere sempre avuto la premura di spiegarmi ogni meme, reel...insomma queste cose da giovani che una vecchia come me fa fatica a capire. Grazie a te ora so la differenza tra "blastare" e "dissare".

Matte, grazie per avermi sempre ascoltata e avermi trasmesso la tua sensibilità. Sei una persona splendida ricordalo sempre.

Fresco, la tua premura nei confronti degli altri è impressionante, riesci ogni volta a trovare la parola giusta da dire e a rassicurare tutti. Grazie per esserci.

Grazie ad Albi, forse una delle prime persone che ho conosciuto in uni. Ho sempre ammirato la tua crescente passione per ciò che studiavi, forse a volte ne ero addirittura invidiosa. Grazie per la tua simpatia e per ribadire ogni volta quanto io sia boomer.

Grazie a TEA, una parte di questa associazione la sento un pò mia. Ha permesso di mettermi alla prova continuamente, di conoscere moltissime persone e cercare di nascondere la parte introversa di me, e tirare fuori una parte che pensavo di non avere.

Gio, sto ascoltando Ben Böhmer... Grazie per essere entrato nella mia vita in punta di piedi tra una birra del birrificio Lambrate bevuta e una rovesciata nel mio zaino. Grazie per aver passato le serate degli ultimi mesi a vagare in piazza Leo ascoltando le mie perplessità, lamentele e cercare di farmi capire che oltre il lavoro c'è altro. Grazie per puntualmente farmi la domanda: "ma tu lo conosci ..?" tirando fuori film, canzoni a me sconosciute, costringendomi a deluderti ogni volta! Te lo prometto ora che non ho più la scusa della laurea proverò a recuperare. Ma

soprattutto grazie per riuscire ogni volta a farmi ridere. TVBXSNLDM.

Marifrancesca ed Enrica, siete state le prime persone che ho conosciuto arrivata a Milano. Marifra, grazie per tutte le giornate di studio intense passate in Bicocca ad aspettare quei cinque minuti di pausa che poi minimo ne diventavano trenta. Grazie perché so che anche se non ci sentiamo spesso, ogni volta che ci ritroviamo siamo sempre complici. Enrica, abbiamo condiviso il fantastico collegio facendoci forza l'una con l'altra. Sappiamo quanto questa vita universitaria sia stata intensa per entrambe, e forse anche per quello siamo sempre riuscite a capirci. Grazie per avermi sempre supportato.

Insomma, grazie a tutti! Vi voglio bene!

Ah dimenticavo, ci tenevo a lasciare qui i ringraziamenti autoprodotti (forse avevano paura di non essere ringraziati:).

I would like to thank Giovanni for his valuable support during this work. Tutto quello che è in questa tesi non sarebbe stato possibile senza il suo supporto continuo e la sua grandissima simpatia nei miei confronti. È e sarà sempre una guida, pratica e spirituale.

Poi ringrazio gli altri soprattutto Liv. Thank you Liv! <3

Giovanni "Prego"

